

Warm-start sampling with square-root condition-number dependence

Abstract

For each fixed dimension exponent $\eta > 0$, fixed total variation accuracy, and polynomial squared- L^2 warmness bound, we give a finite positive sampler using at most $C\sqrt{\kappa}d^\eta$ original-gradient queries. The bound holds on every execution, assumes a supplied warm sample and sufficiently large dimension, and requires $\kappa \leq d^{100}$. The algorithm stores a latent position and momentum, makes a noisy Hamiltonian move, and draws one original-potential posterior per phase; its last posterior sample is the output. The proof combines a modified energy estimate for the latent chain with a simultaneous affine Picard iteration. A retained-noise mean estimate supplies the joint coupling and the stationary curvature bound. Complete component proofs and the analytic original-gradient construction are included.

Preliminary draft. The arguments and exposition remain subject to further scrutiny. No complete independent verification of the proofs is claimed; corrections and suggestions are welcome.

Source and contribution boundary. The analytic noisy-sampling and conditional-mean construction used here is adapted from OpenAI, *Subpolynomial query complexity for well-conditioned log-concave sampling* [Ope26]. Its Theorem 1.1 gives an every-execution $C_\eta d^\eta$ value-and-gradient query bound, for each fixed $\eta > 0$, at total variation accuracy $1/10$ when the minimizer is known and $I \preceq \nabla^2 V \preceq 2I$. The main improvement here is an extension to explicit square-root condition-number dependence in the supplied-warm setting: at fixed total variation accuracy and polynomial squared- L^2 warmness, the present sampler has an every-execution $C\sqrt{\kappa}d^\eta$ original-gradient query cap for each fixed $\eta > 0$, sufficiently large d , and $\kappa \leq d^{100}$. Preparation of the warm input is excluded. The underlying derivative estimates, stationary noisy-mean construction, baseline finite numerical formulas and recursive evaluation routines, and sampling Gaussian reserve are inherited. Explicit result citations and [Section M](#) distinguish that source material from the retained-noise strengthening, simultaneous-iteration organization, and latent-chain mixing argument developed here.

Contents

1	Main result	3
2	The sampling algorithm	4
3	Mixing of the latent chain	6
4	Affine numerical approximation	7
5	Error bounds and query complexity	10
A	A retained-noise mean at exact stationary centers	12

A.1	Statement and assumptions	12
A.2	A local absorption identity and the root support	12
A.3	The required posterior covariance inequality	13
A.4	One exact stationary-center reference	13
A.5	Simultaneous covariance and finite-error induction	14
B	The latent Hamiltonian kernel with its dormant transition	15
B.1	The modified quadratic form	16
B.2	Paying the actual dormant loss	17
B.3	Random stopping and the final OU step	17
B.4	The nonlinear curvature certificate	18
C	Simultaneous iteration for conditional means	18
C.1	Deterministic quadrature data	19
C.2	Flatten the free states before iterating	20
C.3	One fixed number of synchronous updates	21
C.4	Explicit substitution into the original-gradient recursion	23
D	Smoothing identities and probability metrics	24
D.1	Probability metrics and smoothing	24
E	Implementation by original gradients	25
E.1	Noisy sampling and noisy means	25
E.2	A finite affine composition	26
E.3	Removing the noise from a small posterior	28
F	Conditional gradient means and noisy sampling	29
F.1	Convexity, moments, and comparisons	30
F.2	Coupling and smoothing conventions	31
G	Gaussian conditional jets and probability transport	32
G.1	Cumulants from the Poincaré inequality	33
G.2	The transport equation	36
H	Adjoint divergence and material derivatives	36
H.1	Material derivatives of the conditional jets	37
H.2	Contractions and the expressions they generate	38
H.3	One final vector dimension factor	42
I	A stationary representation of a noisy mean	44
I.1	Transport Jacobians and the zero endpoint	44
I.2	A kernel for the centered observable	45
I.3	Stationarity and the path integral	46
I.4	Derivatives on a stationary trajectory	48
J	Finite Picard formulas for conditional means	49
J.1	Probability transport by global Picard iteration	50
J.2	The kernel action through harmonic trajectories	51
J.3	The zero endpoint of the kernel integral	53

J.4	A globally Lipschitz approximate centering velocity	54
J.5	The mean formula and the order of parameter choices	55
J.6	Bounds for the finite formulas	57
K	Recursive evaluation using original gradients	59
K.1	Why Gaussian center noise helps	60
K.2	The numerical input and the implementation task	60
K.3	Two routines and two different kinds of depth	61
K.4	Evaluating an expression at a noisy center	63
K.5	Query-preserving shifts and the deterministic cap	63
K.6	Terminal errors and propagation of scales	65
K.7	Stability for arbitrary fixed seeds	66
K.8	Standalone accuracy and the primitive theorem	66
L	The mean-root specialization	68
L.1	The recursive computation	69
L.2	Starting at a mean routine	69
L.3	Reserving a fresh Gaussian	69
M	Source correspondence and attribution	71

1 Main result

For a potential F , let π_F be the probability measure proportional to e^{-F} . Write

$$V_t = -\log(e^{-V} * \mathcal{N}(0, tI)), \quad \gamma_d = \mathcal{N}(0, I_d).$$

Additive constants in a potential are immaterial. Warmness means the squared L^2 norm of a relative density, namely $1 + \chi^2$.

Theorem 1.1 (Warm start original gradient complexity). *Fix $\eta > 0$, $C_w > 0$, and $\epsilon \in (0, 1/4)$. There are constants C, d_0 , depending only on these choices, with the following property. Let $d \geq d_0$, $1 \leq \kappa \leq d^{100}$, and let $V : \mathbb{R}^d \rightarrow \mathbb{R}$ be C^2 with supplied curvature bounds*

$$\kappa^{-1}I \preceq \nabla^2 V \preceq I.$$

Choose a sufficiently small fixed $e = e(\eta) > 0$, put $h = d^{-e}$ and $a = h^2$, and suppose a sample (Y_0, p_0) is supplied from a law ν with

$$\nu \ll \pi_{V_{2a}} \otimes \gamma_d, \quad \int \left(\frac{d\nu}{d(\pi_{V_{2a}} \otimes \gamma_d)} \right)^2 d(\pi_{V_{2a}} \otimes \gamma_d) \leq d^{C_w}.$$

A finite positive randomized algorithm, using only the original gradient ∇V , returns a sample within total variation ϵ of π_V . On every execution its original-gradient query count is at most

$$C\sqrt{\kappa} d^\eta.$$

The cap includes the initial posterior lift and every subsequent posterior call. The algorithm has no gradient gate or failure branch.

Corollary 1.2 (Original target warm input). *The same conclusion holds if a sample A_0 is supplied with*

$$\int \left(\frac{d\text{Law}(A_0)}{d\pi_V} \right)^2 d\pi_V \leq d^{C_w}.$$

In this case initialize directly by $X_0 = A_0 + \sqrt{a} Z_0$ and independent $p_0 \sim \gamma_d$. No initial posterior call is needed.

Proof. Gaussian convolution sends π_V to π_{V_a} and cannot increase the squared L^2 norm of a relative density. Adding independent reference momentum also preserves this bound. The proof below starts from this latent law. $\square$

Oracle convention and scope. A query evaluates ∇V at a real vector. Potential values, Hessians, smoothed gradients, exact posteriors, and proximal maps are not supplied oracles. Independent Gaussian and uniform draws and finite exact-real arithmetic, including deterministic quadrature constants and Gaussian seed transformations, are free in this query model. The claim is not an arithmetic-runtime or bit-complexity bound.

The warm sample is supplied and its preparation is not counted. No approximate minimizer, center-proximity promise, or density evaluation is needed afterward. All orders, tree shapes, and caps are chosen before oracle answers. The result concerns fixed accuracy, polynomial L^2 warmth, sufficiently large dimension, and $\kappa \leq d^{100}$. It does not provide a cold initialization, a small-dimension or large- κ fallback, or uniform logarithmic inverse-accuracy dependence.

Comparison with high-accuracy warm-start sampling. High-accuracy warm-start sampling offers a different trade-off. Proximal BPS [CCLZ26a, Theorem 4.3 and Corollary 4.4] uses only original gradients and, from a supplied input with bounded order-2 Rényi divergence, attains ϵ -total-variation accuracy with expected $\tilde{O}(\sqrt{\kappa} d^{1/4})$ queries, where the hidden factors are polylogarithmic in d , κ , and $1/\epsilon$; the corollary also gives explicit dependence on a general warmth bound. The present Theorem 1.1 instead fixes total-variation accuracy and polynomial squared- L^2 warmth, and gives an every-execution $C\sqrt{\kappa} d^\eta$ cap for each fixed $\eta > 0$, sufficiently large d , and $\kappa \leq d^{100}$. Warm-start preparation is excluded. Thus the arbitrarily small dimension exponent is a fixed-accuracy result in the stated regime, not a uniform high-accuracy replacement.

Proof organization. The sampler keeps one latent position and momentum. Each phase makes a noisy Hamiltonian move and one original-potential posterior draw; the last such draw is the answer. Its proof uses one latent modified energy, one retained-noise mean lemma, and one disjoint affine compiler. The numerical construction uses the same simultaneous affine Picard iteration at both the outer trajectory and the inner conditional-mean levels. The appendices prove these interfaces and retain the analytic construction of the original-gradient primitives. The simultaneous formulation in Appendix C replaces the nested numerical schedule when those primitives are used here.

2 The sampling algorithm

Set

$$\begin{aligned} U &= V_a, & \mu &= \pi_U \otimes \gamma_d, & \mu_{\text{obs}} &= \pi_{V_{2a}} \otimes \gamma_d, \\ m_a &= (\kappa + a)^{-1}, & \gamma &= \sqrt{m_a}, & \Gamma &= \gamma h, & s &= \frac{1}{2}, & n_0 &= \frac{\sqrt{3}}{2}. \end{aligned}$$

Increase d_0 so that $h \leq 1/16$. For the general input in [Theorem 1.1](#), first draw a finite approximation to $A_0 \sim R_{2a, Y_0}^V$ and set

$$X_0 = (Y_0 + A_0)/2 + \sqrt{a/2} Z_0, \quad (2.1)$$

retaining the supplied p_0 . Here $R_{w,y}^V$ denotes the density proportional to $\exp[-V(x) - |x - y|^2/(2w)]$. In the exact comparison, [\(2.1\)](#) is the posterior channel from μ_{obs} to μ and does not increase warmth. It preserves the actual momentum, including its correlation with Y_0 . For [Corollary 1.2](#), use the direct Gaussian initialization instead.

Algorithm 2.1 (One latent cycle). Starting from (X, p) , repeat the following steps for a predetermined number N of phases.

1. Draw $T \sim \text{Unif}[0, 1]$. With T fixed, form the two deterministic affine output formulas and their disjoint evaluation forests. Compute the momentum forest’s unit zero-oracle coefficient vector $c(T)$, including its parent momentum anchor. Draw its raw Gaussian tape conditionally by

$$\Xi = c(T) \otimes p + (I - c(T)c(T)^\top)Z.$$

Thus $c(T)^\top \Xi = p$ pathwise. The position tape is separate.

2. Execute the position and momentum forests using original gradients. Apply a fresh final momentum Ornstein–Uhlenbeck Gaussian. Call the resulting pair (Y, p_{new}) .
3. Execute the finite posterior supplier for $A_{\text{new}} \sim R_{2a, Y}^V$.
4. If another phase follows, draw fresh $Z_B \sim \gamma_d$ and retain

$$X_{\text{new}} = (Y + A_{\text{new}})/2 + \sqrt{a/2} Z_B, \quad p_{\text{new}}.$$

On the last phase, return A_{new} directly.

The algorithm calls the original posterior exactly N times, plus one initial call for the general observed input. It never draws an unused final bridge Gaussian. Each call is a finite original-gradient procedure with a predetermined cap, valid at every center.

The exact comparison law. The zero-oracle tape splits the entering momentum as $p = sG_0 + n_0H_0$. At stationarity, G_0, H_0 are independent standard Gaussians, independent of X . Solve, for $0 \leq t \leq T$,

$$\dot{X}_t = hG_t, \quad \dot{G}_t = -h\nabla U(X_t). \quad (2.2)$$

Conditional on this exact parent trajectory, apply the Gaussian-invariant dormant transition supplied by the retained-noise compiler to H_0 , obtaining H_1 . The ideal pair in step 2 is

$$Y = X_T + \sqrt{a} Z_X, \quad p_{\text{new}} = e^{-\Gamma}(sG_T + n_0H_1) + \sqrt{1 - e^{-2\Gamma}} Z_{\text{OU}}. \quad (2.3)$$

The observation Z_X is fresh relative to the parent and momentum-reference tape; Z_{OU} is independent of all earlier variables. Exact means, trajectories, and posterior laws in this description are comparison objects. The executed program uses their finite implementations.

3 Mixing of the latent chain

The general modified-energy hypocoercivity framework has earlier antecedents, including Dolbeault, Mouhot, and Schmeiser [DMS15]. The gap-shifted corrector and modified energy used here follow Fan, Li, and Lu [FLL26, Section 2, equations (22)–(23), and Section 3, Lemma 1]; the discrete latent-cycle and dormant-transition estimates are established in this manuscript.

Let $\mathcal{K}$ be the latent kernel consisting of (2.2), the dormant transition, and the final OU step, before the position observation. It preserves μ . Appendix B proves that its density operator satisfies, for centered f ,

$$\mathcal{E}(\mathcal{K}^*f) \leq \mathcal{E}(f) - \frac{\Gamma}{512} \|\mathcal{K}^*f\|_2^2, \quad \frac{31}{64} \|f\|_2^2 \leq \mathcal{E}(f) \leq \frac{33}{64} \|f\|_2^2. \quad (3.1)$$

Here, with Π denoting momentum averaging,

$$\mathcal{E}(f) = \frac{1}{2} \|f\|_2^2 + \varepsilon_0 \langle Af, f \rangle, \quad \varepsilon_0 = \gamma/16, \quad A = (m_a + L)^{-1} (\mathcal{T}\Pi)^*,$$

where $L = -\Delta + \nabla U \cdot \nabla$ and $\mathcal{T} = p \cdot \nabla_x - \nabla U \cdot \nabla_p$. All norms in this section use μ unless specified otherwise.

Lemma 3.1 (The latent Gaussian refresh). *Let $B : L^2(\mu_{\text{obs}}) \rightarrow L^2(\mu)$ be the posterior conditional expectation operator for $Y = X + \sqrt{a}Z$, leaving momentum unchanged, and let B^* be its adjoint. Set $R = BB^*$. The complete latent cycle has density operator $P^* = R\mathcal{K}^*$ and*

$$\mathcal{E}(P^*f) \leq (1 + \Gamma/528)^{-1} \mathcal{E}(f). \quad (3.2)$$

Consequently, if the initial squared L^2 norm is at most B_0 ,

$$\chi^2(\nu_j | \mu) \leq \frac{33}{31} (1 + \Gamma/528)^{-j} (B_0 - 1).$$

Proof. The operator R is a positive self-adjoint contraction. Put $C = I - R$. Conditional variance and the curvature $a^{-1}I + \nabla^2 U \succeq a^{-1}I$ of $X | Y$ give

$$0 \preceq C \preceq I, \quad \langle v, Cv \rangle \leq a \|\nabla_x v\|_2^2.$$

The resolvent and integrated Bochner bounds give $\|\nabla_x A\|, \|\nabla_x A^*\| \leq 1$; hence $\|C^{1/2}A\|, \|C^{1/2}A^*\| \leq \sqrt{a}$. For g , write $r = \|g\|_2$ and $s_0 = \|C^{1/2}g\|_2$. Spectral calculus yields

$$\|g\|_2^2 - \|Rg\|_2^2 = \langle g, (2C - C^2)g \rangle \geq s_0^2.$$

Moreover,

$$\langle ARg, Rg \rangle - \langle Ag, g \rangle = -\langle ACg, Rg \rangle - \langle Ag, Cg \rangle,$$

whose absolute value is at most $2\sqrt{a}s_0r$. Thus

$$\mathcal{E}(Rg) - \mathcal{E}(g) \leq -\frac{1}{4}s_0^2 + 4\varepsilon_0^2 ar^2 = -\frac{1}{4}s_0^2 + \frac{\gamma^2 a}{64} \|g\|_2^2.$$

Use $g = \mathcal{K}^*f$, $\gamma^2 a = \Gamma^2$, and $\Gamma \leq 1/16$. Equation (3.1) then gives

$$\mathcal{E}(P^*f) \leq \mathcal{E}(f) - \frac{\Gamma}{1024} \|\mathcal{K}^*f\|_2^2 \leq \mathcal{E}(f) - \frac{\Gamma}{528} \mathcal{E}(P^*f).$$

Iterate and use norm equivalence only at the initial and final endpoints. $\square$

Let H be the observation–posterior channel, with $\mathcal{K}$ applied separately: it draws the fresh observation $Y = X + \sqrt{a}Z_X$, then $A \sim R_{2a,Y}^V$, and outputs only A . It sends μ exactly to π_V , so its density operator is an L^2 contraction between these reference spaces. The last phase is $\mathcal{K}H$.

After $N - 1$ complete latent cycles, apply the last $\mathcal{K}$. Equation (3.1) implies $\mathcal{E}(\mathcal{K}^*f) \leq (1 + \Gamma/264)^{-1}\mathcal{E}(f)$. Contract L^2 through H and use energy equivalence only at the initial and final endpoints. The exact final error is therefore

$$\begin{aligned} \text{TV}(\text{Law}(A_N), \pi_V) &\leq \frac{1}{2} \sqrt{\frac{33}{31}(1 + \Gamma/264)^{-1}(1 + \Gamma/528)^{-(N-1)}(B_0 - 1)} \\ &\leq \frac{1}{2} \sqrt{\frac{33}{31}(1 + \Gamma/528)^{-N}(B_0 - 1)}. \end{aligned} \quad (3.3)$$

Thus $N = \max\{1, \lceil C\Gamma^{-1} \log((B_0 + 1)/\epsilon^2) \rceil\}$ suffices. All N Hamiltonian phases count toward mixing. The argument controls the actual final return channel; it does not infer the law of a hidden posterior draw from a close latent marginal.

4 Affine numerical approximation

The analytic primitives supply conditional means only as mathematical nodes. Their finite implementation, including the strengthened retained-noise conclusion below, ultimately terminates in original-gradient evaluations.

Lemma 4.1 (Simultaneous affine iteration). *Suppose $\Phi(E)_i = b_i + \sum_j a_{ij}u_j(E_j)$, every u_j is L_u -Lipschitz, and $q = L_u \max_i \sum_j |a_{ij}| < 1$. Define $E^{[0]} = b$ and $E^{[n+1]} = \Phi(E^{[n]})$. For a same-anchor benchmark E^* , if*

$$\max_i \|E_i^* - b_i\|_2 \leq rC_{\text{init}}D, \quad \max_i \|E_i^* - \Phi(E^*)_i\|_2 \leq r\epsilon_{\text{res}}D,$$

then

$$\max_i \|E_i^{[N]} - E_i^*\|_2/r \leq \left(q^N C_{\text{init}} + \frac{\epsilon_{\text{res}}}{1 - q} \right) D.$$

Proof. Subtract $\Phi(E^*)$, apply the deterministic row bound and Minkowski’s inequality, and sum the resulting geometric recurrence. The norm is the maximum of individual RMS norms, not the RMS of a maximum; no number-of-nodes factor appears. $\square$

For the outer phase, eliminate momentum to obtain

$$X_t = X + htG_0 - h^2 \int_0^t (t - r) \nabla U(X_r) dr. \quad (4.1)$$

Use $n = \lceil d^w \rceil$ cells, fixed interpolation order k , and J simultaneous layers. The center rows have coefficient mass at most $C_k h^2$. Final position and momentum correction masses are at most $C_k h^2$ and $C_k h$, respectively. The dependency depth is at most J plus a fixed number of final layers, independent of the mesh size.

Writing $Z = X/h$, the stationary-derivative estimates apply to the force $h\nabla U(hZ)$ with radius one. Its positive-order spatial jets have all-split norm at most $C_j h^2$. Thus its fixed-order time derivatives on the stationary trajectory have RMS at most $C_k h^2 \sqrt{d}$, with no h^{-k} loss. Apply Lemma 4.1 with row factor $C_k h^2$. First choose a fixed J for the requested error order q , next a small $w > 0$, and finally a sufficiently large fixed k . The resulting same-anchor errors satisfy

$$\text{RMS}(X^{\text{num}} - X_T) \leq h\sqrt{d}h^q, \quad \text{RMS}(G^{\text{num}} - G_T) \leq \sqrt{d}h^q. \quad (4.2)$$

Indeed, put $W(z) = U(hz)$. Since $a = h^2$, this is, up to an additive constant, $-\log(e^{-V(h\cdot)} * \gamma_d)(z)$. The rescaled dynamics are $\dot{Z}_t = G_t$, $\dot{G}_t = -\nabla W(Z_t)$, with invariant density proportional to $e^{-W(z)-|g|^2/2}$, where g is the dummy momentum coordinate. The conditional-cumulant argument of [Proposition G.2](#), now at Gaussian variance one and with $\text{Lip}(\nabla[V(h\cdot)]) \leq h^2$, gives the stated positive-order force-jet bounds. Apply [Lemma I.4](#) with $R = 1$, the constant symplectic coefficient matrix (so $\alpha = 1$), and $\phi_0 = \nabla W$, $a_0 = h^2$. The product potential has the required score-derivative bounds, and converting a one-slot split to Hilbert–Schmidt norm costs only $\sqrt{d}$. This gives the stated time-derivative bound without any inverse power of h .

The same argument applies to all requested intermediate output times. [Appendix C](#) applies the identical iteration to the physical centers inside each mean primitive, replacing nested probability, harmonic, and centering Picard solvers by one global update.

The strengthened mean interface. To target $\nabla U(z)$ use the primitive with

$$F(x) = V(\sqrt{h}x), \quad \delta = \lambda_F = h, \quad x_F = z/\sqrt{h}, \quad r_F = \sqrt{h}, \quad \sigma_F = \sqrt{h}\sigma.$$

Divide its output by $\sqrt{h}$. Each query to ∇F is one query to ∇V . Fix a base order K_0 larger than every requested accuracy order before taking $t_0 < 1/[320(K_0 + 2)]$. Under the size conditions verified below, [Appendix A](#) gives:

1. A deterministic query cap $B_d h^{-t_0}$, a fixed-seed center Lipschitz bound B_d , and a query-preserving shift $M(z + \Delta; \Xi + v \otimes \Delta/h) = M(z; \Xi)$, with $|v| \leq B_d h^{-t_0}$.
2. A deterministic zero-oracle coefficient C , $|C| = \sigma$, $C \cdot v = 0$, vanishing on private center and sampling-anchor subtrees.
3. A coupling retaining $P_0 = C^\top \Xi / \sigma$ identically, with comparison output $\nabla U(z) + \sigma P_1$, standard Gaussian endpoint marginals, and RMS error at most

$$B_d \sigma h^q D(z), \quad D(z) = \sqrt{d} + \sqrt{h} |\nabla V(z)|.$$

4. The exact stationary-center reference satisfies

$$\mathbb{E}[(P_1 - P_0)(P_1 - P_0)^\top] \preceq \frac{4a}{\sigma^2} \nabla^2 U(z).$$

Here $B_d = C(1 + \log d)^C$, with fixed-order constants. The input bookmark and output Gaussian generally differ.

The common disjoint compiler. Expand each output into an occurrence tree. Every mean occurrence has its own center supplier and Gaussian tape, even when two exact expressions are equal. The position and momentum trees are disjoint and share only (X, G_0, T) . Set

$$D_* = B_d h^{-t_0} (C_k n)^{J+2}, \quad \rho_c = h/(2D_*).$$

Choose $w(J + 2) \leq e/8$; with $t_0 \leq 1/8$, fixed logarithms are absorbed for large d , so $D_* \leq h^{-1/3}$. For a scalar row $E = b_{\text{free}} + \sum_i c_i M(\text{center}_i)$, a positive declared mass $A_E \geq \sum_i |c_i|$, and desired output width ν :

1. Request each mean at noise scale $\sigma_i = \nu/(2A_E)$.

2. Recursively supply its center at width ρ_c , on a disjoint tape.
3. Execute the mean with fresh raw seeds transformed by

$$P_i = (I - v_i v_i^\top / (4D_*^2))^{1/2}.$$

4. Add an independent residual Gaussian of variance $\nu^2 - \sum_i c_i^2 \sigma_i^2 \geq 3\nu^2/4$.

Mean-free rows return their affine value plus the declared Gaussian. Use $(\nu, b_{\text{free}}) = (h, X + hTG_0)$ at the position root and (n_0, sG_0) at the momentum root, with their respective coefficient masses. Positive declared masses handle vanishing coefficients and $T = 0$ without division by zero. Gaussian absorption is a proof identity; the program applies the predeclared P_i , not an unobserved comparison noise.

At an internal center row, $\sigma_i \geq c/(D_*h)$, so

$$h/\sigma_i \leq Ch^2 D_* \leq Ch^{5/3}.$$

The primitive parameters are $l = h^2$ and $L_F/\sigma_F = h/\sigma_i$; hence every call satisfies the mean-root conditions. Final output calls have scale comparable to $1/h$ and also qualify. The finite cap for both forests is

$$Q_{\text{forests}} \leq B_d(C_k n)^{J+2} h^{-t_0}. \quad (4.3)$$

Retaining the actual entering momentum. The momentum root has zero-oracle output

$$sG_0 + \sum_i b_i \sigma_i P_{0,i} + \beta_P G_P, \quad \beta_P^2 = n_0^2 - \sum_i b_i^2 \sigma_i^2 \geq 3n_0^2/4.$$

Its coefficient vector is the unit $c(T)$ in [Algorithm 2.1](#). All position slots and private center-supplying slots have zero coefficient. Child absorption preserves the active coefficients by $C \cdot v = 0$. Hence the entire finite/reference coupling retains (X, p) , even after conditioning the tape on an arbitrary prescribed p . Such a conditioned tape is not asserted to have independent coordinates.

After the private-center comparisons, move each output-active momentum child to the exact parent position X_{t_i} with its seeds fixed, using (4.2). Define

$$n_0 H_0 = \sum_i b_i \sigma_i P_{0,i} + \beta_P G_P, \quad n_0 H_1 = \sum_i b_i \sigma_i P_{1,i} + \beta_P G_P.$$

Conditional on the exact parent trajectory, the child pairs are independent and each increment is centered. The strengthened mean lemma gives

$$\mathbb{E}[(n_0(H_1 - H_0))(n_0(H_1 - H_0))^\top \mid \text{parent}] \preceq 4a \sum_i b_i^2 \nabla^2 U(X_{t_i}). \quad (4.4)$$

For $u = (m_a + L)^{-1} \varphi$, stationary transport and Bochner imply $\|\nabla^2 U(X_{t_i})^{1/2} \nabla u(X_T)\|_2 \leq (1+h)\|\varphi\|_2$. Thus the actual dormant kernel Q_T obeys

$$\|(Q_T - I)A^*\| \leq 2\sqrt{a} \left(\sum_i b_i^2 \right)^{1/2} (1+h) \leq C_k h^2 \leq h/32. \quad (4.5)$$

This is the hypothesis of [Appendix B](#); no linear conditional-mean assumption is used.

5 Error bounds and query complexity

Under stationary latent input, draw T first and a standalone Gaussian momentum tape, then define $p = c(T)^\top \Xi$. This has law μ and is the coupling used for all stationary estimates. The conditional tape law in [Algorithm 2.1](#) is exactly its conditional law given (X, p, T) .

The affine error induction propagates center errors with factor $B_d C_k h^2$. Intrinsic errors sum to the output noise budget times $B_d h^q D$. At each exact numerical or stationary replacement center, $\mathbb{E}D^2 = O(d)$: ∇V is Lipschitz, stationary scores have second moment $O(d)$, and the small-mass center rows have RMS displacements $O(h\sqrt{d})$. With fixed-order slack, the retained-input coupling satisfies

$$\text{RMS}(e_X) \leq Ch\sqrt{d}h^q, \quad \text{RMS}(e_p) \leq C\sqrt{d}h^q.$$

The position residual has variance at least $3a/4$ and is query-unused. The final OU seed supplies independent covariance $1 - e^{-2\Gamma} \geq c\Gamma$. Tilt the retained-input coupling by an entering relative density $f(X, p)$ with $\|f\|_2^2 \leq B_0$. Gaussian smoothing followed by Cauchy–Schwarz gives the averaged forest defect

$$\text{TV}(f\mu\hat{F}, f\mu F) \leq C\sqrt{B_0 d}h^q/\sqrt{\Gamma}. \quad (5.1)$$

The density does not depend on either fresh reserve. This bound concerns the actual entering joint law; it is not a marginal-TV estimate promoted to a joint assertion.

All-center posterior accuracy. The supplier in [Proposition E.3](#) has the pointwise error, for $w \leq h$, $r = \sqrt{w/h} \leq 1$, and predetermined depth L_0 ,

$$e_R(y) \leq CL_0 B_d h^{K_M} D(y) + Cr2^{-L_0/2} D(y) + Chr^2 2^{-L_0} \sqrt{d}, \quad (5.2)$$

and cap $L_0 B_d h^{-t_0}$ at every center. All calls here have $w = 2h^2 \leq h$. At stationarity write $Y = A + \sqrt{2a}Z$, with $A \sim \pi_V$. Integration by parts and $\nabla^2 V \preceq I$ give $\mathbb{E}|\nabla V(A)|^2 \leq d$, and hence

$$\mathbb{E}|\nabla V(Y)|^2 \leq (2 + 4a)d, \quad \mathbb{E}D(Y)^2 \leq Cd.$$

Every exact latent prefix, and its observed output, has squared L^2 norm at most B_0 , because its transitions are invariant Markov channels. Therefore

$$\mathbb{E}_{\text{warm}} D(Y) \leq C\sqrt{B_0 d}.$$

This also applies to the initial observed input in [\(2.1\)](#). Averaging [\(5.2\)](#) directly proves the required posterior error. No gradient gate, fallback, or moment bound on an erroneous implemented prefix is needed.

Chronological comparison. Replace the actual phases one at a time, using an exact prefix and a common implemented suffix. Split a one-phase comparison at its exact observed output: the forest defect is [\(5.1\)](#), and the posterior defect is [\(5.2\)](#) averaged under that exact output. Subsequent posterior, bridge, and suffix channels contract TV. Add the separately paid initial lift error when present. Together with [\(3.3\)](#), the total error is at most

$$\frac{1}{2} \sqrt{\frac{33}{31} (1 + \Gamma/528)^{-N} (B_0 - 1) + NC\sqrt{B_0 d}h^q/\sqrt{\Gamma} + (N + 1)\bar{e}_R}.$$

The initial-lift term is absent for [Corollary 1.2](#).

Choosing fixed orders and counting every call. Since $\kappa \leq d^{100}$, $\Gamma^{-1} \leq Cd^{50+e}$ and $B_0 \leq d^{C_w}$. First choose $e < \min\{\eta/4, 1/16\}$. Choose q, K_M , and $L_0 = C_L \log d$ with sufficiently large fixed exponents to make the last two errors at most $\epsilon/2$; for example the forest exponent is made negative with slack by taking $eq > 76 + (C_w + 1)/2 + 2e$. Choose the outer depth J and primitive base order K_0 accordingly, then choose t_0 and w as above, and finally the interpolation orders. Every choice precedes d and all oracle answers. The numerical and recursive thresholds are then enforced by increasing d_0 .

There are N forest pairs and at most $N + 1$ posteriors. Thus

$$Q_{\text{total}} \leq CNB_d h^{-t_0} ((C_k n)^{J+2} + L_0 + 1), \quad N \leq C_{\epsilon, C_w} \sqrt{\kappa} h^{-1} \log d.$$

Take $w(J + 2) \leq e/8$ and $t_0 \leq 1/8$. Absorb all fixed powers of $\log d$ into, for example, $d^{e/2}$. The remaining exponent is below $2e < \eta$, proving the every-execution cap in [Theorem 1.1](#). All finite calls remain capped at inaccurate centers as well. This completes the warm proof.

A A retained-noise mean at exact stationary centers

This appendix develops the retained-noise joint-coupling strengthening needed by the present latent construction. It builds on the inherited mean and sampling recursion of OpenAI [Ope26, Sections 6–7]; this strengthened statement is not quoted as an existing theorem of that source.

The only extra property needed beyond the standalone mean construction is a coupling that keeps its input Gaussian coordinate and controls the displacement of its output coordinate. We prove that property directly for one reference at exact stationary parent centers.

A.1 Statement and assumptions

Use the primitive notation

$$\nu_{x,r}(\mathrm{d}z) \propto e^{-|z|^2/2 - F(x+rz)} \mathrm{d}z, \quad u_{x,r} = \mathbb{E}_{\nu_{x,r}} \nabla F(x + rZ), \quad D_F(x) = \sqrt{d} + |\nabla F(x)|.$$

Assume $F \in C^2$ is convex, $0 \preceq \nabla^2 F \preceq \lambda I$, $\lambda = \delta = d^{-e} \leq 1$, $0 < r \leq 1$, and $\sigma > 0$. No upper bound on σ is imposed. Fix the primitive orders and labels as in Sections K and L, including

$$40(K_0 + 2)t < \frac{1}{4}, \quad R = \delta^t, \quad b_* = (1 - R^2)^{1/2}, \quad l = \lambda r^2, \quad L = \lambda r,$$

and the routine size conditions (K.2). Use the finite formulas of Appendix C, evaluated with the same uncached occurrence-tree recursion, labels, shifts, residuals, and terminal programs as in Section K. That appendix verifies the interface substitution. All thresholds are fixed-order thresholds.

Lemma A.1 (Strengthened mean primitive). *Every admissible mean call $\mathbf{M}(x, r, \sigma; b, p)$ has the inherited deterministic cap, query-preserving shift, and all-fixed-seed center stability. Freeze its scalar parameters and replace only its gradient oracle by zero. Its output is $C^\top \Xi$, where C is center independent, $|C| = \sigma$, and $C \cdot v = 0$ for its shift vector v . The coefficient C is zero on every private center-supplying and private sampling-anchor subtree.*

There is a recursively defined reference pair (P_0, P_1) with standard Gaussian marginals and

$$P_0 = C^\top \Xi / \sigma, \quad \|\mathbf{M} - (u_{x,r} + \sigma P_1)\|_2 \leq B_d \sigma \delta^p D_F(x),$$

*in a coupling that retains P_0 identically. With $F_a = -\log(e^{-F} * \mathcal{N}(0, aI))$ and $a = r^2$, this reference also satisfies*

$$\mathbb{E}[(P_1 - P_0)(P_1 - P_0)^\top] \preceq \frac{4r^2 \lambda}{\sigma^2} \nabla^2 F_a(x). \quad (\text{A.1})$$

The RMS and covariance statements use the standalone stationary Gaussian seed law. They retain the center, scale, and legitimate outside parameters, but do not expose private query transcripts or sampling-anchor outputs.

A.2 A local absorption identity and the root support

At a terminal mean, $\mathbf{M} = \nabla F(x + rZ) + \sigma G$, so its zero-oracle coefficient is σe_G and its shift vector is $-e_Z$. At a nonterminal mean, the direct term is $(\sigma/2)G_0$. Zero-oracle child outputs do not depend on their centers. Induction therefore makes the whole zero-oracle output a center-independent linear Gaussian expression. Applying its exact shift identity gives $C \cdot v = 0$.

For a child of scale r_i , let $D_s = \delta^{-t}$ be the inherited shift budget and supply its center at width $r_i/(2D_s)$. For its shift v , put $u = v/(2D_s)$ and $P = (I - uu^\top)^{1/2}$. If Z is a standard Gaussian center error independent of the child tape G , then

$$H = PG - u \otimes Z \sim \gamma, \quad C^\top H = C^\top G. \quad (\text{A.2})$$

Indeed $PP^\top + uu^\top = I$, while $C \cdot u = 0$ implies $PC = C$. The primitive shift identity identifies its value at the ideal noisy center $x_i + r_i Z / (2D_s)$ with its value at the exact center and tape H . This is a conditional identity under the parent anchor law, not a claim conditional on the center error Z .

The support assertion also follows by induction. A center supplier has its own disjoint slots; a zero-oracle mean invoked at that center ignores the center, so none of those slots enters the invoked output. The direct mean term ignores its private position anchor. Consequently the whole position-anchor subtree, including its sampling routine's reserved noise, has zero root coefficient. Final-expression mean children are the only active descendants: on each of their tapes the root coefficient is a deterministic scalar multiple of that child's C . Absorption fixes this coefficient by (A.2). Every absorption within a private subtree fixes the root because its coefficient there is zero. Direct momentum and residual slots are unchanged.

This does not apply a mean shift to a sampling output. A sampling routine has shift law $S(x + \Delta; \Xi + v_S \otimes \Delta/r) = S(x; \Xi) - \Delta/r$; its own zero-oracle coefficient satisfies $C_S \cdot v_S = -1$. It causes no problem here precisely because the entire private sampling subtree has zero coefficient in the mean bookmark. Its approximation error is still paid.

The variance identity at each row gives the declared Gaussian budget. Since $PC = C$, transforming a child's tape does not change its zero-oracle coefficient or variance. At the mean root, the direct momentum variance plus the final evaluator budget is $\sigma^2/4 + 3\sigma^2/4 = \sigma^2$. This proves $|C| = \sigma$.

A.3 The required posterior covariance inequality

Let A have the width- a posterior under F , and write $H = \nabla^2 F(A)$ and $\Sigma = \text{Cov}(\nabla F(A))$. Brascamp–Lieb, applied to each scalar projection of ∇F , gives

$$\Sigma \preceq \mathbb{E}[H(a^{-1}I + H)^{-1}H] \preceq \frac{a\lambda}{1 + a\lambda} \mathbb{E}H.$$

The second inequality is pointwise spectral calculus for $0 \preceq H \preceq \lambda I$. Since $\nabla^2 F_a = \mathbb{E}H - \Sigma$, rearrangement yields

$$\text{Cov}(\nabla F(A)) \preceq a\lambda \nabla^2 F_a(x). \quad (\text{A.3})$$

The posterior Hessian is strictly positive, so its inverse is legitimate; no inverse of H or $\nabla^2 F_a$ is used. Smooth approximation extends the calculation to the stated C^2 assumptions. This is the only posterior-gradient curvature input required here.

Now put $A = x + rZ$ and observe $Y = b_*Z + RN$, with N independent Gaussian. Gaussian conditioning gives the child center and variance $x + rb_*Y$ and aR^2 . The total-covariance identity then gives

$$\begin{aligned} \mathbb{E}_Y \nabla^2 F_{aR^2}(x + rb_*Y) &= \nabla^2 F_a(x) + \text{Cov}(\mathbb{E}[\nabla F(A) \mid Y]) \\ &\preceq (1 + a\lambda) \nabla^2 F_a(x). \end{aligned} \quad (\text{A.4})$$

This is a posterior-averaged relation, not a pointwise comparison at unrelated centers.

A.4 One exact stationary-center reference

At a terminal mean put $P_1 = P_0$. At a nonterminal parent, draw ideal independent anchors $Y_0 \sim \xi_*$ and $G_0 \sim \gamma_d$, where ξ_* is the law of $b_*Z + RN$ above. With $s = \sigma/2$, run the exact stationary centering flow of Section I:

$$\dot{Y}_t = -K(Y_t)^\top G_t/s, \quad \dot{G}_t = (M_*(Y_t) - u_{x,r})/s.$$

It preserves $\xi_* \otimes \gamma_d$ and obeys

$$sG_0 + \int_0^1 M_*(Y_t) dt = u_{x,r} + sG_1. \quad (\text{A.5})$$

At every final quadrature time t_i , put $x_i^* = x + rb_* Y_{t_i}$ and use child scale rR . Conditional on the parent anchors, draw independent child reference pairs $(P_{0,i}, P_{1,i})$ at these exact centers. Let ω_i be the final positive weights, with $\sum_i \omega_i = 1$, and use the actual program's child noise scales σ_i and residual β_E . Define

$$\begin{aligned} P_0 &= \sigma^{-1} \left(sG_0 + \sum_i \omega_i \sigma_i P_{0,i} + \beta_E G_E \right), \\ P_1 &= \sigma^{-1} \left(sG_1 + \sum_i \omega_i \sigma_i P_{1,i} + \beta_E G_E \right). \end{aligned} \quad (\text{A.6})$$

The variance identity is $s^2 + \sum_i \omega_i^2 \sigma_i^2 + \beta_E^2 = \sigma^2$. Each endpoint collection, separately, is conditionally product Gaussian with a law independent of the parent anchors. The child output noise is therefore independent of the entire exact parent path. Parent stationarity proves that both endpoints in (A.6) are standard Gaussian. Its first line is exactly the actual zero-oracle coefficient recursion. Disintegration defines the required Gaussian-invariant kernel, which need not have a linear conditional mean.

A.5 Simultaneous covariance and finite-error induction

We prove the covariance bound and retained-input RMS estimate by induction on the same finite routine height. Terminal displacement is zero, and terminal accuracy is the original terminal estimate with the output Gaussian fixed.

After idealizing the private anchor, condition only on the parent anchors, without additionally conditioning on a private center Gaussian. Each ideal noisy numerical center uses a Gaussian independent of the invoked child tape. Absorption therefore gives the standalone child-tape law conditionally on the parent anchors and preserves the bookmark $C^\top \Xi$. With that tape fixed, move the child to the exact-path center, then apply the strengthened child coupling independently across disjoint occurrences. Each child endpoint collection separately has a conditional product-standard-Gaussian law independent of the parent anchors, although the joint pair law may depend on those anchors. Thus child increments are centered conditionally on the parent path; centering conditionally on the input bookmark is not required.

For the covariance step, condition on parent anchors. Every child displacement is centered, and different children are independent. Thus all cross terms between children, and between a child and the parent increment, vanish. For a deterministic vector v , stationarity and (A.3) give

$$\mathbb{E} \left| v \cdot \frac{s(G_1 - G_0)}{\sigma} \right|^2 \leq \sigma^{-2} \text{Var}(v \cdot M_*(Y)) \leq \frac{r^2 \lambda}{\sigma^2} v^\top \nabla^2 F_a(x) v.$$

If a child covariance coefficient is at most c , its contribution, after (A.4), is at most

$$c \omega_i^2 \frac{r^2 R^2 \lambda}{\sigma^2} (1 + l) v^\top \nabla^2 F_a(x) v.$$

Since $\sum_i \omega_i^2 \leq 1$, the scalar covariance coefficients satisfy

$$c_{\text{parent}} \leq 1 + R^2 (1 + l) c_{\text{child}}.$$

Taking $R \leq 1/2$ and $l \leq 1/2$ bounds all coefficients by $8/5$, hence by the stated 4. This bound uses exact stationary child centers and the positive GL final weights. For the historical signed-weight formula the same argument uses its squared absolute mass and a correspondingly smaller fixed R .

For the finite-error step, make the following comparisons, preserving the root bookmark throughout.

1. Couple the private sampling anchor to its ideal law. Its entire block has zero root coefficient; use the inherited fixed-seed anchor stability.
2. Conditional on the parent anchors, couple each private center supplier to its ideal Gaussian-noised numerical center, keeping the independent invoked child tape fixed. The all-fixed-seed center bound pays the supplied-center error.
3. Absorb the ideal center noise with (A.2). The invoked child now has its standalone Gaussian tape and the same input bookmark.
4. With that tape fixed, move each output-active numerical center $x_i^{\text{num}} = x + rb_* y_i$ to $x_i^* = x + rb_* Y_{t_i}$. Appendix C gives the same-anchor state estimate $\|y_i - Y_{t_i}\|_2 \leq B_d \delta^J D_F(x)$ at a prescribed order J exceeding the desired error order. The child's fixed-seed center bound $B_d \lambda$ pays at most $B_d \lambda r \delta^J D_F(x)$ after summing the final weights. The routine size bound on $\lambda r / \sigma$ places this within $B_d \sigma \delta^p D_F(x)$. Center independence leaves its bookmark unchanged.
5. Apply the child induction at each exact center using independent conditional couplings. Then replace the exact-path quadrature by (A.5), retaining the same child endpoint noise and final residual.

The center-moment bound $\|D_F(x_i^*)\|_2 \leq C D_F(x)$ follows from the stationary observation law and the primitive moment estimates. All other scale, label, and height estimates are exactly those of the original two-type error induction in Section K. The new numerical formula provides its same coefficient and error interfaces by Appendix C. Thus the same finite error recurrence closes, with only fixed-order enlargement of B_d .

Private subtrees require their marginal error guarantees, while active final children require this strengthened induction. No query is added by an analysis coupling or an exact-center replacement. An external density tilt is applied only later, to the whole retained-input coupling with its fresh output reserves, as in (5.1). The result does not identify P_1 with P_0 , expose private transcripts, or provide a cold-source guarantee.

B The latent Hamiltonian kernel with its dormant transition

We prove the additive estimate used by the main argument for the actual dormant-inclusive kernel. Let $mI \preceq \nabla^2 U \preceq I$, $\gamma = \sqrt{m}$, $0 < h \leq 1/16$, $\Gamma = \gamma h$, and $\mu = \pi_U \otimes \gamma_d$.

Split physical momentum by $G = sp + ne$, $H = np - se$, where $s = 1/2$, $n = \sqrt{3}/2$, and e is independent standard Gaussian. At stationarity (x, G, H) has product law. Evolve (x, G) by Hamiltonian dynamics for time hT , T uniform on $[0, 1]$. Given the exact parent trajectory, apply a Gaussian-invariant kernel Q_T to H , obtaining H_1 . Recombine $sG_T + nH_1$, discard the hidden momentum and stopping time, and apply the momentum OU kernel for time Γ .

For fixed T , invertibility of the parent Hamiltonian flow makes the entire parent path a deterministic function of its terminal (X_T, G_T) ; consequently Q_T is a kernel on H conditional on that current pair, holding the pair fixed and preserving the conditional standard Gaussian law.

Proposition B.1 (Dormant-inclusive latent dissipation). *Let A be the resolvent corrector defined below. Suppose, uniformly in the fixed stopping time,*

$$\|(Q_t - I)A^*\| \leq b \leq h/32 \quad (\text{B.1})$$

at the stationary parent law. Then the resulting invariant kernel $\mathcal{K}$ satisfies

$$\mathcal{E}(\mathcal{K}^*f) \leq \mathcal{E}(f) - \frac{\Gamma}{512} \|\mathcal{K}^*f\|_2^2 \leq \mathcal{E}(f) - \frac{\Gamma}{264} \mathcal{E}(\mathcal{K}^*f)$$

for centered $f \in L^2(\mu)$, where $\mathcal{E}$ has norm-equivalence constants $31/64$ and $33/64$. No reversibility or linear conditional mean is assumed for Q_t .

B.1 The modified quadratic form

The gap-shifted corrector and modified-energy method follow Fan, Li, and Lu [FLL26, Section 2, equations (22)–(23), and Section 3, Lemma 1]. The discrete latent-cycle and dormant-transition estimates used below are established in this manuscript.

Work first in the extended stationary space with physical Gaussian coordinates (P, E) . Let Π average both momenta, let

$$T_P = P \cdot \nabla_x - \nabla U \cdot \nabla_P, \quad T_E = E \cdot \nabla_x - \nabla U \cdot \nabla_E, \quad T_G = sT_P + nT_E,$$

and put

$$K = -\Delta_x + \nabla U \cdot \nabla_x, \quad A = (m + K)^{-1}(T_P \Pi)^*.$$

The norms below are in the corresponding stationary L^2 spaces. For a mean-zero extended density f , write

$$M = \|\Pi f\|_2, \quad R = \|(I - \Pi)f\|_2.$$

The resolvent and the Poincaré inequality for π_U give

$$\|A\| \leq \frac{1}{2\gamma}, \quad \langle AT_G \Pi f, \Pi f \rangle \geq \frac{s}{2} M^2, \quad \|T_G A\| \leq 1. \quad (\text{B.2})$$

The middle inequality uses $K(m + K)^{-1} \succeq I/2$ on mean-zero macroscopic functions.

To estimate the remaining transport term, let $u = (m + K)^{-1}g$. The centered adjoint transport field is, up to its irrelevant overall sign,

$$s(PP^\top - I) : \nabla^2 u + nP^\top \nabla^2 u E.$$

The two terms are orthogonal Gaussian chaoses. Their total squared norm is

$$(2s^2 + n^2) \|\nabla^2 u\|_2^2 = (1 + s^2) \|\nabla^2 u\|_2^2.$$

Integrated Bochner and the resolvent bound consequently imply

$$\|AT_G(I - \Pi)\| \leq \sqrt{1 + s^2} = \frac{\sqrt{5}}{2}. \quad (\text{B.3})$$

For the Hamiltonian density evolution $f'_t = -hT_G f_t$, $0 \leq t \leq 1$, define $J(t) = \langle A f_t, f_t \rangle$. Equations (B.2)–(B.3) yield

$$J'(t) \leq h \left(-\frac{1}{4} M_t^2 + \frac{\sqrt{5}}{2} M_t R_t + R_t^2 \right)$$

$$\leq h \left(-\frac{1}{8} M_t^2 + \frac{7}{2} R_t^2 \right). \quad (\text{B.4})$$

The last inequality is Young's inequality, with $M_t^2/8$ allocated to the mixed term.

Set $\varepsilon_0 = \gamma/16$ and introduce

$$\mathcal{E}(f) = \frac{1}{2} \|f\|_2^2 + \varepsilon_0 \langle Af, f \rangle. \quad (\text{B.5})$$

Its norm equivalence is

$$\frac{31}{64} \|f\|_2^2 \leq \mathcal{E}(f) \leq \frac{33}{64} \|f\|_2^2. \quad (\text{B.6})$$

Hamiltonian transport is unitary, so $N_0 = \|f_t\|_2^2$ is constant. Multiplying (B.4) by ε_0 gives

$$\frac{d}{dt} \mathcal{E}(f_t) \leq -\frac{\Gamma}{128} N_0 + \frac{29\Gamma}{128} R_t^2. \quad (\text{B.7})$$

B.2 Paying the actual dormant loss

The conditional kernel fixes macroscopic functions and preserves their projection. Put $q_t = Q_t^* f_t$ and

$$D_t = \|f_t\|_2^2 - \|q_t\|_2^2 \geq 0.$$

By adjunction, (B.1) bounds $A(Q_t^* - I)$ by b ; that operator annihilates macroscopic functions. Thus

$$\mathcal{E}(q_t) - \mathcal{E}(f_t) \leq -\frac{1}{2} D_t + \varepsilon_0 b M_t R_t \leq -\frac{1}{2} D_t + \frac{1}{2} \varepsilon_0 b \|f_t\|_2^2. \quad (\text{B.8})$$

Because the macroscopic projection is unchanged, the microscopic norm obeys the exact identity

$$R_t^2 = \|(I - \Pi)q_t\|_2^2 + D_t.$$

This equality is why a contraction depending on the parent history is controlled by its actual loss rather than by invariance alone.

B.3 Random stopping and the final OU step

Let $v_t = \Pi_E q_t$, $z_t = q_t - v_t$, and $\bar{f} = \int_0^1 v_t dt$. Projection onto the visible momentum leaves the corrector unchanged. The quadratic projection and averaging identities give

$$\mathcal{E}(\bar{f}) \leq \int_0^1 \mathcal{E}(q_t) dt - \frac{1}{2} \int_0^1 \|z_t\|_2^2 dt - \frac{31}{64} \int_0^1 \|v_t - \bar{f}\|_2^2 dt. \quad (\text{B.9})$$

Integrate (B.7) with survival weight $1 - t$, and then use (B.8). The positive coefficient of D_t is at most $29\Gamma/128$, so it is absorbed by $D_t/2$. Write the visible microscopic part of v_t as its average plus a fluctuation. The bound $\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$, together with $\int_0^1 (1 - t) dt = 1/2$, leaves at most $29\Gamma/128$ on the averaged microscopic norm; its fluctuation coefficient is at most $58\Gamma/128 < 31/64$. The hidden fluctuation is absorbed by its $1/2$ loss. With $N_0 = \|f_t\|_2^2$ and $\bar{R} = (I - \Pi_P)\bar{f}$, this gives

$$\mathcal{E}(\bar{f}) \leq \mathcal{E}(f_0) - \left(\frac{\Gamma}{256} - \frac{\varepsilon_0 b}{2} \right) N_0 + \frac{29\Gamma}{128} \|\bar{R}\|_2^2. \quad (\text{B.10})$$

The coefficient in parentheses is at least $3\Gamma/1024$.

For the final OU step, $1 - e^{-2\Gamma} \geq 15\Gamma/8$; half its squared-norm loss is at least $15\Gamma\|\bar{R}\|_2^2/16$. Its corrector change has absolute value at most

$$\frac{\Gamma}{32}\bar{M}\|\bar{R}\|_2 \leq \frac{\Gamma}{1024}\bar{M}^2 + \frac{\Gamma}{4}\|\bar{R}\|_2^2, \quad \bar{M} = \|\Pi_P \bar{f}\|_2.$$

Use $N_0 \geq \bar{M}^2 + \|\bar{R}\|_2^2$ in (B.10). The remaining coefficients are at most $-\Gamma/512$ on $\bar{M}^2$ and $-475\Gamma/1024$ on $\|\bar{R}\|_2^2$. Since OU is an L^2 contraction, this proves

$$\mathcal{E}(f_{\text{out}}) \leq \mathcal{E}(f_0) - \frac{\Gamma}{512}\|f_{\text{out}}\|_2^2 \leq \mathcal{E}(f_0) - \frac{\Gamma}{264}\mathcal{E}(f_{\text{out}}).$$

All calculations hold first on a smooth core. Approximation preserving Hessian bounds and lower semicontinuity extend them to bounded-Hessian C^2 potentials and arbitrary centered L^2 densities.

B.4 The nonlinear curvature certificate

For $u = (m + K)^{-1}g$, stationary Hamiltonian transport and integrated Bochner yield, at physical times $r, t \in [0, h]$,

$$\|\nabla u(X_t) - \nabla u(X_r)\|_2 \leq |t - r|\|\nabla^2 u\|_2, \quad \|\nabla^2 U(X_r)^{1/2}\nabla u(X_r)\|_2 \leq \|g\|_2.$$

Since $\nabla^2 U \preceq I$,

$$\|\nabla^2 U(X_r)^{1/2}\nabla u(X_t)\|_2 \leq (1 + h)\|g\|_2. \quad (\text{B.11})$$

The same Bochner calculation and resolvent spectral calculus also give $\|\nabla_x A\|, \|\nabla_x A^*\| \leq 1$, as used in Lemma 3.1. For $A^*g = p \cdot \nabla(m + K)^{-1}\Pi g$ the assertion is the Hessian resolvent bound; for A it follows by the polar decomposition of $\mathcal{T}\Pi$ and $\|K(m + K)^{-1}\| \leq 1$.

For the actual dormant pair, $A^*g = (sG_t + nH) \cdot \nabla u(X_t)$, up to an irrelevant sign. Conditional Jensen gives

$$\|(Q_t - I)A^*g\|_2^2 \leq \mathbb{E} \left[\nabla u(X_t)^\top \mathbb{E}[(n\Delta H)(n\Delta H)^\top \mid \text{parent}] \nabla u(X_t) \right].$$

If the conditional covariance is bounded by $4a \sum_i c_i^2 \nabla^2 U(X_{r_i})$, where the times and weights are deterministic conditional on the independently drawn clock and $\sum_i |c_i| \leq Ch$, then (B.11) gives

$$\|(Q_t - I)A^*\| \leq 2\sqrt{a} \left(\sum_i c_i^2 \right)^{1/2} (1 + h) \leq 2C\sqrt{a}h(1 + h).$$

With $a = h^2$, this is $O(h^2)$ and satisfies (B.1) beyond a fixed threshold. This applies to (4.4) at the exact parent centers. It does not substitute approximate centers into an unbounded resolvent test or replace the nonlinear kernel by an affine Gaussian channel.

C Simultaneous iteration for conditional means

The simultaneous-iteration organization below is the present manuscript's replacement for the nested numerical schedule in OpenAI [Ope26, Section 6], used with the inherited recursion of its Section 7.

This appendix gives the finite numerical formulas executed in the present sampler. The nested formulas proved in Section J remain a useful analytic antecedent; they are not executed in addition. We replace their numerical interface by the simultaneous formula below and then use the same original-gradient recursion.

C.1 Deterministic quadrature data

Fix a primitive base order $K_0 \geq 2$, $40(K_0 + 2)t < 1/4$, and

$$R = \delta^t, \quad b_* = (1 - R^2)^{1/2}, \quad \psi = \delta^{4t}, \quad h_0 = \delta^w, \quad s = \sigma/2.$$

The quantities $l = \lambda r^2$ and $L = \lambda r$ satisfy the original numerical size assumptions, with $b \geq 1/2$. All constants below may depend on the fixed orders, but not on the center, r , σ , or oracle values.

Let $x_j \in (0, 1)$ be the m Gauss–Legendre nodes, $\omega_j > 0$ their weights, and ℓ_j their cardinal polynomials. These are fixed real constants in the stated exact-real query model. We use the following data.

1. Probability meshes have equal steps of size at most h_0 in $-\log(1 - \rho)$. On each cell use the m GL source nodes. Include source nodes and endpoints among state targets. For backward transport on $[0, b_*]$, also include zero and the $m - 2$ special first-cell targets defined below. For each forward transport from ρ to b_* use its own deterministic mesh. Sampling transport uses the mesh from zero to $\rho_e = (1 + \tau^2/4)^{-1/2}$.
2. Harmonic transport on $[0, z]$ uses GL source nodes and the endpoint z as a state target, retaining its free sine and cosine solution exactly. Angle differentiation uses the m GL nodes on $[0, \psi]$, differentiating their interpolating polynomial at zero.
3. The centering flow uses the m GL nodes on $[0, 1]$ as both source and state targets. Its final mean output uses the full positive weights ω_j .
4. For the kernel integral of $f'(\rho)/\rho$, integrate the derivative of the ordinary GL interpolant separately on each ordinary cell. On the first cell $[0, \ell]$, let $n = m - 2$, use its n GL nodes z_j , and set

$$Q(x) = f(0) + x^2 I_n \left[\frac{f(\ell z_j) - f(0)}{z_j^2} \right] (x).$$

This interpolates the specified values and enforces $Q'(0) = 0$ for every supplied value list. Integrate $\ell^{-1}Q'(\rho/\ell)/\rho$ on that cell. There is no derivative oracle.

The kernel uses cellwise polynomial derivatives, not distributional derivatives of a possibly discontinuous piecewise interpolant. Thus no artificial endpoint jump is included.

Here are the coefficient estimates needed for the finite construction. Orthogonal-polynomial reproduction gives, for $a_j = \int_0^1 g(x) \ell_j(x) dx$,

$$\sum_j a_j^2 / \omega_j \leq \|g\|_2^2, \quad \sum_j |a_j| \leq \|g\|_2.$$

Indeed, in an orthonormal polynomial basis the first sum is the squared norm of the degree- $(m - 1)$ projection of g . Cauchy–Schwarz and $\sum_j \omega_j = 1$ give the second inequality. Partial integration rows and sine-weighted rows consequently have mass at most the interval length. The final weights are positive because Gaussian exactness gives $\omega_j = \int \ell_j^2 > 0$.

For $q = 0, 1, 2$, shifted Legendre derivative bounds and the same reproducing kernel give

$$\sup_{x \in [0, 1]} \sum_j |\ell_j^{(q)}(x)| \leq m^{2q+1}.$$

These follow from $|P_k| \leq 1$, the endpoint formula $P_k^{(q)}(1) = (k+q)!/[2^q q!(k-q)!]$, and the nonnegative-coefficient derivative expansion of Legendre polynomials. Thus the angle derivative row has mass at most m^3/ψ .

The smallest n -node GL point satisfies $z_{\min} \geq 1/[n(n+1)]$, by the mean-value theorem between the endpoint and first root and $\sup |P'_n| = n(n+1)/2$. If $P = I_n[(f(\ell z_j) - f(0))/z_j^2]$, the first-cell integral equals

$$\ell^{-1} \left(P(1) + \int_0^1 P(x) dx \right).$$

Its row mass is at most $2m^5/\ell$. On the geometric mesh, the first cell has length at least $h_0/4$, and every other cell length is comparable to $h_0(1-\rho)$. Summing the derivative rows against $1/\rho$ gives

$$\sum_{\rho} |q_{\rho}| \leq 32m^5 h_0^{-1} G, \quad G = 1 + |\log R| + |\log h_0|. \quad (\text{C.1})$$

Taylor subtraction proves the corresponding error bounds: a weighted integration row with kernel bounded by one has residual at most $2\ell^{m+1}D_m/m!$; an ordinary derivative interpolant has residual at most $\ell^{m-1}D_m[1/(m-1)! + m^3/m!]$; the first-cell singular integral, when $f'(0) = 0$, has residual at most $\ell^{m-1}D_m[1/(m-2)! + 7(m-2)^5/m!]$. Here D_m is the supremum of the individual L^2 derivative norms on that deterministic cell. These are nodewise RMS estimates, not random-path supremum estimates.

C.2 Flatten the free states before iterating

Write $M_{\rho}(v) = u_{x+r\rho v, r\sqrt{1-\rho^2}}$ and $M_0 = u_{x,r}$. The following are uniterated collocation equations; they will specify one finite affine system, not an implicit runtime solve. At centering nodes a, b , put

$$y_a = y_0 + \sum_b w_{ab} V_b, \quad g_a = G_0 + s^{-1} \sum_b w_{ab} [M_*(y_b) - M_0].$$

For each centering node b , probability node ρ , and angle node z , introduce backward, harmonic, and forward states:

$$\begin{aligned} B_{b,\eta} &= y_b - r \sum_j w_{\eta,j}^{\text{back}} M_{\eta_j}(B_{b,\eta_j}), \\ W_{b,\rho,z,v} &= \cos(v) B_{b,\rho} + \sin(v) g_b - r \rho \sum_j w_{v,j}^{\text{sin}} M_{\rho}(W_{b,\rho,z,v_j}), \\ F_{b,\rho,z,\eta} &= W_{b,\rho,z,z} - r \sum_j w_{\eta,j}^{\text{forw}} M_{\eta_j}(F_{b,\rho,z,\eta_j}), \\ V_b &= s^{-1} \sum_{\rho} q_{\rho} \sum_z d_z \sum_j w_{b*,j}^{\text{forw}} M_{\eta_j}(F_{b,\rho,z,\eta_j}). \end{aligned}$$

Backward weights integrate from b_* to the indicated target, and forward weights from ρ to the indicated target. Harmonic weights include the sine kernel. The final line is the source cancellation of $-1/(rs)$ against the forward correction $-r$ times its mean integral; no approximate matrix is multiplied by an approximate momentum.

Attach a physical center variable E_i to every mean occurrence and replace it by u_{E_i, r_i} . Substitute only explicit free-state terms: centering states into backward inputs, then backward and momentum states into harmonic inputs, then harmonic endpoints into forward inputs. Stop at every mean

value. This triangular free-state elimination has at most three local stages, although the resulting mean-dependence system may be cyclic. It gives

$$E_i = b_i + \sum_j a_{ij} u_{E_j, r_j}. \quad (\text{C.2})$$

The direct terms are exactly

$$b_i = x + r(\ell_i y_0 + m_i G_0), \quad (\ell_i, m_i) = (\rho_i, 0) \text{ or } (\rho_i \cos z_i, \rho_i \sin z_i), \quad 0 \leq z_i \leq \psi.$$

For mean routines $\rho_i \leq b_*$ and $r_i \geq rR$; for sampling routines $\rho_i \leq \rho_e$ and $r_i \geq rR/\sqrt{5}$. A node for M_0 has center x . Every directly final mean child has scale exactly rR .

The integration-row estimates above and (C.1) give positive declared coefficient bounds

$$\begin{aligned} A_M &= 128m^8 G(r/\sigma) \delta^{-4t-w} + 3r^2, & A_S &= r^2, \\ A_{f,M} &= 1, & A_{f,S} &= 2r. \end{aligned} \quad (\text{C.3})$$

For example, the outer position velocity has mass at most $64m^8 G\sigma^{-1}\psi^{-1}h_0^{-1}$, the outer momentum velocity at most $4/\sigma$, and each local flow contributes at most one r -times-mean integral. There is no division by an actual row mass.

There are at most $P = 16mh_0^{-1}(1 + |\log R|)$ probability nodes on a mesh. One direct center expression has at most $m^2 P^2$ outer terms plus its backward, harmonic, forward, and momentum lists. A convenient bound is

$$F_0 \leq 2^{14} m^4 (1 + |\log R|)^2 \delta^{-2w}. \quad (\text{C.4})$$

A mean's center is not expanded while counting one direct row.

C.3 One fixed number of synchronous updates

Let $U(E)_i = u_{E_i, r_i}$ and define

$$E^{[0]} = b, \quad E^{[j+1]} = b + AU(E^{[j]}), \quad 0 \leq j < N_P. \quad (\text{C.5})$$

Use the layer- N_P final means in

$$\tilde{C}_M = sG_0 + \sum_i \omega_i u_{E_i^{[N_P]}, rR}.$$

For $\tilde{C}_S$ use the analogous uniterated probability system, with standard Gaussian anchor, and divide its endpoint affine output by ρ_e . Neither formula adds an extra Gaussian. After unrolling (C.5), these are finite acyclic formulas of mean-dependency depth at most $N_P + 1$.

Conditional means are $C\lambda$ -Lipschitz in their physical centers. Under the original size assumptions and for sufficiently large d , (C.3) gives

$$q = C\lambda \max_i \sum_j |a_{ij}| \leq \delta^{1/3} \leq \frac{1}{2}.$$

We apply Lemma 4.1 to the exact stationary benchmark. At each centering time use the exact (Y_t, G_t) ; along each local branch use exact backward probability transport, exact harmonic evolution, and exact forward probability transport. Every segment has the stationary marginal used in Sections I and J. Branches need not be independent because only individual RMS norms are maximized.

Write these exact physical centers as E_i^* . The exact centering displacement is at most $C(L/\sigma)\sqrt{d}$ in RMS, and each local probability or harmonic correction is at most $CrD_F(x)$. After free-state elimination,

$$\max_i \|E_i^* - b_i\|_2/r \leq C[(L/\sigma)\sqrt{d} + rD_F(x)] \leq CD_F(x). \quad (\text{C.6})$$

The factor r is essential: the cruder coefficient-mass bound would lose uniformity for small r or unrestricted $\sigma > 0$.

Evaluate the uniterated equations on this same exact benchmark. Their residuals are precisely the centering quadrature, the kernel derivative and angle quadratures, and the three local flow quadratures. The derivative estimates in [Sections I and J](#), together with the Taylor bounds above, give

$$\max_i \|E_i^* - \Phi(E^*)_i\|_2/r \leq B_d D_F(x) \left[\delta^{wm} + \delta^{w(m-2)} + \delta^{w(m-1)-4t} + \delta^{2tm-4t-w} + \delta^{(m+1)/3} \right]. \quad (\text{C.7})$$

After factoring out $B_d D_F(x)$ from the normalized physical-center residual, the five contributions in [\(C.7\)](#) can be recorded before substituting $h_0 = \delta^w$, $\psi = \delta^{4t}$, and $R = \delta^t$:

Residual source	Bound before substitution	Dimension power
Direct probability quadrature	h_0^m	δ^{wm}
Kernel derivative quadrature, including the first-cell rule	h_0^{m-2}	$\delta^{w(m-2)}$
Forward probability quadrature with kernel and angle row bounds	$h_0^{m-1}\psi^{-1}$	$\delta^{w(m-1)-4t}$
Angle truncation and a conservative harmonic-remainder envelope	$h_0^{-1}\psi^{m-1}R^{-2m}$	$\delta^{2tm-4t-w}$
Centering-state quadrature	$((L/s)R^{-2})^{m+1}$	$\leq \delta^{(m+1)/3}$

Fixed-order constants and geometric-mesh logarithms are included in B_d ; the prefactors l and L/s are bounded under the size assumptions. These dimensionless factors bound the individual RMS contributions under the common exact stationary anchors after the stated normalization.

For this calculation, every mean in $\Phi(E^*)$ is evaluated at its exact benchmark branch center. Free-state elimination substitutes only the explicit starting-state terms and stops at every mean value. Therefore a physical-center residual is $r\rho_i$ times a sum of a fixed number of local state residuals, with bounded sine and cosine coefficients, together with the already-accounted-for centering-velocity quadrature residual. It does not propagate a local state approximation through all downstream mean evaluations and kernel weights. Those mean dependencies are instead controlled by the single contraction estimate for Φ . This preserves the factor r and introduces neither an additional inverse-scale loss nor a fan-out multiplier.

Choose

$$N_P = \lceil 10(K_0 + 4) \rceil, \quad w(m-3) > K_0 + 12, \quad t(m-3) > K_0 + 12, \quad m > 4(K_0 + 12).$$

With $0 < w < t$, all exponents in [\(C.7\)](#) exceed $K_0 + 11$, as does $N_P/3$. Equations [\(C.6\)](#)–[\(C.7\)](#) and [Lemma 4.1](#) prove

$$\max_i \|E_i^{[N_P]} - E_i^*\|_2/r \leq B_d \delta^{K_0+2} D_F(x). \quad (\text{C.8})$$

For a directly final mean center, division by rb_* yields the same-anchor centering-position error. If the centering momentum is requested, reconstruct it from its displayed affine row and residual;

no division by $\sin z$ is used. In the final mean output the perturbation has multiplier $C\lambda r$; since $L/\sigma \leq 1$, (C.8) and stationary final quadrature give

$$\|\tilde{\mathcal{C}}_M - (u + sG_1)\|_2 \leq B_d \sigma \delta^{K_0+2} D_F(x).$$

The sampling endpoint has the same order against its exact probability-transport target, with the inherited Gaussian anchor. Thus both the state and output accuracy required by the retained-noise induction hold on the same anchors.

C.4 Explicit substitution into the original-gradient recursion

Proposition C.1 (Compatible replacement of the numerical interface). *In the compiler and mean-root construction of Sections K and L, replace every use of $\mathcal{C}_M, \mathcal{C}_S$ from Theorem J.7 by $\tilde{\mathcal{C}}_M, \tilde{\mathcal{C}}_S$ defined above. The requested numerical accuracy $J = K_0 + 2$ in those proofs is supplied by (C.8) and its output bounds. The scalar size, fixed-seed stability, shift, routine-height, and error inductions remain valid, with fixed-order constants enlarged if needed. The executed numerical depth is $D_0 = N_P + 1$; the old bound $3N_P^2 + 1$ remains a valid conservative bound, but no nested iteration is executed.*

Proof. The required inputs of the original recursion are deterministic affine centers, direct anchor coefficients, smoothing-scale lower bounds, declared center and output masses, direct expression counts, and ideal-anchor numerical accuracy. Each has been proved above. The new bounds (C.3)–(C.4) are no larger than the original interface’s generous bounds after fixed-order enlargement: its exponent $c_1 = 10$ dominates the new exponents 1 and 2. The direct trigonometric coefficients have the same form, now with $0 \leq z \leq \psi$, and final mean children still have scale rR .

For completeness, fix a routine-height upper bound $H = \lceil K_0/t \rceil + 60K_0 + 10$ and take $w = t/[100D_0(H+1)]$ before choosing m . The unchanged label rules (K.3) imply this height bound; expression dependencies do not repeatedly decrease a parent’s labels. Expanding disjoint occurrences, a work-and-shift bound is

$$[8(D_0 + 2)(1 + F_0)^{D_0}]^{H+1} \leq B_d \delta^{-2wD_0(H+1)} = B_d \delta^{-t/50}.$$

The shift ratio at a mean node is bounded by 3, since for $\tau_i = \sqrt{1 - \rho_i^2} \geq R$,

$$0 \leq 1 - \rho_i \cos z_i/b_* \leq 2\tau_i + \psi^2/2, \quad \psi^2/\tau_i \leq R^7.$$

The sampling ratio is at most 2; inverse seed transformations have norm at most 2, and the anchor-reserve coefficient is at most 2. Hence the universal factor 8 suffices in the same bottom-up shift count. For large d the fixed B_d is absorbed by the remaining $\delta^{-49t/50}$, proving the predeclared shift bound δ^{-t} before the positive covariance matrices are formed. Their eigenvalues are then at least $\sqrt{3}/2$.

All size descents use the same labels, scales, and the original dominating mass bound. The fixed-seed induction is deterministic and consequently also holds for the conditioned and transformed arrays used here. The two-type RMS induction uses the same ideal-anchor moments and numerical error order, now supplied by (C.8). This verifies the substitution at each invocation of the old numerical interface, including the final accuracy and mean-root proofs, without using a new stochastic DAG law. Every syntactic mean occurrence and its supplier retain disjoint seeds. No exact mean, implicit fixed point, or stationary flow becomes a runtime oracle. $\square$

The construction establishes the fixed-order theorem proved here. Although the shorter depth also permits sharper quantitative order schedules, no growing-order prefactor or optimized dimension exponent is asserted in this manuscript.

D Smoothing identities and probability metrics

The normalized potential satisfies

$$\kappa^{-1}I \preceq \nabla^2 V \preceq I. \quad (\text{D.1})$$

Only original-gradient queries are counted. All auxiliary probability laws below are analytic definitions, not additional oracles.

D.1 Probability metrics and smoothing

For probability measures μ, ν , write $\text{KL}(\mu \parallel \nu) = \int \log(d\mu/d\nu) d\mu$, with value $+\infty$ if μ is not absolutely continuous with respect to ν , and $\text{TV}(\mu, \nu) = \sup_A |\mu(A) - \nu(A)|$. The quadratic Wasserstein distance is $W_2^2(\mu, \nu) = \inf_{\lambda \in \mathcal{C}(\mu, \nu)} \int \|x - y\|^2 \lambda(dx, dy)$, where $\mathcal{C}$ is the set of couplings. For a differentiable U , write $D_U(z, x) = U(z) - U(x) - \nabla U(x) \cdot (z - x)$. Vector norms are Euclidean; matrix operator and Hilbert–Schmidt norms carry subscripts op and HS . All optimal maps below are quadratic-cost Brenier maps, not arbitrary transports.

Let $\gamma_d = \mathcal{N}(0, I)$, and define

$$\pi_a = \pi * \mathcal{N}(0, aI), \quad V_a = -\log \pi_a, \quad g_a = \nabla V_a, \quad \Pi_a = \pi_a \otimes \gamma_d. \quad (\text{D.2})$$

We use a for the fixed smoothing variance, reserving η for the arbitrarily small dimension exponent. For $b > 0$ and $y \in \mathbb{R}^d$, the restricted Gaussian distribution is

$$R_{b,y}^V(dx) \propto \exp\left(-V(x) - \frac{\|x - y\|^2}{2b}\right) dx. \quad (\text{D.3})$$

We also write R_b^V for the Markov kernel $y \mapsto R_{b,y}^V$. These densities and smoothed scores are analytic objects; the model does not provide them as oracles.

We use the logarithmic Sobolev and transport inequalities for a c -strongly log-concave reference ν :

$$\text{KL}(\mu \parallel \nu) \leq \frac{I(\mu \parallel \nu)}{2c}, \quad W_2^2(\mu, \nu) \leq \frac{2}{c} \text{KL}(\mu \parallel \nu). \quad (\text{D.4})$$

We also write $H(\mu \mid \nu) = \text{KL}(\mu \parallel \nu)$. The relative score is $\ell = \nabla \log(d\mu/d\nu)$. Here $I(\mu \parallel \nu) = \int \|\nabla \log(d\mu/d\nu)\|^2 d\mu$ is relative Fisher information when defined, and is $+\infty$ otherwise. We also use Pinsker’s inequality, $\text{TV}^2 \leq \text{KL}/2$, and entropy duality, $\mathbb{E}_\mu f \leq \text{KL}(\mu \parallel \nu) + \log \mathbb{E}_\nu e^f$. These classical facts are cited in [BGL14, OV00]; their applications to nonsmooth densities are by approximation or the variational definition of entropy.

Lemma D.1 (Smoothing and posterior identities). *Under (D.1), V_a is smooth and*

$$(\kappa + a)^{-1}I \preceq \nabla^2 V_a \preceq I, \quad g_a(y) = \mathbb{E}_{R_{a,y}^V} \nabla V(X). \quad (\text{D.5})$$

Moreover,

$$\|\nabla V(y) - g_a(y)\| \leq a\|g_a(y)\| + \sqrt{ad}. \quad (\text{D.6})$$

For $s > 0$, an exact draw from $R_{s,c}^{V_a}$ is obtained by independently drawing $Z \sim R_{a+s,c}^V$ and $G \sim \gamma_d$, then setting

$$X = \frac{ac + sZ}{a + s} + \sqrt{\frac{as}{a + s}} G. \quad (\text{D.7})$$

In this joint draw, $\text{Law}(Z \mid X, c) = R_{a,X}^V$. This remains true for an arbitrary law of c and any retained external anchors, provided the displayed draws are fresh conditionally on them. Finally, $Y \sim \pi_a$ followed by $X \sim R_{a,Y}^V$ gives $X \sim \pi$.

Proof. Differentiation of the convolution gives the mean identity and $\nabla^2 V_a = a^{-1}I - a^{-2} \text{Cov}(X | Y = y)$. The posterior curvature is at least $\kappa^{-1} + a^{-1}$, so Poincaré gives the lower bound. Differentiating through the original potential instead gives $\nabla^2 V_a = \mathbb{E}[\nabla^2 V | Y] - \text{Cov}(\nabla V | Y) \preceq I$. The convolution is smooth because V has quadratic growth and the Gaussian factor allows differentiation locally to every order. The posterior mean is $y - ag_a(y)$ and its covariance has trace at most ad . Lipschitzness of ∇V and Jensen’s inequality prove (D.6).

Let ϕ_b be the centered Gaussian density of covariance bI . The product identity

$$\phi_a(x - z)\phi_s(c - x) = \phi_{a+s}(c - z)\phi_{as/(a+s)}\left(x - \frac{ac + sz}{a + s}\right) \quad (\text{D.8})$$

shows that the joint density in (D.7), conditional on any c , is proportional to $e^{-V(z)}\phi_a(x - z)\phi_s(c - x)$. Marginalizing z proves the claimed law of X . Conditioning on x cancels the c -dependent factor and proves $\text{Law}(Z | X, c) = R_{a,X}^V$, also after mixing over c and external anchors. The final assertion is the same Gaussian conditioning identity without the second observation. $\square$

E Implementation by original gradients

The finite phase can be written as a triangular scalar-affine computation in conditional gradient means. Replacing each mean independently at its exact center would not implement that computation: the centers are themselves outputs of earlier approximate evaluations. We therefore supply each center with Gaussian noise and absorb that noise into the random seeds of the next mean evaluation. An independent final Gaussian converts a Wasserstein comparison of the whole output into total variation.

Throughout this section, fix $e \in (0, 1)$, an integer $K_M \geq 2$, and $t > 0$ with $40(K_M + 2)t < 1/4$, and put $\delta = d^{-e}$. Take d above the fixed threshold in Theorem F.1, in particular large enough that $\delta \leq 1/2$. We write $B_d = C(1 + \log d)^C$, allowing the fixed constants to depend on these orders but not on the input center or oracle answers.

E.1 Noisy sampling and noisy means

For $F \in C^2(\mathbb{R}^d)$ with $\text{Lip}(\nabla F) \leq \delta$, define

$$\nu_{x,r}(\mathrm{d}z) \propto e^{-\|z\|^2/2 - F(x+rz)} \mathrm{d}z, \quad m(x) = \mathbb{E}_{\nu_{x,r}} \nabla F(x + rZ), \quad D_F(x) = \sqrt{d} + \|\nabla F(x)\|, \quad (\text{E.1})$$

where $0 < r \leq 1$. In the technical appendices the same mean is denoted $u_{x,r}$ to retain its scale argument. No minimizer is supplied. When invoking the technical construction we set its order $K_0 = K_M$. The noisy near-quadratic sampling theorem, Theorem F.1, returns an approximation to $Z + G$, $Z \sim \nu_{x,r}$, with pointwise total variation error at most $B_d \delta^{K_M} D_F(x)$ and every-execution query cap $B_d \delta^{-t}$. It retains a final independent Gaussian reserve of covariance $I/4$, with a pre-reserve W_2 bound of the same order against $\nu_{x,r} * \mathbf{N}(0, 3I/4)$.

The next version starts the recursion at its mean routine. Its proof is in Section L; the change of root is separate from the sampling statement in Theorem F.1.

Construction provenance. This is the mean-root specialization of OpenAI’s conditional-mean construction [Ope26, Sections 6–7]; the additional mean-root statement is proved in Section L.

Proposition E.1 (A standalone noisy mean). *Fix $C_0, q_0 \geq 0$ and $\sigma > 0$. Suppose*

$$\max\{\delta r^2, \delta r/\sigma\} \leq C_0(1 + \log d)^{q_0} \delta. \quad (\text{E.2})$$

There is a finite gradient procedure $M_\sigma(x; G)$ with deterministic seed slots and query cap $B_d \delta^{-t}$ such that

$$W_2(\text{Law}(M_\sigma(x)), \mathbf{N}(m(x), \sigma^2 I)) \leq B_d \sigma \delta^{K_M} D_F(x). \quad (\text{E.3})$$

Its fixed-seed Lipschitz constant in x is $L_M \leq B_d \delta$. It has a deterministic query-preserving shift vector v_σ , $\|v_\sigma\| \leq D_s := \delta^{-t}$, satisfying

$$M_\sigma(x + \Delta; z + v_\sigma \otimes \Delta/r) = M_\sigma(x; z). \quad (\text{E.4})$$

It can reserve a final independent Gaussian of covariance $\sigma^2 I/4$, with pre-reserve W_2 error at most $B_d \sigma \delta^{K_M} D_F(x)$ against $\mathbf{N}(m(x), 3\sigma^2 I/4)$. Its total variation error is therefore at most $2B_d \delta^{K_M} D_F(x)$. The constants in B_d and the dimension threshold may also depend on C_0, q_0 .

E.2 A finite affine composition

Condition on an external anchor record, which may include the initial state and the clock U . Suppose the exact centers form a triangular computation

$$x_i = b_i + \sum_{j < i} a_{ij} m(x_j), \quad \sum_j |a_{ij}| \leq A.$$

All coefficients are deterministic given the anchors. Let the dependency depth be at most L , a fixed positive integer, and maximum number of direct mean dependencies (fan-in) at most M , including the final expression. We seek only the declared output

$$b + \sum_i C_i m(x_i) + \mathbf{N}(0, \Sigma), \quad \Sigma \succ 0. \quad (\text{E.5})$$

Internal query records are not included in this law. Put $R_\Sigma = \sum_i \|\Sigma^{-1/2} C_i\|_{\text{op}}$, using a deterministic upper bound when necessary.

Construction provenance. Gaussian center-noise absorption and recursive evaluation use OpenAI [Ope26, Lemma 7.1 and Section 7]. The outer affine composition and its joint covariance accounting are the specialization stated here.

Proposition E.2 (Affine implementation). *For fixed $C_0, q_0 \geq 0$, assume that all mean calls use the common scale r and satisfy (E.2) with these same constants, including the output call with $\sigma_{\text{out}} = 1/(4R_\Sigma)$ when $R_\Sigma > 0$. Then (E.5) has a finite original-gradient implementation with conditional total variation error at most*

$$B_d \delta^{K_M} \max_i D_F(x_i) \quad (\text{E.6})$$

and every-execution query cap $B_d(1 + M)^{L+1} \delta^{-t}$. The Gaussian comparison law retains the full covariance Σ in (E.5), including its cross-covariances. No equality between Σ and the covariance of the approximate output is asserted. The fixed constants in B_d may also depend on C_0, q_0 and the depth L .

Proof. If $R_\Sigma = 0$, draw the displayed Gaussian directly. Otherwise put $s_c = r/(2D_s)$ and supply each internal center with target noise $\mathbf{N}(0, s_c^2 I)$. Expand every repeated mean occurrence as a separate copy of its entire recursive center computation. For a scalar row with coefficient sum $A_E \leq A$, request each mean with noise $\sigma_E = s_c/(2A)$, ignoring term-free rows. For its child shift vector v , use fresh seeds transformed by

$$P_v = \left(I - \frac{vv^\top}{4D_s^2} \right)^{1/2}.$$

Add residual Gaussian variance $s_c^2 - \sum_j a_j^2 \sigma_E^2 \geq 3s_c^2/4$ to the row. Term-free rows return the direct term plus $s_c G$.

At a center with ideal supplied value $x + s_c Z$, identity (E.4) gives

$$M_\sigma(x + s_c Z; P_v G) = M_\sigma \left(x; P_v G - \frac{v \otimes Z}{2D_s} \right) \stackrel{\text{law}}{=} M_\sigma(x; G).$$

The last equality uses independence of Z and G , and covariance $P_v P_v^\top + vv^\top/(4D_s^2) = I$. It is an averaged identity, not one conditional on the auxiliary Z .

Couple an actual supplied center to the ideal noisy center first. Fixed-seed stability costs $L_M e_i$ in W_2 , where e_i is its center error. Then use seed absorption, followed by (E.3). Independent tree copies admit product couplings. Put $P_d = C_0(1 + \log d)^{q_0}$. If all centers are term-free, then $e_i = 0$ and we set $\chi = 0$. Otherwise the internal noise $\sigma_E = r/(4AD_s)$ and its root condition give $4AD_s \leq P_d$. Consequently $\chi = L_M A \leq B_d P_d \delta^{1+t}/4$ is bounded beyond a fixed threshold. Put $G_L(\chi) = \sum_{j=0}^{L-1} \chi^j$. The row errors satisfy

$$e_i \leq L_M \sum_j |a_{ij}| e_j + B_d \delta^{K_M} D_* \sum_j |a_{ij}| \sigma_E \leq \chi \max_{j < i} e_j + \frac{1}{2} B_d s_c \delta^{K_M} D_*,$$

where $D_* = \max_i D_F(x_i)$ over exact centers. Term-free supplied centers have zero error. Induction on the fixed dependency depth gives $e_i \leq \frac{1}{2} B_d s_c \delta^{K_M} D_* G_L(\chi)$. Because L is fixed and χ is bounded, this finite geometric polynomial is bounded. No independent contraction or stability assumption is required.

For the final output, request each mean with $\sigma_{\text{out}} = 1/(4R_\Sigma)$ using independent recursive center computations and transformed seed blocks. Put $S = \sum_i \sigma_{\text{out}}^2 C_i C_i^\top \preceq \Sigma/16$. Add independent noise of covariance $3\Sigma/4 - S$, then an independent final reserve of covariance $\Sigma/4$. Before that reserve the whitened W_2 error is bounded by

$$\begin{aligned} & \sum_i \|\Sigma^{-1/2} C_i\|_{\text{op}} \{L_M e_i + B_d \sigma_{\text{out}} \delta^{K_M} D_*\} \\ & \leq B_d \delta^{K_M} D_* \left\{ \frac{1}{4} + \frac{1}{2} R_\Sigma L_M s_c G_L(\chi) \right\}. \end{aligned}$$

The output root condition gives $4rR_\Sigma \leq P_d$. Since $L_M \leq B_d \delta$ and $s_c = r/(2D_s)$, it follows that $R_\Sigma L_M s_c \leq B'_d \delta^{1+t}$, which is bounded beyond a fixed dimension threshold. Together with the fixed-depth estimate this absorbs the bracket into B_d . The ideal pre-reserve law is Gaussian of covariance $3\Sigma/4$. Convolution by the independent reserve bounds total variation by a universal constant times the whitened W_2 error: couple the means and use the Gaussian shift bound, then average. This proves (E.6). A depth- L , fan-in- M expanded tree has at most $C(1+M)^{L+1}$ terms. Every term has a deterministic query cap, so the claimed total cap holds even on erroneous intermediate histories. $\square$

E.3 Removing the noise from a small posterior

The primitive in [Theorem F.1](#) returns a posterior plus independent Gaussian noise. The following finite descent removes that noise at arbitrary supplied centers.

Construction provenance. The following routine adapts the conditional-descent construction and ideal-prefix replacement argument of OpenAI [[Ope26](#), Section 8.2]. The arbitrary-center bounds and the terminal Gaussian rule are analyzed here; related posterior descent appears in [[CCLZ26c](#), Algorithm 3.3 and Section 6.3].

Proposition E.3 (Small-width posterior sampling). *Let $0 < w \leq \delta$, and let V satisfy [\(D.1\)](#). For every fixed $A_0 > 0$ and input-center distribution with*

$$\mathbb{E}[\sqrt{d} + \sqrt{\delta}\|\nabla V(Y)\|] \leq Cd^{M_0}, \quad (\text{E.7})$$

there is an original-gradient procedure approximating $R_{w,Y}^V$ with averaged total variation error at most d^{-A_0} and cap $B_d\delta^{-t}$ times a fixed logarithmic factor, provided $eK_M > A_0 + M_0 + 2$. The orders and threshold depend on the fixed parameters, not on w, Y , or κ .

Proof. Set $F(z) = V(\sqrt{\delta}z)$, $x = Y/\sqrt{\delta}$, and $r = \sqrt{w/\delta}$. Then $\text{Lip}(\nabla F) \leq \delta$ and $D_F(x) = \sqrt{d} + \sqrt{\delta}\|\nabla V(Y)\|$. Let $(x_0, r_0) = (x, r)$. For $j = 0, \dots, L_0 - 1$, apply the noisy sampling routine of [Theorem F.1](#) to (x_j, r_j) , obtaining Y_j , and set

$$x_{j+1} = x_j + r_j Y_j / 2, \quad r_{j+1} = r_j / \sqrt{2}.$$

At the terminal center, draw $Z_{L_0} \sim \gamma_d$ independently. Return the physical variable

$$\sqrt{\delta}(x_{L_0} + r_{L_0} Z_{L_0}).$$

The exact conditioning identity is $Z = Y_j/2 + Z_{j+1}/\sqrt{2}$, with $Z_{j+1} \mid Y_j \sim \nu_{x_j + r_j Y_j/2, r_j/\sqrt{2}}$. Thus the reconstruction is exact if all noisy observations and the terminal conditional sample are exact.

We use a Gaussian comparison valid at every center and scale:

$$\text{TV}(\gamma_d, \nu_{x,r}) \leq C\{r\|\nabla F(x)\| + \delta r^2 \sqrt{d}\}, \quad (\mathbb{E}_{\nu_{x,r}} \|Z\|^2)^{1/2} \leq C\{\sqrt{d} + r\|\nabla F(x)\|\}. \quad (\text{E.8})$$

Indeed the target curvature is at least $1/2$, and the relative score under $G \sim \gamma_d$ is $r\nabla F(x + rG)$, with L^2 norm at most $r\|\nabla F(x)\| + \delta r^2 \sqrt{d}$. Logarithmic Sobolev followed by Pinsker gives the first bound. Logarithmic Sobolev followed by T_2 bounds $W_2(\gamma_d, \nu_{x,r})$ by the same score norm; a coupling and Minkowski give the second bound.

Condition on a deterministic initial center x . Realize the ideal descent using one latent $W = x + rZ$, $Z \sim \nu_{x,r}$, and fresh independent Gaussian observations

$$Y_j = (W - x_j)/r_j + N_j, \quad x_{j+1} = (x_j + W + r_j N_j)/2.$$

Gaussian conditioning gives the required posterior at each level. Expanding the linear recursion yields, marginally for every j ,

$$x_j \stackrel{\text{law}}{=} x + (1 - 2^{-j})rZ + r\sqrt{2^{-j} - 4^{-j}} G_j, \quad G_j \sim \gamma_d, \quad G_j \perp Z.$$

The G_j need not be independent across levels. Using the second bound in [\(E.8\)](#), uniformly in j ,

$$(\mathbb{E}\|\nabla F(x_j)\|^2)^{1/2} \leq \|\nabla F(x)\| + C\delta r(\sqrt{d} + r\|\nabla F(x)\|) \leq CD_F(x).$$

The first bound in (E.8) controls the terminal Gaussian approximation. Averaging it with the preceding ideal-center bound requires no terminal optimization.

A forward replacement of noisy observations, using a common exact prefix each time, then gives the pointwise total error

$$CL_0 B_d \delta^{K_M} D_F(x) + Cr 2^{-L_0/2} D_F(x) + C \delta r^2 2^{-L_0} \sqrt{d}. \quad (\text{E.9})$$

For example, take $L_0 = \lceil 2(A_0 + \max\{M_0, 1/2\} + 3) \log_2 d \rceil$. Average (E.9) using (E.7). The specified accuracy order proves the assertion beyond a fixed dimension threshold. The deterministic work is $L_0 B_d \delta^{-t}$. No moment bound on an implemented erroneous prefix is used. $\square$

F Conditional gradient means and noisy sampling

This and the following five appendices prove the noisy sampling construction and its analytic ingredients, adapting OpenAI [Ope26]. [Theorem F.1](#) states the sampling guarantee; its proof concludes in [Section K](#). The mean-root specialization used in the mean construction is proved in [Section L](#).

We retain the law $\nu_{x,r}$ and envelope $D_F(x)$ from (E.1), and write $u_{x,r} = m(x)$ to display the mean's scale. The Gaussian interpolation variance is $v_\rho = 1 - \rho^2$; the sampling noise standard deviation is τ . For tensor-valued random variables, $\|\cdot\|_2$ is the L^2 norm of the displayed pointwise norm, and $\|\cdot\|_{\text{HS},2}$ uses the Hilbert–Schmidt norm. All kernels and oracle algorithms below are measurable in the exact-real model of [Section D](#).

Fix $0 < e < 1$, and put $\delta = d^{-e}$. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}$ be C^2 , with gradient $g = \nabla F$ of Lipschitz constant at most the declared bound $\lambda = \delta$. Values of F need not be available. For $x \in \mathbb{R}^d$ and $0 < r \leq 1$, define

$$\nu_{x,r}(dz) = \frac{\exp(-|z|^2/2 - F(x + rz)) dz}{\int_{\mathbb{R}^d} \exp(-|v|^2/2 - F(x + rv)) dv}, \quad f(z) = g(x + rz), \quad u = \mathbb{E}_{\nu_{x,r}} f. \quad (\text{F.1})$$

We often write $\nu = \nu_{x,r}$, and use

$$L = \lambda r, \quad l = \lambda r^2 = rL, \quad D_F(x) = \sqrt{d} + |g(x)|. \quad (\text{F.2})$$

Here x is the actual input center; it need not be a minimizer of F . The positive number λ is a declared upper bound, so the notation also covers affine or constant F . The full potential in (F.1) has Hessian between $(1 - l)I$ and $(1 + l)I$. Since $l \leq \delta < 1$ for $d \geq 2$, the normalizing integral is finite, even when F is not convex. For the quantitative construction we take d large enough that $l \leq 1/2$.

Throughout the construction, $\text{polylog}(d)$ denotes a factor bounded by $C(1 + \log d)^C$. Its constants may depend on the fixed parameters e , accuracy orders, and other fixed orders introduced below, but never on F, x, r , or the oracle responses. Different occurrences may have different constants. Every sufficiently-large-dimension threshold has the same uniformity.

Source attribution. Adapted from OpenAI [Ope26, Theorem 2.1].

Theorem F.1 (Noisy near-quadratic sampling). *Fix $0 < e < 1$, an integer $K_0 \geq 2$, and $t > 0$ satisfying*

$$40(K_0 + 2)t < \frac{1}{4}. \quad (\text{F.3})$$

There are constants and a dimension threshold depending only on e, K_0, t with the following property. For every integer d above that threshold, put $\delta = d^{-e}$. One measurable randomized oracle procedure works for every $F \in C^2(\mathbb{R}^d)$ with $\text{Lip}(\nabla F) \leq \delta$, every $x \in \mathbb{R}^d$, and every $0 < r \leq 1$. Given evaluation access to $g = \nabla F$, it uses at most $\text{polylog}(d)\delta^{-t}$ evaluations of g on every execution and returns

$$X = X_{\text{pre}} + \frac{1}{2}G,$$

where $G \sim \mathcal{N}(0, I_d)$ is independent of X_{pre} and of all queries, and

$$W_2(\text{Law}(X_{\text{pre}}), \nu_{x,r} * \mathcal{N}(0, \frac{3}{4}I_d)) \leq \text{polylog}(d)\delta^{K_0}D_F(x). \quad (\text{F.4})$$

Consequently both

$$W_2(\text{Law}(X), \nu_{x,r} * \mathcal{N}(0, I_d)) \quad \text{and} \quad \|\text{Law}(X) - \nu_{x,r} * \mathcal{N}(0, I_d)\|_{\text{TV}}$$

are at most $\text{polylog}(d)\delta^{K_0}D_F(x)$. All constants in these bounds are uniform in F, x, r . The procedure's scalar parameters, shapes of its computations, and query cap are independent of x and of the primitive. It uses no evaluations of F itself.

Theorem F.1 is proved at the end of Section K. Its final Gaussian is generated independently after the query computation. A common copy of that Gaussian preserves the coupling error, and Lemma F.3 converts the pre-noise W_2 bound to TV. The comparison law is $\nu_{x,r} * \mathcal{N}(0, I_d)$. The posterior application in Proposition E.3 removes this added noise by conditioning.

The factor $D_F(x)$ is part of the guarantee. Uniformity of the constants in x does not make the error uniformly small over centers. At every random-center application, we will either retain this factor or establish a moment bound for it.

F.1 Convexity, moments, and comparisons

We will use the following consequences of uniform convexity. We include the C^2 approximation argument to record the regularity needed later; see [BGL14] for the curvature method.

Source attribution. Adapted from OpenAI [Ope26, Lemma 2.2].

Lemma F.2 (Uniformly convex laws). *Let $H \in C^2(\mathbb{R}^n)$ satisfy $mI \preceq \nabla^2 H \preceq MI$, with $0 < m \leq M < \infty$. Write $\mu(dx) \propto e^{-H(x)}dx$, and let z_0 be its unique mode. Then*

$$\mathbb{E}_\mu |X - z_0|^2 \leq n/m, \quad \text{Var}_\mu h \leq m^{-1} \mathbb{E}_\mu |\nabla h|^2.$$

The latter applies, in particular, to C^1 functions whose value and gradient have polynomial growth. The diffusion

$$dX_t = -\nabla H(X_t) dt + \sqrt{2} dB_t$$

has invariant law μ , and synchronous solutions contract in distance by e^{-mt} .

If $\bar{H}$ is another strongly convex C^2 potential with bounded Hessian and law χ , then

$$W_2(\mu, \chi) \leq \frac{1}{m} (\mathbb{E}_\chi |\nabla H - \nabla \bar{H}|^2)^{1/2}. \quad (\text{F.5})$$

Proof. We first prove the moment bound and contraction. Strong convexity gives Gaussian tails and a unique minimizer. Integration by parts yields $\mathbb{E}_\mu(X - z_0) \cdot \nabla H(X) = n$; strong monotonicity of ∇H gives the moment bound. The globally Lipschitz drift gives a nonexplosive diffusion. For synchronous solutions, differentiating their squared distance and using strong monotonicity gives the contraction.

To establish invariance and Poincaré, first assume that H is smooth and all derivatives of order at least two are bounded. For a smooth compactly supported h , let $P_t h(x) = \mathbb{E}[h(X_t) \mid X_0 = x]$. Differentiating the synchronous flow in the initial state gives bounded first and second derivatives on bounded time intervals. Itô's formula and the semigroup identity yield

$$\partial_t P_t h = P_t \mathcal{A} h = \mathcal{A} P_t h, \quad \mathcal{A} = \Delta - \nabla H \cdot \nabla.$$

The equality with $\mathcal{A} P_t h$ also follows by differentiating $P_{t+s} h = P_s(P_t h)$ at $s = 0$. Its at-most-linear growth is integrable against μ . Integration by parts gives $\int \mathcal{A} P_t h d\mu = 0$, hence invariance. Coupling to an invariant starting point and using contraction shows $P_t h \rightarrow \mu h$ in $L^2(\mu)$. Therefore

$$\text{Var}_\mu h = 2 \int_0^\infty \int |\nabla P_t h|^2 d\mu dt.$$

The flow derivative has operator norm at most e^{-mt} . Jensen's inequality and invariance thus bound the inner integral by $e^{-2mt} \int |\nabla h|^2 d\mu$, proving the Poincaré inequality.

For C^2 potentials, convolve with smooth compactly supported mollifiers of radii tending to zero. Both Hessian bounds are preserved; all higher derivatives are bounded for each mollification. The normalized laws converge in total variation by the uniform quadratic tail bounds (subtract the potential's value at zero). The gradients converge uniformly since the original gradient is globally Lipschitz. Synchronous solutions of the mollified and original diffusions consequently converge uniformly in their common starting point on each bounded time interval, by Gronwall's inequality. Testing invariance against bounded Lipschitz functions passes it to the limit. Poincaré first passes on smooth compact tests, then by cutoff and approximation to the stated class.

For (F.5), start the two diffusions with their respective invariant marginals, with any initial coupling, and use common Brownian motion. Writing $a(t) = (\mathbb{E}|X_t - Y_t|^2)^{1/2}$ and $\beta = (\mathbb{E}_\chi |\nabla H - \nabla \bar{H}|^2)^{1/2}$, strong monotonicity and Cauchy-Schwarz give, in the usual upper-derivative sense,

$$a'(t) \leq -ma(t) + \beta.$$

The marginals remain fixed, so $W_2(\mu, \chi) \leq a(t)$. Letting t tend to infinity proves the claim. $\square$

For the auxiliary law, the quadratic term supplies strong convexity even when F is nonconvex. Its mode solves $z_0 = -rg(x + rz_0)$. In particular, for $l \leq 1/2$,

$$|z_0| \leq \frac{r|g(x)|}{1-l}, \quad (\mathbb{E}_\nu |Z|^2)^{1/2} \leq C(\sqrt{d} + r|g(x)|), \quad |u| \leq CD_F(x). \quad (\text{F.6})$$

Indeed, $|g(x + rz_0) - g(x)| \leq L|z_0|$ proves the first bound, and the lemma gives $\|Z - z_0\|_2 \leq \sqrt{d/(1-l)}$. The triangle inequality and $|u| \leq |g(x)| + L\|Z\|_2$ give the remaining bounds. Thus $D_F(x)$ controls the required moments uniformly over all centers, including those with very large gradients.

F.2 Coupling and smoothing conventions

We measure numerical error by the L^2 norm under the stated input law. We compare randomized output laws in W_2 . The following rules pass from conditional couplings to unconditional laws and from a pre-noise coupling to total variation.

Source attribution. Adapted from OpenAI [Ope26, Lemma 2.3].

Lemma F.3 (Conditional comparisons and smoothing). *Let (μ_z) and (χ_z) be probability kernels on finite-dimensional Euclidean spaces, and let Z have law ζ . Whenever the relevant second moments are finite,*

$$W_2^2\left(\int \mu_z d\zeta(z), \int \chi_z d\zeta(z)\right) \leq \int W_2^2(\mu_z, \chi_z) d\zeta(z).$$

For finite families, independent sums satisfy

$$W_2\left(\text{Law}\left(\sum_i a_i X_i\right), \text{Law}\left(\sum_i a_i Y_i\right)\right) \leq \sum_i |a_i| W_2(\text{Law}(X_i), \text{Law}(Y_i)),$$

where independence is required within each of the two families, not between them. For $s > 0$,

$$\|\mu * \mathbf{N}(0, s^2 I) - \chi * \mathbf{N}(0, s^2 I)\|_{\text{TV}} \leq s^{-1} W_2(\mu, \chi). \quad (\text{F.7})$$

Moreover, applying a common probability kernel cannot increase total variation, and

$$\left\| \int \mu_z d\zeta(z) - \int \chi_z d\zeta(z) \right\|_{\text{TV}} \leq \int \|\mu_z - \chi_z\|_{\text{TV}} d\zeta(z).$$

Proof. For the first inequality, condition on Z and use near-optimal couplings. For independent sums, take their product and apply the L^2 triangle inequality. Measurability of such conditional comparisons can also be seen without an exact optimal-kernel selection: quantize each output law on a finite grid inside a box, sending the complement to zero. A minimizing plan on the finite sets can be chosen measurably by a fixed ordering of the finitely many basic feasible solutions. Lift it using the conditional measures in the cells, and refine the grids and boxes. Finite unconditional second moments and the triangle inequality give the asserted limit.

For deterministic centers x, y , the relative entropy of $\mathbf{N}(x, s^2 I)$ from $\mathbf{N}(y, s^2 I)$ is $|x - y|^2 / (2s^2)$. Pinsker's inequality gives TV at most $|x - y| / (2s)$. Integrate this estimate over any coupling and apply Cauchy–Schwarz to obtain (F.7), with room in its constant. The total-variation rules follow directly by testing measurable functions taking values in $[0, 1]$. $\square$

We will also repeatedly combine deterministic root-mean-square errors with absolute coefficient sums. This uses the ordinary L^2 triangle inequality and does not require independence.

G Gaussian conditional jets and probability transport

The conditional-gradient mean is smooth after Gaussian observation even when F is only C^2 . We establish all-split derivative bounds, then identify its probability flow. These estimates supply both the finite noisy-mean computation and the derivative bounds needed by the inherited primitive analysis. The argument in this section is adapted from [Ope26].

Retain the primitive notation from Section F, and assume $l \leq 1/2$, as holds for all sufficiently large d . For independent $Z \sim \nu$ and $N \sim \mathbf{N}(0, I_d)$, put

$$v_\rho = 1 - \rho^2, \quad Y = \rho Z + \sqrt{v_\rho} N, \quad \xi_\rho = \text{Law}(Y), \quad 0 \leq \rho < 1.$$

Write $\xi_\rho(dy) = e^{-H_\rho(y)} dy$, incorporating the normalizing constant in H_ρ , and define

$$M_\rho(y) = \mathbb{E}[f(Z) \mid Y = y].$$

We use the version defined at every y by the conditional density. Completing its quadratic gives

$$\text{Law}(Z \mid Y = y)(dz) \propto \exp\left(-\frac{|z - \rho y|^2}{2v_\rho} - F(x + rz)\right) dz. \quad (\text{G.1})$$

Under the change of variables $z = \rho y + \sqrt{v_\rho} w$, $M_\rho(y)$ is exactly the primitive mean with center $x + r\rho y$ and scale $r\sqrt{v_\rho}$. In particular, $M_0 = u$. Thus conditioning preserves the auxiliary form and specifies its new center and scale. Equation (G.1) identifies the mean exactly. Its finite-query implementation is deferred to Section K.

G.1 Cumulants from the Poincaré inequality

We use Poincaré to control cumulants across every split of their coordinate slots. This strengthens the type of derivative estimate needed here beyond bounds tested only on products of directions. Related Gaussian-smoothing estimates appear in [CCLZ26c, Appendix C.2].

A tensor $T \in (\mathbb{R}^D)^{\otimes n}$ can be viewed as a matrix by partitioning its n slots into two nonempty sets A and B . Contraction on the A slots then defines a linear map $(\mathbb{R}^D)^{\otimes |A|} \rightarrow (\mathbb{R}^D)^{\otimes |B|}$, where both tensor spaces carry their Euclidean, or Hilbert–Schmidt, norm. We call the operator norm of this matrix a *split norm*. Equivalently, it is the supremum of $|T : (V \otimes W)|$ over coefficient tensors V, W of Hilbert–Schmidt norm at most one, with slots in A, B , respectively. The colon denotes contraction over all paired slots. A bound on $|T|_{\text{split}}$ means that bound for every such partition. Permuting slots simply permutes these norms. The Hilbert–Schmidt norm of a tensor itself is denoted by $|T|_{\text{HS}}$.

We use the multivariate Appell polynomials described in [Ans04, Section 2.3]. Their vanishing expected lower derivatives connect Poincaré inequalities to tensor covariance bounds; see [AW15, Proposition 3.2 and Theorem 3.3] and [KS17, Section 4.2]. We give the direct L^2 argument, with its dependence on the polynomial degree, and then use a logarithm with two groups of variables to bound every split of the cumulant tensor.

We use the following consequence of Poincaré, including for random vectors whose laws need not have densities in their ambient spaces.

Source attribution. Adapted from OpenAI [Ope26, Lemma 3.1].

Lemma G.1 (All-split cumulant bound). *Let S be a random vector in $\mathbb{R}^D$ whose moment-generating function is finite in a neighborhood of zero. Suppose that, for some $\beta \geq 0$,*

$$\text{Var}(P(S)) \leq \beta \mathbb{E}|\nabla P(S)|^2$$

for every polynomial P on $\mathbb{R}^D$. For each integer $n \geq 2$, define its cumulant tensor by $\kappa_n(S) = \nabla_\theta^n \log \mathbb{E} e^{\theta \cdot S} \big|_{\theta=0}$. Then, for an absolute constant C ,

$$|\kappa_n(S)|_{\text{split}} \leq C^n n^{Cn} \beta^{n/2}$$

for every nontrivial split. The constant is independent of D and of the mean of S .

Proof. We reduce the cumulant estimate to a covariance estimate for Appell polynomials. If $\beta = 0$, the coordinate variances vanish and S is almost surely constant. All cumulants in the statement are then zero. Assume henceforth that $\beta > 0$.

Define the symmetric Appell tensor polynomials by

$$P_j(s) = \nabla_\theta^j \frac{\exp(\theta \cdot s)}{\mathbb{E} \exp(\theta \cdot S)} \bigg|_{\theta=0}.$$

For a coefficient tensor T of order j , let $Q_T(s) = T : P_j(s)$, where the colon denotes contraction on all slots. Differentiating the generating function in s multiplies it by copies of θ . Consequently

$$\mathbb{E} \nabla^h Q_T(S) = 0 \quad (0 \leq h < j), \quad \nabla^j Q_T = j! \text{Sym}(T).$$

Here $\text{Sym}(T)$ is the average over all permutations of the slots; hence $|\text{Sym}(T)|_{\text{HS}} \leq |T|_{\text{HS}}$. Apply Poincaré to each component of $\nabla^h Q_T$ and sum. Iterating from $h = 0$ to $h = j - 1$ yields

$$\mathbb{E} |T : P_j(S)|^2 \leq \beta^j (j!)^2 |T|_{\text{HS}}^2. \quad (\text{G.2})$$

Only first derivatives in the displayed Poincaré inequality are used at each application.

In particular, if T and V have orders $j, k \geq 1$, then

$$|\mathbb{E}[(T : P_j(S))(V : P_k(S))]| \leq \beta^{(j+k)/2} j! k! |T|_{\text{HS}} |V|_{\text{HS}}. \quad (\text{G.3})$$

The coefficient tensors in (G.3) are arbitrary. Consequently the estimate controls the operator norm of the indicated matrix unfolding.

Fix a split of the slots. Assign one group of variables to each side and define

$$\mathcal{G}(\theta, \omega) = \frac{\mathbb{E} \exp((\theta + \omega) \cdot S)}{\mathbb{E} \exp(\theta \cdot S) \mathbb{E} \exp(\omega \cdot S)}.$$

Its derivative of orders $j \geq 1$ in θ and $k \geq 1$ in ω at $(0, 0)$ is the cross-moment tensor $\mathbb{E}[P_j(S) \otimes P_k(S)]$. Across the split between the θ and ω slots its operator norm is at most $\beta^{(j+k)/2} j! k!$ by (G.3).

Both $\mathcal{G}(\theta, 0)$ and $\mathcal{G}(0, \omega)$ are identically one. Thus every nonzero term of $\mathcal{G} - 1$ contains variables from both groups. In the formal identity

$$\log \mathcal{G} = \sum_{m \geq 1} \frac{(-1)^{m+1}}{m} (\mathcal{G} - 1)^m,$$

only $m \leq n$ can contribute at total order n . Every differentiated product is a tensor product of the preceding cross-moment operators, up to permutations within the two sides. Its norm is therefore at most the product of their norms. To bound the total coefficient, distribute the n labeled derivative slots among at most n ordered nonempty factors. There are at most n^{n+1} such assignments; the product of the factorials within any assignment is at most $n!$. Enlarging an absolute constant bounds their total by $C^n n^{C^n}$.

The mixed derivatives of $\log \mathcal{G}$ are precisely the corresponding derivatives of $\log \mathbb{E} \exp((\theta + \omega) \cdot S)$, since the other two logarithms have only one variable group. They are the cumulants, and the two groups can represent any prescribed nontrivial split of the n slots. This proves the claim. $\square$

The dependence on y in (G.1) is a linear tilt with coefficient ρ/v_ρ . We normalize the spatial derivatives by ρ^q . The cumulant formula below defines their extension at zero.

Source attribution. Adapted from OpenAI [Ope26, Proposition 3.2].

Proposition G.2 (Gaussian jet bounds). *Under the assumptions above, in particular $l \leq 1/2$, define for each integer $q \geq 1$*

$$U_q(\rho, y) = \rho^{-q} \nabla^q M_\rho(y) \quad (\rho > 0).$$

These tensors extend smoothly to $\rho = 0$. There is an absolute constant C such that, for every $0 \leq \rho < 1$, $y \in \mathbb{R}^d$ and every split,

$$|U_q(\rho, y)|_{\text{split}} \leq C^q q^{Cq} L(\sqrt{v_\rho})^{-(q-1)}. \quad (\text{G.4})$$

Here and below the vector-output slot is included among the tensor slots. The constant is uniform in the dimension, the primitive, its center and scale, and the derivative order q . Moreover, DM_ρ and U_1 are symmetric, and

$$\nabla H_\rho(y) = y + r\rho M_\rho(y), \quad \|\nabla^2 H_\rho\|_{\text{op}} \leq C. \quad (\text{G.5})$$

For $j \geq 1$, every split of $\nabla^j \nabla^2 H_\rho$ is bounded by

$$C^{j+1} (j+1)^{C(j+1)} (\sqrt{v_\rho})^{-j}. \quad (\text{G.6})$$

Proof. We apply the cumulant bound to a joint vector containing the state and the gradient. The conditional potential has Hessian between $(v_\rho^{-1} - l)I$ and $(v_\rho^{-1} + l)I$, so Lemma F.2 gives Poincaré constant at most Cv_ρ . Under this conditional law the map

$$z \mapsto S(z) = (z, f(z)/L)$$

has derivative norm at most $\sqrt{2}$, since f is L -Lipschitz. Pulling back an ambient test function shows that the possibly singular law of S satisfies Poincaré with constant Cv_ρ . It has exponential moments of every linear functional: f has at most linear growth and the conditional law has Gaussian tails. Lemma G.1 therefore applies. This use of repeated Poincaré does not differentiate f repeatedly. Each application pulls back a new ambient polynomial and uses only Df .

In the conditional density, the linear tilt parameter for z is $\rho y / v_\rho$. Repeated differentiation of an expectation with respect to a linear tilt gives joint cumulants. In coordinates this says

$$(U_q)_{i,k_1,\dots,k_q} = v_\rho^{-q} \text{cum}(f_i(Z), Z_{k_1}, \dots, Z_{k_q} \mid Y = y). \quad (\text{G.7})$$

It can also be checked directly by differentiating the logarithm of the conditional moment-generating function with one auxiliary parameter coupled to f_i . Restrict the cumulant of S to one f/L coordinate and q ordinary coordinates. Lemma G.1 gives a factor $Lv_\rho^{(q+1)/2}$, and the prefactor v_ρ^{-q} gives (G.4).

All the conditional integrals used here are smooth in (ρ, y) for $\rho < 1$. This follows by differentiating their densities: on compact parameter sets, every derivative inserts a polynomial with an integrable Gaussian majorant. Formula (G.7) gives the smooth extension at zero without division by ρ . For later integrations by parts we record a slightly stronger regularity fact. On each compact interval in $\rho < 1$, conditional moments of any fixed order grow at most polynomially in y . Indeed, the conditional mode satisfies

$$z_0 = \rho y - v_\rho r g(x + rz_0),$$

so $|z_0| \leq C(|y| + r|g(x)|)$. The upper and lower conditional Hessian bounds compare the density and its normalizing integral around this mode to Gaussian integrals. This proves the claimed moment

growth. The same argument applies to time derivatives of the conditional integrals. Dimension-dependent constants are harmless for this regularity justification; the quantitative estimates above remain dimension-uniform.

Conditional integration by parts gives $\mathbb{E}[Z \mid Y = y] = \rho y - v_\rho r M_\rho(y)$. Differentiating the marginal density then yields the first identity in (G.5). For $\rho > 0$ it implies symmetry of DM_ρ , hence of U_1 ; continuity handles zero. Its derivative, together with (G.4), gives the bounded Hessian and (G.6). In particular, all these conclusions use only the original C^2 regularity of F . $\square$

G.2 The transport equation

Source attribution. Adapted from OpenAI [Ope26, Proposition 3.3].

Proposition G.3 (Probability transport). *The ordinary differential equation*

$$\frac{dY_\rho}{d\rho} = -rM_\rho(Y_\rho) \tag{G.8}$$

transports $\xi_0 = \mathcal{N}(0, I_d)$ to ξ_ρ for every $\rho < 1$. Its velocity is globally Cl -Lipschitz in space. On every compact probability-time interval it has a smooth flow in both directions.

Proof. We identify the conditional velocity of the interpolation and then use uniqueness of the induced transport. Differentiation gives

$$\frac{d}{d\rho}(\rho Z + \sqrt{v_\rho} N) = \frac{Z - \rho Y}{v_\rho}.$$

The conditional mean of this derivative is $-rM_\rho(Y)$ by the conditional integration-by-parts identity above. Thus, for compactly supported smooth ϕ ,

$$\frac{d}{d\rho} \int \phi d\xi_\rho = -r \int M_\rho \cdot \nabla \phi d\xi_\rho.$$

On the other hand, (G.4) with $q = 1$ gives $r\|DM_\rho\|_{\text{op}} \leq Cl$. The velocity has at most linear growth, so its smooth ordinary differential equation has a global-in-space flow and inverse on every compact interval in $\rho < 1$. To identify its pushforward, fix a smooth compactly supported terminal test and transport it backward along this flow. The transported test has zero material derivative. Its supports remain in a bounded set on the interval, by the linear-growth bound. The last display then shows its expectation under ξ_ρ is constant, which proves the pushforward assertion. $\square$

The flow is now defined and its spatial derivatives are controlled. For high-order quadrature we next need repeated derivatives along its trajectories. We keep the dimension dependence visible in that step.

H Adjoint divergence and material derivatives

We now differentiate the conditional mean along its probability flow. The tensor calculation retains matrix split norms until the last vector-valued divergence; only that last step incurs a square-root dimension factor. We give the complete argument of [Ope26], including its high-moment trace estimate, since ordinary directional derivative bounds would not suffice for the numerical formulas.

Retain the primitive parameters $L = \lambda r$ and $l = rL \leq 1/2$, and the interpolation $\xi_\rho = \text{Law}(\rho Z + \sqrt{v_\rho} N)$, where $v_\rho = 1 - \rho^2$ and $0 \leq \rho < 1$. Expectations and L^p norms in this section use ξ_ρ at the displayed probability time. Along a solution of $Y'_\rho = -rM_\rho(Y_\rho)$, the chain rule gives

$$\frac{d}{d\rho}Q(\rho, Y_\rho) = (\mathcal{D}Q)(\rho, Y_\rho), \quad \mathcal{D} = \partial_\rho - rM_\rho \cdot \nabla.$$

Thus the required path derivatives are iterates of $\mathcal{D}$.

We express the material identities using the spatial adjoint under ξ_ρ . For a tensor field V with distinguished slot k , define its adjoint divergence on that slot by

$$(\nabla^*V)_I = \sum_{k=1}^d \{-\partial_k V_{I,k} + (\partial_k H_\rho) V_{I,k}\}.$$

Here I lists the slots that remain; the distinguished slot need not have been last before a permutation. For a compactly supported smooth test function ϕ , the defining integration-by-parts relation is

$$\int (\nabla^*V)_I \phi d\xi_\rho = \int \sum_{k=1}^d V_{I,k} \partial_k \phi d\xi_\rho.$$

The next identities express material derivatives using this operation and contractions of the normalized spatial jets $U_q = \rho^{-q} \nabla^q M_\rho$ from Proposition G.2.

H.1 Material derivatives of the conditional jets

Source attribution. Adapted from OpenAI [Ope26, Lemma 4.1].

Lemma H.1. *The mean and its jets satisfy*

$$\mathcal{D}M = \nabla^*U_1, \tag{H.1}$$

$$(\mathcal{D}U_q)_{i,I} = (\nabla^*U_{q+1})_{i,I} + 2r\rho \sum_{\emptyset \neq S \subseteq I} (U_{|S|})_{k,S} (U_{q-|S|+1})_{i,k,I \setminus S}, \quad q \geq 1. \tag{H.2}$$

In the second identity I is a labeled list of q derivative slots; subsets refer to positions in that list and the index k is summed. For any smooth tensor field V ,

$$\mathcal{D}(\nabla^*V) = \nabla^*(\mathcal{D}V) + r\nabla^*((DM_\rho)V), \tag{H.3}$$

where the matrix acts on the slot removed by the divergence. These identities hold also at $\rho = 0$ by continuous extension.

Proof. Differentiate $\mathbb{E}[f_i(Z)\phi(Y)] = \int (M_\rho)_i \phi d\xi_\rho$. The right side differentiates using the transporting velocity from Proposition G.3. On the left the random velocity is $(Z - \rho Y)/v_\rho$. Subtract its conditional mean from the flux multiplying $\nabla \phi$. The remaining coefficient is $\text{Cov}(f_i(Z), Z_k | Y)/v_\rho = (U_1)_{i,k}$ by (G.7). Integration by parts proves (H.1).

For the higher identity, coordinate differentiation gives

$$\mathcal{D}\partial_I M_i = \partial_I \mathcal{D}M_i + r \sum_{\emptyset \neq S \subseteq I} (\partial_S M_k) \partial_k \partial_{I \setminus S} M_i.$$

Insert (H.1) and multiply by ρ^{-q} . When all derivatives pass inside the adjoint divergence, the result is $\nabla^* U_{q+1}$. In the other terms they hit the score $y + r\rho M_\rho$. A derivative hitting its linear part contributes qU_q/ρ , canceled by differentiating the prefactor ρ^{-q} . The nonlinear score terms and the material/spatial commutator terms are identical after scaling: each is

$$r\rho \sum_{\emptyset \neq S \subseteq I} (U_{|S|})_{k,S} (U_{q-|S|+1})_{i,k,I \setminus S}.$$

This proves (H.2). Each contraction retains at least one free slot in each tensor.

Finally differentiate the adjoint identity against a transported scalar test, componentwise in the other slots. The terms involving $\mathcal{D}\phi$ cancel. The gradient commutator is

$$\mathcal{D}\partial_k \phi - \partial_k \mathcal{D}\phi = r(\partial_k M_j) \partial_j \phi.$$

Moving it to the other side by the adjoint relation gives (H.3). All coefficients in the final formulas are nonsingular at zero, so the extensions follow from Proposition G.2. $\square$

H.2 Contractions and the expressions they generate

The product terms in the material identities contract one slot from each of two tensors. We need this operation to preserve split-norm bounds. The next lemma gives the matrix-composition interpretation; it requires both factors to retain a free slot.

Source attribution. Adapted from OpenAI [Ope26, Lemma 4.2].

Lemma H.2 (A single contraction). *Let A and B be tensors, and contract one slot of A with one slot of B , leaving at least one free slot in each. For any nontrivial split of the resulting free slots there are nontrivial splits of A and B such that*

$$|A * B|_{\text{split}} \leq |A|_{\text{split}} |B|_{\text{split}}.$$

Proof. Fix the desired split of the resulting tensor. We orient the contracted index so that each factor acts between nonempty input and output groups. If all free slots of one factor lie on the same side, its contracted slot must be placed on the opposite side of that factor's unfolding. These requirements are compatible: incompatibility would put every free slot of the result on one side, contrary to the hypothesis. If a factor already has free slots on both sides, it imposes no requirement on the orientation. With the resulting orientation, the contraction is composition of the two matrix actions, with identities on the unused input and output coordinates and with coordinate permutations. Identities and these permutations have operator norm one. Submultiplicativity proves the assertion. $\square$

We now define the class of expressions generated by the material identities. Its contraction rule does not permit a trace within a single factor.

Source attribution. Adapted from OpenAI [Ope26, Definition 4.3].

Definition H.3 (Tensor expressions and their weights). A tensor expression is a finite formula built by the following rules.

1. For each integer $q \geq 1$, U_q/L is an expression with $q + 1$ free slots and weight $q - 1$.
2. A permutation of the free slots preserves the weight.

3. Two expressions may be contracted on one chosen slot each, provided each factor retains at least one free slot. The resulting weight is the sum of their weights.
4. An adjoint divergence on a chosen free slot is allowed when at least two free slots remain. It increases the weight by one.

The formula specifies its leaves, their orders, and the slots used at each operation. This finite syntactic data is independent of d, ρ, x, r and of the potential. Repeated occurrences are treated as separate factors, even when their values agree.

For example, $\nabla^*(U_2/L)$ has two free slots and weight two, whereas $\nabla^*(U_1/L)$ is not an expression under these rules: it has vector output. The latter is precisely the operation we will reserve for the final $\sqrt{d}$ estimate. If E is an expression, $\nabla^j E$ has j additional free derivative slots, and a split may place any of these slots on either side. The next theorem bounds all such splits simultaneously. Its dependence on j and p is explicit because its proof must later take a moment order growing with $\log d$.

For a related Gaussian result, [CCLZ26b, Theorem 1.1 and Corollary 1.4] prove sharp divergence inequalities with Hilbert–Schmidt derivative norms. Here we require estimates under the generally non-Gaussian law ξ_ρ that retain every matrix split through the contractions and adjoint divergences in Definition H.3.

Source attribution. Adapted from OpenAI [Ope26, Theorem 4.4].

Theorem H.4 (Tensor divergence bound). *Let E be a tensor expression of weight k in Definition H.3. There is a constant $C_E \geq 1$, depending only on that finite formula, such that for every integer $j \geq 0$, every real $2 \leq p < \infty$, every $0 \leq \rho < 1$, and every nontrivial split,*

$$\|\nabla^j E\|_{\text{split}} \leq C_E^{j+1} (p + j + \log(d+1))^{C_E(j+1)} (\sqrt{v_\rho})^{-(k+j)}. \quad (\text{H.4})$$

The same C_E works for all j, p, d, ρ, x, r . In particular it is not restricted to fixed derivative or moment orders.

Proof. We induct on the construction of E , simultaneously in the derivative and moment orders. The only new estimate is the adjoint-divergence step. For that step, we expand a high trace moment and remove the adjoints by integration by parts. We will arrange each resulting contraction as a matrix composition followed by one trace.

For a normalized jet, $\nabla^j U_q = \rho^j U_{q+j}$, so Proposition G.2 proves the assertion after enlarging a constant depending on q . Permutations cause no change. For a contraction, distribute the derivatives by Leibniz, use Lemma H.2 on each summand, and use the inductive estimates at exponent $2p$ on both factors. The number of Leibniz summands is at most 2^j . Their weights and derivative orders add, giving (H.4) with an enlarged fixed constant.

It remains to consider $E = \nabla^* V$, where V has weight k_V and E has at least two slots. Commute the j derivatives past the divergence. Apart from

$$T = \nabla^*(\nabla^j V),$$

the terms are single contractions of $\nabla^{j-h} V$ with an order- h derivative of the score, $1 \leq h \leq j$. Both factors retain slots. The score derivative is bounded pointwise by a constant of the form $C^h h^{Ch} (\sqrt{v_\rho})^{-(h-1)}$, by (G.5)–(G.6). Lemma H.2, the inductive hypothesis, and Leibniz multiplicities therefore give the required bound for all these terms, even with exponent $k_V + j - 1$ in place of the larger allowed exponent $k_V + j + 1$. It remains to bound T .

Step 1: integration by parts in a trace moment. Fix an unfolding of T , with both sides nonempty, and an integer $s \geq 1$. Pointwise,

$$\|T\|_{\text{op}}^{2s} \leq \text{tr}((TT^T)^s).$$

Expand the trace by coordinate indices. It contains $2s$ labeled primary factors, each an adjoint divergence of $\nabla^j V$. Their free slots are joined in the alternating trace cycle; the two bundles incident to each primary factor are nonempty.

Remove adjoint divergences one at a time by integration by parts. For a chosen primary, its adjoint differentiates the product of the other factors, never its own inside field. If a derivative hits a remaining adjoint, use

$$\partial_i(\nabla^* W) = \nabla^*(\partial_i W) + (\partial_i \partial_k H_\rho) W_k.$$

The first term leaves that adjoint to be processed later. The second consumes it and creates a Hessian factor joining the two divergence indices. A later derivative may also hit a previously created Hessian factor. After all adjoints have been removed, a typical term has primary factors $\nabla^{j+e_i} V$, $1 \leq i \leq 2s$, and n_H factors $\nabla^{h_v} \nabla^2 H_\rho$. The bookkeeping identity is

$$\sum_{i=1}^{2s} e_i + \sum_{v=1}^{n_H} h_v + 2n_H = 2s. \quad (\text{H.5})$$

Indeed, a removed adjoint either adds one derivative or pairs with another adjoint to create a Hessian. Each step has at most Cs possible target factors and at most two commutator terms, so there are at most $(Cs)^{Cs}$ resulting terms.

We use a graph only to describe these explicit coordinate sums. Its primary vertices are the $2s$ labeled tensor factors; a Hessian vertex represents one factor $\nabla^{h_v} \nabla^2 H_\rho$. An edge represents a pair of tensor slots summed over a common coordinate, and a bundle represents several such edges. Three properties will control its contraction. All original trace bundles remain. Every new direct edge connects distinct primary factors, because an adjoint never differentiates its own inside field. Finally, each Hessian factor has at least two distinct primary neighbors, the pair that created it, and there are no Hessian-to-Hessian edges. Later derivatives only add further primary-to-Hessian edges. These statements remain true if some derivatives had already been commuted inside an unprocessed adjoint.

Step 2: one closing trace. Choose one bundle in the original trace cycle and cut it, leaving its $b \geq 1$ indices unsummed at the two ends. All remaining original bundles form a chain through the $2s$ primary factors. Order these factors along that chain. For each Hessian factor, its two distinct creating primaries have different positions, so we can place it strictly between its earliest and latest primary neighbors. No edge joins two Hessians, so these placements impose no competing condition. Order Hessians occupying the same interval arbitrarily, and orient every uncut edge from its earlier factor to its later factor. The b cut indices enter the first primary and leave the last one.

Every tensor has both an input and an output in this ordering. An interior primary receives one original chain bundle and sends the next; at the two endpoints the cut bundle supplies the missing side. A Hessian receives an edge from an earlier neighbor and sends an edge to a later one. Thus all the matrix unfoldings used below are nontrivial splits, exactly as required by the induction hypothesis.

Here is an explicit matrix-composition interpretation. Immediately before processing a factor, retain one copy of $\mathbb{R}^d$ for each index whose earlier endpoint has been processed but whose later endpoint has not. The initial space is $(\mathbb{R}^d)^{\otimes b}$, belonging to the cut inputs. Permute these tensor

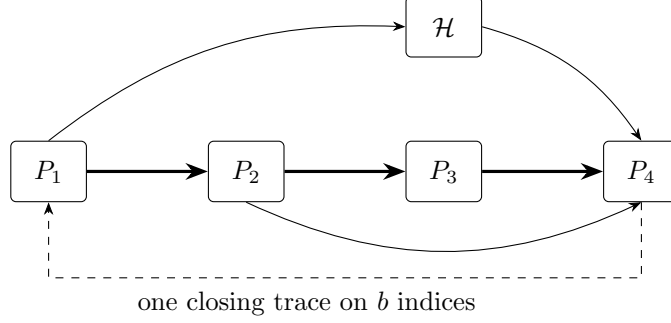


Figure 1: A schematic term with four primary tensors $P_i = \nabla^{j+e_i} V$ after removing the adjoint divergences. Thick arrows are the surviving trace-chain bundles; thin arrows are new contracted indices. Here $\mathcal{H} = \nabla^2 H_\rho$ is the Hessian factor, inserted between its first and last primary neighbors. The arrows specify matrix inputs and outputs, so every factor acts across a nontrivial split. The dashed closure is the one trace in (H.6); its cost d^b becomes bounded after taking a sufficiently high moment root.

coordinates so that the incoming indices of the next factor are adjacent. Apply that factor's matrix unfolding to them and the identity to all other retained coordinates; then permute the outputs into the new retained list. Matrix multiplication performs precisely the sums over the incoming indices. Induction over the factors therefore accounts for every uncut index once and leaves only the b cut outputs at the end. If Q denotes the resulting map from $(\mathbb{R}^d)^{\otimes b}$ to itself, then

$$\|Q\|_{\text{op}} \leq \prod_{i=1}^{2s} |\nabla^{j+e_i} V|_{\text{split}} \prod_{v=1}^{n_H} |\nabla^{h_v} \nabla^2 H_\rho|_{\text{split}}.$$

Indeed coordinate permutations and the intervening identity maps have operator norm one. Closing the original cut is now exactly a single matrix trace, so the absolute value of this completely contracted term is at most

$$|\text{tr } Q| \leq d^b \|Q\|_{\text{op}}. \quad (\text{H.6})$$

This argument explains why other undirected cycles introduce no further dimension factor: their coordinate sums have already been performed by the matrix compositions. Figure 1 shows a possible ordering.

Step 3: choice of the moment order. The coordinate expansion and its contraction estimate are now complete. It remains to verify that their constants allow s to grow with d and j . Apply the induction hypothesis to every primary factor with moment exponent $2s$, and use the pointwise Hessian estimates on all other factors. There are exactly $2s$ random primary factors, so Hölder's inequality uses no larger exponent. The power of $1/\sqrt{v_\rho}$ before taking the $2s$ -th root is at most

$$2s(k_V + j) + \sum_i e_i + \sum_v h_v \leq 2s(k_V + j + 1).$$

Since $e_i, h_v \leq 2s$, all polynomial bases in the inductive estimates are at most $C(s + j + \log(d + 1))$. The total exponent of those bases is at most

$$C \left(2s(j + 1) + \sum_i e_i + \sum_v (h_v + 1) \right) \leq C' s(j + 1),$$

using (H.5). The term count $(Cs)^{Cs}$ is absorbed in the same bound. The cut bundle uses only the original free slots of $T = \nabla^*(\nabla^j V)$, not the new indices introduced by integration by parts. Consequently $b \leq j + C_V$, where C_V depends only on the fixed number of slots of V , and not on s . We have proved

$$\mathbb{E}\|T\|_{\text{op}}^{2s} \leq d^{j+C_V} [C(s+j+\log(d+1))]^{Cs(j+1)} (\sqrt{v_\rho})^{-2s(k_V+j+1)}.$$

Choose s to be the smallest positive integer satisfying $2s \geq p$ and $s \geq (j+C_V)\log(d+1)$, increasing C_V if necessary. Then $d^{(j+C_V)/(2s)} \leq e^{1/2}$ and

$$s \leq C_V(p + (j+1)\log(d+1)).$$

The expression on the right is bounded by a fixed multiple of $(p+j+\log(d+1))^2$. Taking the $2s$ -th root, using monotonicity of probability-space L^p norms, and increasing C_E now gives (H.4) with $k = k_V + 1$.

All integration-by-parts identities above are legitimate for each finite j, s, d and each $\rho < 1$. Indeed, the conditional-integral regularity in Proposition G.2, together with the score formula, gives polynomial growth for every field and derivative appearing in the finite expansion. The laws ξ_ρ have Gaussian tails. Cutoffs therefore justify the identities with vanishing boundary terms. Uniformity in growing s is supplied by the explicit estimates after the identities, not by an unproved uniform regularity assertion. This completes the induction. $\square$

H.3 One final vector dimension factor

We now pass from split norms to the vector output. All preceding operations retained at least two slots. The last divergence is estimated in Hilbert–Schmidt norm, using the following L^2 identity.

Source attribution. Adapted from OpenAI [Ope26, Lemma 4.5].

Lemma H.5 (Hilbert–Schmidt divergence estimate). *Let $\mu(dy) \propto e^{-H(y)}dy$, where $H \in C^2(\mathbb{R}^d)$ has bounded Hessian and μ has Gaussian tails. Let V be a smooth tensor field with a distinguished divergence slot and with $V, \nabla V \in L^2(\mu)$ in Hilbert–Schmidt norm. Then*

$$\|\nabla^* V\|_{\text{HS},2} \leq C(\|V\|_{\text{HS},2} + \|\nabla V\|_{\text{HS},2}), \quad (\text{H.7})$$

where C depends only on $\|\nabla^2 H\|_{\text{op},\infty}$, and is independent of d and the number of remaining tensor slots. For $H = H_\rho$ it is an absolute constant.

Proof. First take a compactly supported smooth vector field, so that the output is scalar. Integrating by parts twice gives

$$\int (\nabla^* V)^2 d\mu = \int \sum_{i,j} [(\partial_i V_j)(\partial_j V_i) + V_i(\partial_{ij} H)V_j] d\mu.$$

The first term is bounded in absolute value by $|\nabla V|_{\text{HS}}^2$, and the second by $\|\nabla^2 H\|_{\text{op}}|V|^2$. For tensor output, apply the scalar identity to each fixed collection of remaining indices and sum. Taking a square root proves (H.7) for compactly supported fields. For general V , multiply by smooth cutoffs χ_R equal to one on the ball of radius R , with $|\nabla \chi_R| \leq C/R$. Then $\chi_R V \rightarrow V$ and $\nabla(\chi_R V) \rightarrow \nabla V$ in $L^2(\mu)$, while their adjoint divergences converge pointwise. Fatou’s lemma proves the stated estimate. All fields to which we apply it have polynomially growing derivatives and hence satisfy the required integrability. $\square$

Source attribution. Adapted from OpenAI [Ope26, Corollary 4.6].

Corollary H.6 (Derivatives along transported paths). *For each fixed integer $m \geq 1$,*

$$\|\mathcal{D}^m M_\rho\|_{L^2(\xi_\rho)} \leq \text{polylog}(d) L \sqrt{d} v_\rho^{-m}. \quad (\text{H.8})$$

For fixed integers $q \geq 1$, $m \geq 0$, and fixed real $2 \leq p < \infty$, every nontrivial split satisfies

$$\| |\mathcal{D}^m U_q|_{\text{split}} \|_{L^p(\xi_\rho)} \leq \text{polylog}(d) L (\sqrt{v_\rho})^{-(q-1+2m)}. \quad (\text{H.9})$$

The constants hidden in $\text{polylog}(d)$ may depend on m, q, p when these parameters occur, but are uniform in ρ , the primitive center, scale, and oracle.

Proof. We first prove (H.9), and then isolate the final divergence needed for (H.8). Iterate Lemma H.1 and use the product rule on each contraction. Replacing U_q by $\nabla^* U_{q+1}$ raises weight by two. Replacing it by one of the contracted products in (H.2) does not raise weight: the two jet weights sum to $q - 1$. The commutator (H.3) inserts $DM_\rho = \rho U_1$ of weight zero on the divergence slot. Every additional jet factor L is accompanied by a factor r , so, after retaining one overall L , the extra factors are powers of $l = rL \leq 1$. Scalar coefficients are fixed polynomials in ρ and remain bounded under the fixed number of differentiations. Starting from a jet U_q , all divergences retain at least two free slots, and all contractions retain a slot of each factor. The resulting terms are therefore expressions of Theorem H.4, of weight at most $q - 1 + 2m$. Its case $j = 0$ proves (H.9).

Starting instead from M , the first identity is $\mathcal{D}M = \nabla^* U_1$. Its outermost divergence remains a divergence to a vector under further material differentiations. Inside it, the same construction gives finitely many normalized expressions of weight at most $2(m - 1)$, each with one overall factor L . Apply Lemma H.5 at this outermost operation. For a tensor with at least two slots, its Hilbert–Schmidt norm is at most $\sqrt{d}$ times the operator norm of an unfolding that puts one coordinate slot on one side: that matrix has at most d singular values. Thus Theorem H.4, for $j = 0, 1$ and $p = 2$, bounds the right side of (H.7) by

$$\text{polylog}(d) L \sqrt{d} (\sqrt{v_\rho})^{-(2m-1)} \leq \text{polylog}(d) L \sqrt{d} v_\rho^{-m}.$$

This is (H.8). □

To see the quantitative connection with quadrature, start the transport at its correct marginal and put $Q_\rho = M_\rho(Y_\rho)$. Proposition G.3 ensures that $Y_\rho \sim \xi_\rho$, so (H.8) bounds $\|Q_\rho^{(m)}\|_2$ for every probability time. On a cell $[\rho_0, \rho_1] \subset [0, 1)$ of length h_c , Taylor’s integral remainder and the L^2 triangle inequality give, for every $\rho \in [\rho_0, \rho_1]$,

$$\left\| Q_\rho - \sum_{h=0}^{m-1} \frac{(\rho - \rho_0)^h}{h!} Q_{\rho_0}^{(h)} \right\|_2 \leq \frac{h_c^m}{m!} \sup_{t \in [\rho_0, \rho_1]} \|Q_t^{(m)}\|_2 \leq C_m \text{polylog}(d) L \sqrt{d} h_c^m (1 - \rho_1^2)^{-m}.$$

In particular, for a mesh ratio $\theta > 0$, the condition $h_c \leq \theta(1 - \rho_1)$ makes this remainder at most $C_m \text{polylog}(d) L \sqrt{d} \theta^m$. This is a supremum of deterministic RMS bounds; it does not require an expected supremum over a random trajectory. The proof of Proposition J.1 uses this cancellation between cell length and endpoint singularity in polynomial quadrature. Before constructing those finite formulas, we use the material identity itself to produce a stationary flow whose path integral represents a noisy mean.

I A stationary representation of a noisy mean

The mean formula comes from an invariant flow whose integrated observable has exactly the desired Gaussian law. We construct its centering kernel, prove global existence and invariance, and derive the stationary time-derivative estimates used by Picard quadrature and the inherited primitive analysis. This section develops the construction of [Ope26] with the conditional notation fixed above.

Retain the primitive and interpolation notation of Sections F–H. Thus $L = \lambda r$, $l = \lambda r^2 \leq 1/2$, $M_0 = u$, and the symmetric first jet satisfies $DM_\rho = \rho U_1$. Fix

$$0 < R < 1/2, \quad b_* = (1 - R^2)^{1/2},$$

and use a star for evaluation at $\rho = b_*$. The positive smoothing radius R keeps all coefficients smooth, although F itself is only C^2 . In this section ρ is the Gaussian-interpolation parameter; the new stationary flow will use a separate time variable t .

I.1 Transport Jacobians and the zero endpoint

Let $T_\rho = T_{\rho \rightarrow b_*}$ denote the probability transport (G.8) from time ρ to time b_* . Given a terminal position y , define

$$Y_\rho(y) = T_\rho^{-1}(y), \quad B_\rho(y) = DT_\rho(Y_\rho(y)) = (DY_\rho(y))^{-1}.$$

In particular, $Y_{b_*}(y) = y$. The backward path satisfies $Y'_\rho = -rM_\rho(Y_\rho)$ with this terminal condition. Its spatial Jacobian $J_\rho = DY_\rho$ solves

$$J'_\rho = -rDM_\rho(Y_\rho)J_\rho, \quad J_{b_*} = I_d.$$

Differentiating its inverse gives the order of multiplication in

$$B'_\rho = rB_\rho DM_\rho(Y_\rho) = r\rho B_\rho U_1(\rho, Y_\rho), \quad B'_0 = 0. \tag{I.1}$$

The equality $B'_0 = 0$ will be used in the quadrature at zero. Gronwall also gives $\|B_\rho\|_{\text{op}} \leq C$, since $r\|DM_\rho\|_{\text{op}} \leq Cl$ on an interval of length at most one. We record the higher spatial and time derivative bounds together.

Source attribution. Adapted from OpenAI [Ope26, Lemma 5.1].

Lemma I.1 (Transport derivatives). *For each fixed $k \geq 1$, the order- k spatial derivatives of any transport between two times in $[0, b_*]$, after subtracting the identity map, have all nontrivial split norms at most*

$$C_k l R^{-(k-1)}.$$

For each fixed $m \geq 1$ and fixed finite $p \geq 1$, uniformly in $0 \leq \rho \leq b_$,*

$$\| \|B_\rho^{(m)}(y)\|_{\text{op}} \|_{L^p(y \sim \xi_*)} \leq \text{polylog}(d) l (1 - \rho^2)^{-m}.$$

The constants and the polylogarithmic factor may depend on the fixed orders and on p , but not on the primitive center or scale.

Proof. For the spatial estimate, differentiate the transport in its initial position. Gronwall bounds its first derivative and gives an $O(l)$ bound for the difference from I_d . For order $k > 1$, the chain rule has terms consisting of an order- j derivative of the velocity and derivatives of the state of orders $i_1, \dots, i_j \geq 1$, where $i_1 + \dots + i_j = k$. By (G.4), the outer tensor has every split norm at most $C_j l R^{-(j-1)}$. Attach the inner tensors one at a time. Each attachment is a single contraction retaining the outer output and the inner derivative slots, so Lemma H.2 applies without a dimension factor. Bound a first state derivative by a constant and use the induction hypothesis for all higher state derivatives. Since $l \leq 1$, the total power of R^{-1} is at most

$$(j-1) + \sum_{\nu=1}^j (i_\nu - 1) = k - 1,$$

and each forcing term retains the outer factor l . The term involving the unknown order- k state derivative is linear, with coefficient bounded by Cl , and this derivative starts at zero. Gronwall proves the assertion. The same argument with absolute interval lengths applies to transport in the reverse direction.

For the second assertion, write $C_\rho = rDM_\rho(Y_\rho) = r\rho U_1(\rho, Y_\rho)$, so that $B'_\rho = B_\rho C_\rho$. The product rule expresses $B_\rho^{(m)}$ as a finite sum of ordered products of B_ρ and $k \geq 1$ factors $C_\rho^{(j_1)}, \dots, C_\rho^{(j_k)}$, with $j_1 + \dots + j_k = m - k$. The zero-order jet bound and (H.9) imply, for every fixed moment order,

$$\|C_\rho^{(j)}\|_{\text{op}} \|_{L^p} \leq \text{polylog}(d) l (1 - \rho^2)^{-j}.$$

Here the path estimates apply because inverse transport sends $y \sim \xi_*$ to $Y_\rho(y) \sim \xi_\rho$. Apply Hölder with sufficiently high fixed moments to each product. Using $l^k \leq l$ and $(1 - \rho^2)^{-(m-k)} \leq (1 - \rho^2)^{-m}$ gives the claimed conservative exponent. Moments below one, if desired, follow from the L^1 bound. $\square$

I.2 A kernel for the centered observable

For a matrix field K , define its row-wise adjoint divergence by

$$(\nabla_{\xi_*}^* K)_i = \sum_j \{-\partial_j K_{ij} + (\partial_j H_*) K_{ij}\}.$$

Thus $\nabla_{\xi_*}^* K = h$ means $\mathbb{E}_{\xi_*}[h_i \phi] = \mathbb{E}_{\xi_*} \sum_j K_{ij} \partial_j \phi$ for smooth compactly supported scalar tests ϕ . We seek this identity with $h = M_* - u$.

This is a Stein-type integration-by-parts identity. A classical coordinate Stein kernel for a law ξ represents the observable $y - \mathbb{E}_\xi y$ instead. Courtade, Fathi and Pananjady construct such kernels from spectral-gap assumptions [CFP19, arXiv version 2, Definition 2.1 and Theorem 2.4]. Our observable is $M_* - u$; its mean is zero by conditional expectation. The transport formula below supplies its particular adjoint-divergence representation together with pointwise derivative bounds. Those bounds are needed for quadrature and do not follow from the cited existence statement. We do not require K to be symmetric or positive.

Define

$$K(y)^T = \int_0^{b_*} B_\rho(y) U_1(\rho, Y_\rho(y)) d\rho = \frac{1}{r} \int_0^{b_*} B'_\rho(y) \frac{d\rho}{\rho}. \quad (\text{I.2})$$

The first integral defines the integrand at zero in the second formula: $B'_\rho/(r\rho) = B_\rho U_1(\rho, Y_\rho)$ extends smoothly there. The transpose is essential: the adjoint divergence will act on the rows of K , while the position velocity will use K^T .

Source attribution. Adapted from OpenAI [Ope26, Lemma 5.2].

Lemma I.2 (The centering kernel). *The field K is smooth and satisfies*

$$\nabla_{\xi_*}^* K = M_* - u, \quad \|K(y)\|_{\text{op}} \leq CL.$$

For each fixed $j \geq 1$, every nontrivial split norm of $\nabla^j K$ is bounded pointwise by $C_j LR^{-j}$.

Proof. We prove the adjoint-divergence identity against a scalar test. Fix an output coordinate i and a smooth compactly supported ϕ . The function $\phi \circ T_\rho$ is constant along the probability flow, and therefore has zero material derivative. Differentiate its product with $M_{\rho,i}$ under the transported laws. Lemma H.1 gives $\mathcal{D}M_\rho = \nabla_{\xi_\rho}^* U_1$, so integration by parts and the chain rule yield

$$\frac{d}{d\rho} \mathbb{E}_{\xi_\rho} [M_{\rho,i} \phi \circ T_\rho] = \mathbb{E}_{\xi_\rho} \sum_{j,k} (U_1)_{ik} (DT_\rho)_{jk} (\partial_j \phi) \circ T_\rho.$$

Smooth transport on the compact time interval and Gaussian tails justify differentiation; alternatively the test can first be cut off and the resulting identities passed to the limit. Push the expectation forward to $y \sim \xi_*$. Since U_1 is symmetric, its coefficient of $\partial_j \phi(y)$ is

$$\sum_k (U_1)_{ik}(\rho, Y_\rho(y)) (B_\rho(y))_{jk} = (B_\rho(y) U_1(\rho, Y_\rho(y)))_{ji}.$$

The terminal expectation is $\mathbb{E}_{\xi_*} [M_{*,i} \phi]$. At zero, $M_0 = u$ and $(T_0)_\# \xi_0 = \xi_*$, so the initial expectation is $u_i \mathbb{E}_{\xi_*} \phi$. Integrating the displayed derivative in ρ proves

$$\mathbb{E}_{\xi_*} [(M_{*,i} - u_i) \phi] = \mathbb{E}_{\xi_*} \sum_j K_{ij} \partial_j \phi.$$

This proves the adjoint-divergence identity weakly, and smoothness makes it a pointwise identity.

For the bounds, $\|B_\rho\|_{\text{op}} \leq C$ and $\|U_1\|_{\text{op}} \leq CL$ immediately give $\|K\|_{\text{op}} \leq CL$. For a positive-order spatial derivative, use Lemma I.1 on both DT_ρ and Y_ρ . An order- j derivative of DT_ρ has all split norms at most $C_j l R^{-j}$, and an order- j derivative of U_1 has all split norms at most $C_j LR^{-j}$ by (G.4). When these fields are composed with Y_ρ , attach each chain-rule tensor by a single edge as in the preceding lemma. First derivatives of Y_ρ are bounded, and a derivative of order $i > 1$ costs $C_i l R^{-(i-1)}$. Every order- j term in the product has total cost at most $C_j LR^{-j}$, since $l \leq 1$. Each factor retains free slots, so the single-contraction bound preserves every nontrivial split. Integrating over an interval of length at most one proves the assertion. All orders are smooth by Gaussian conditional integration and the smooth transport on $\rho \leq b_* < 1$; this uses no higher derivatives of the original unsmoothed gradient. $\square$

I.3 Stationarity and the path integral

The kernel K defines a skew-matrix flow of the form studied by [MCF15, Section 2.1, equation (3), Theorem 1]. We verify completeness before using the stationary-density identity. This ensures that invariance holds for the full integration interval.

Source attribution. Adapted from OpenAI [Ope26, Proposition 5.3].

Proposition I.3 (A noisy mean from a path integral). *For $s > 0$, let (Y_0, G_0) have law $\xi_* \otimes \mathbf{N}(0, I_d)$ and solve*

$$\dot{Y}_t = -K(Y_t)^T G_t/s, \quad \dot{G}_t = (M_*(Y_t) - u)/s. \quad (\text{I.3})$$

The solution exists for every finite time, and its law remains $\xi_ \otimes \mathbf{N}(0, I_d)$. In particular,*

$$sG_0 + \int_0^1 M_*(Y_t) dt \stackrel{\text{law}}{=} u + sG, \quad G \sim \mathbf{N}(0, I_d). \quad (\text{I.4})$$

Proof. We first establish a complete flow, then invariance, and finally the output law. The vector field v in (I.3) is smooth, so it is locally Lipschitz. The bound on K and the global Lipschitz bound on M_* imply $|v(y, g)| \leq C_{x,r,s}(1 + |y| + |g|)$. Gronwall's inequality therefore prevents a solution from escaping to infinity in finite positive or negative time. Local uniqueness gives a complete smooth flow Φ_t , with inverse Φ_{-t} . We only need local Lipschitz continuity here; the factor G does not in general give a globally Lipschitz joint vector field.

Write the product density as

$$p(y, g) = (2\pi)^{-d/2} \exp(-H_*(y) - |g|^2/2).$$

Differentiating in the position and momentum variables separately, with the row convention of the kernel lemma, gives

$$\operatorname{div}_y(pv_Y) = \frac{p}{s} \sum_i g_i (\nabla_{\xi_*}^* K)_i = \frac{p}{s} g \cdot (M_* - u), \quad \operatorname{div}_g(pv_G) = -\frac{p}{s} g \cdot (M_* - u).$$

Thus $\operatorname{div}(pv) = 0$. The Jacobian formula for the complete flow now yields, pointwise,

$$\frac{d}{dt} \{p(\Phi_t(z)) \det D\Phi_t(z)\} = \operatorname{div}(pv)(\Phi_t(z)) \det D\Phi_t(z) = 0.$$

The determinant starts at one and stays positive. Change of variables therefore proves that Φ_t preserves the probability density p .

For later differentiation it is useful to record the same cancellation as an operator identity. Relative to the product law, the Lie derivative is

$$\mathcal{L} = \nabla^* A \nabla, \quad A = \begin{pmatrix} 0 & K^T/s \\ -K/s & 0 \end{pmatrix}. \quad (\text{I.5})$$

Indeed the second-order part vanishes by skew symmetry. The first-order coefficients are $-K^T g/s$ in position and $\nabla_{\xi_*}^* K/s = (M_* - u)/s$ in momentum, exactly as in (I.3).

Finally, integrating the momentum equation over $[0, 1]$ gives the pathwise identity

$$sG_0 + \int_0^1 M_*(Y_t) dt = u + sG_1.$$

Stationarity makes G_1 a standard Gaussian, proving (I.4). $\square$

Stationarity also makes G_1 independent of Y_1 . It does not assert that G_1 is independent of the initial anchors (Y_0, G_0) . The exact mean $u = M_0$ remains in the momentum velocity, and the proposition assumes an ideal sample from ξ_* . Thus the proposition is a comparison construction, not yet an oracle routine. Section J will express its approximation by finitely many exact mean evaluations; Section K will replace every such evaluation, including M_0 . The displayed output itself needs no separate addition of u .

I.4 Derivatives on a stationary trajectory

For quadrature, it suffices to bound each time derivative in L^2 at a fixed time. Stationarity identifies these norms with norms under the invariant product measure. The following lemma gives the estimate in a form that also records the required coefficient assumptions.

Source attribution. Adapted from OpenAI [Ope26, Lemma 5.4].

Lemma I.4 (Stationary time derivatives). *Fix $0 < R \leq 1$. Let a smooth probability density on $\mathbb{R}^n$, with $n \leq Cd$, have Gaussian tails and negative logarithm H . Assume its Hessian is bounded and that every positive-order score derivative $\nabla^j \nabla H$ has every one-slot split norm at most $C_j R^{-(j-1)}$, for $j \geq 1$. Here a one-slot split isolates one tensor slot as the matrix input and groups all remaining slots as its output. Suppose a stationary autonomous flow has Lie derivative $\mathcal{L} = \nabla^* A \nabla$, where the adjoint is for this density, A is skew symmetric, and*

$$|\nabla^j A|_{\text{one-slot split}} \leq C_j \alpha R^{-j} \quad (j \geq 0)$$

for every one-slot split. Let ϕ_0 be smooth and scalar- or vector-valued, with

$$|\nabla^j \phi_0|_{\text{HS}} \leq C_j a_0 \sqrt{d} R^{-j} \quad (j \geq 1).$$

Assume the fields and all derivatives used have polynomial growth. Then, for each fixed $k \geq 1$, the order- k time derivative of ϕ_0 on the stationary flow has RMS norm at most

$$C_k a_0 \sqrt{d} (\alpha R^{-2})^k. \quad (\text{I.6})$$

Proof. Define $\phi_k = \mathcal{L}^k \phi_0$. We induct on k , proving simultaneously for $j \geq 0$ the stronger estimate

$$\|\nabla^j \phi_k\|_{\text{HS},2} \leq C_{j,k} a_0 \sqrt{d} \alpha^k R^{-(j+2k)}. \quad (\text{I.7})$$

For $k = 0$ the same bound is assumed only for $j \geq 1$, which suffices: the iteration differentiates ϕ_{k-1} before multiplying by A . All L^2 norms below are for the invariant density.

Set $V = A \nabla \phi_{k-1}$. Leibniz's rule and the induction hypothesis give, for each $h \geq 0$,

$$\|\nabla^h V\|_{\text{HS},2} \leq C_{h,k} a_0 \sqrt{d} \alpha^k R^{-(h+1+2(k-1))}.$$

To see why no dimension factor appears, in each coefficient tensor $\nabla^v A$ isolate the slot contracted against the gradient of ϕ_{k-1} . Its one-slot operator norm multiplies the Hilbert–Schmidt norm of the other factor. Summing the finitely many Leibniz terms preserves the displayed estimate.

Now commute j derivatives past the adjoint divergence in $\phi_k = \nabla^* V$. The main term is $\nabla^*(\nabla^j V)$. The Hilbert–Schmidt estimate (H.7) bounds its L^2 norm by a constant times $\|\nabla^j V\|_{\text{HS},2} + \|\nabla^{j+1} V\|_{\text{HS},2}$. These two terms have powers $R^{-(j+2k-1)}$ and $R^{-(j+2k)}$, respectively. Each remaining commutator contracts an order- v derivative of the score, $1 \leq v \leq j$, with $\nabla^{j-v} V$. The coefficient's one-slot norm again suffices; the total exponent of R^{-1} is

$$(v-1) + (j-v+1+2(k-1)) = j+2k-2.$$

All these terms are bounded by (I.7) because $0 < R \leq 1$. This closes the induction, including $j = 0$ for $k \geq 1$. Gaussian tails and polynomial growth justify the integrations by parts through cutoffs.

Along an autonomous trajectory, the k th time derivative of ϕ_0 is ϕ_k evaluated at the current state. Stationarity identifies its RMS norm at every time with the fixed-measure norm in (I.7) for $j = 0$. $\square$

For (I.3), the ambient dimension is $2d$, and (I.5) satisfies the lemma's coefficient bounds with $\alpha = L/s$. The product potential is $H_*(y) + |g|^2/2$; its score-derivative bounds follow from (G.4) and the Gaussian factor. For the joint state take $a_0 = 1$: its first derivative has Hilbert–Schmidt norm $\sqrt{2d}$ and all higher derivatives vanish. For M_* take $a_0 = L$. Indeed, isolating any derivative slot in (G.4) gives a matrix with input dimension d , so passage from its operator norm to its Hilbert–Schmidt norm costs at most $\sqrt{d}$. Consequently, at every stationary time,

$$\left\| \frac{d^k}{dt^k}(Y_t, G_t) \right\|_2 \leq C_k \sqrt{d} ((L/s)R^{-2})^k, \quad \left\| \frac{d^k}{dt^k} M_*(Y_t) \right\|_2 \leq C_k L \sqrt{d} ((L/s)R^{-2})^k.$$

The strict radius choice $0 < R < 1/2$ made for this construction satisfies the abstract lemma's $0 < R \leq 1$ hypothesis.

We also need an initial displacement estimate for the later iteration. Conditional expectation and the primitive Poincaré inequality give

$$\mathbb{E}_{\xi_*} |M_* - u|^2 \leq \mathbb{E}_\nu |f - u|^2 \leq CL^2 d.$$

At every stationary time, G_t is independent of Y_t , hence

$$\mathbb{E} |K(Y_t)^T G_t|^2 = \mathbb{E} \|K(Y_t)\|_{\text{HS}}^2 \leq CL^2 d.$$

Integrate the velocity and apply Minkowski's inequality to obtain

$$\|(Y_t, G_t) - (Y_0, G_0)\|_2 \leq Ct(L/s)\sqrt{d} \leq C(L/s)\sqrt{d} \quad (0 \leq t \leq 1). \quad (\text{I.8})$$

These are fixed-time RMS estimates for the exact stationary flow. They give the derivative and starting-error bounds used in the next section to approximate its path integral by finite formulas.

J Finite Picard formulas for conditional means

For the sampler in this manuscript, the executed numerical formulas are the simultaneous-iteration formulas of Section C. The present appendix records the inherited reference construction; its use in the sampler is through that explicitly stated interface substitution, including the recursive-evaluation inputs and query accounting.

Global Picard iteration and high-order quadrature also build on the numerical frameworks of [CCLZ26c, Appendix E, Algorithm E.1 and Theorem E.2] and [LSV18, Algorithm 2 and Theorem 2.3]. The finite conditional-mean circuits and dependency-depth bounds reproduced below are adapted from OpenAI [Ope26, Section 6].

We turn the probability and stationary flows into deterministic finite formulas. The central requirement is a mean-dependency depth fixed before the mesh and interpolation orders. The resulting coefficient, error, and depth bounds are proved explicitly below, following [Ope26]; they are used directly by the recursive implementation and the mean-root specialization.

Fix an integer $K_0 \geq 2$ and $t > 0$ such that

$$40(K_0 + 2)t < \frac{1}{4}, \quad R = \delta^t, \quad \psi = \delta^{4t}, \quad h_0 = \delta^w, \quad 0 < w < t. \quad (\text{J.1})$$

The iteration count will first fix a depth bound. We then choose w and finally the interpolation orders; increasing those orders will not change the depth bound. For the centering flow set $s = \sigma/2$,

where $\sigma > 0$ is the parent mean-noise parameter. Thus the centering formula itself will approximate noise of standard deviation $\sigma/2$. Throughout its numerical analysis assume

$$\max\{l, L/\sigma\} \leq \text{polylog}(d)\delta^b, \quad b \geq \frac{1}{2}. \quad (\text{J.2})$$

For probability transport alone only the corresponding bound on l is required. All probability-time intervals lie in $[0, (1 - cR^2)^{1/2}]$, for a fixed $c > 0$. Constants may depend on fixed interpolation orders and fixed parameters, but not on the center, dimension, primitive scale r , noise parameter, or oracle. A symbol B_d below denotes a bound $C(1 + \log d)^C$ with precisely this uniformity; the fixed exponent and constant in a size hypothesis are part of the fixed parameters. All assertions are for sufficiently large d .

J.1 Probability transport by global Picard iteration

Partition a probability-time interval using equal steps of size at most h_0 in $-\log(1 - \rho)$. For a nonzero interval take the ceiling of its transformed length divided by h_0 as the number of cells. Cell lengths ℓ_c then obey $\ell_c \leq Ch_0(1 - \rho)$ on that cell, and the number of cells is

$$O(1 + h_0^{-1}(1 + |\log R|)).$$

A zero-length interval requires no operations. On each cell interpolate at $m \geq 2$ distinct equally spaced nodes, including both ends, and integrate the resulting degree- $(m - 1)$ polynomial. Include partial integrals to every state node. By scaling the fixed Lagrange basis on $[0, 1]$, its absolute value sum is at most C_m , and its order- k derivative absolute sum is at most $C_m \ell_c^{-k}$. Consequently every partial-integral row has absolute weight sum at most C_m , uniformly in the number of cells.

Let w_{ij} be these partial-integral weights from the starting time ρ_{in} to node ρ_i , with signed weights for backward integration. Duplicate shared endpoints as labeled nodes if convenient. Given a starting position y_{in} , use simultaneous updates

$$y_i^{[0]} = y_{\text{in}}, \quad y_i^{[k+1]} = y_{\text{in}} - r \sum_j w_{ij} M_{\rho_j}(y_j^{[k]}). \quad (\text{J.3})$$

All node values at layer $k + 1$ use layer k across the full interval. Thus the dependency depth is the number of Picard layers, independent of the number of cells.

Source attribution. Adapted from OpenAI [Ope26, Proposition 6.1].

Proposition J.1 (Probability-flow circuit). *If the input position has marginal law $\xi_{\rho_{\text{in}}}$, then after N_p iterations the maximum, over all state nodes, of the RMS error relative to the exact transport is at most*

$$C_m \text{polylog}(d) l \sqrt{d} h_0^m + (C_m l)^{N_p} C r D_F(x). \quad (\text{J.4})$$

Every computed state is globally Lipschitz in its initial position with constant C_m . After subtracting the initial position, its Lipschitz constant is at most $C_m l$.

Proof. We estimate the quadrature residual on the exact trajectory and then propagate it through Picard iteration. Write Y_ρ for that trajectory and $Q_\rho = M_\rho(Y_\rho)$. The marginal identity in Proposition G.3 and (H.8) give

$$\|Q_\rho^{(m)}\|_2 \leq B_d L \sqrt{d} (1 - \rho^2)^{-m}.$$

On a cell $I = [a_c, b_c]$ of length ℓ_c , let P be the Taylor polynomial of Q of degree $m - 1$ at a_c . Minkowski's inequality applied to the integral remainder gives, at each $v \in I$,

$$\|Q_v - P(v)\|_2 \leq \frac{1}{(m-1)!} \int_{a_c}^v (v-z)^{m-1} \|Q_z^{(m)}\|_2 dz \leq C_m B_d L \sqrt{d} h_0^m. \quad (\text{J.5})$$

The last step uses $\ell_c \leq C h_0(1-z)$ on that cell and $1-z^2 \geq 1-z$. The same estimate holds at the interpolation nodes. Because interpolation reproduces P and its basis has bounded absolute sum, $\|Q_v - \mathcal{I}_c Q(v)\|_2$ obeys the same bound. Integrate this pointwise estimate over the cells, whose total length is at most one. This proves the required quadrature error, for partial integrals and for either orientation. No supremum of the random path has been taken.

Let $e_k = \max_i \|y_i^{[k]} - Y_{\rho_i}\|_2$. The global bound $\text{Lip}(M_\rho) \leq CL$, together with the absolute row sums, gives

$$e_{k+1} \leq C_m l e_k + C_m B_d l \sqrt{d} h_0^m.$$

On each exact transported marginal, conditional expectation and the primitive moment bound give $\|M_\rho\|_2 \leq C D_F(x)$. Integrating the velocity shows $e_0 \leq C r D_F(x)$. Since $C_m l < 1/2$ for large d , summing the geometric recurrence proves (J.4). The norm here is the maximum of the nodewise RMS errors; it is not the RMS of their maximum.

For deterministic inputs the same recurrence, now in maximum node norm, gives a bounded Lipschitz constant for every iterate. In the correction $y_i^{[k+1]} - y_{\text{in}}$, the factor r and the CL -Lipschitz mean give the stronger constant $C_m l$. $\square$

J.2 The kernel action through harmonic trajectories

We need a globally Lipschitz approximation of the vector $K(y)^T G$. An approximation of K alone would leave its error multiplied by an unbounded momentum. We therefore approximate the action directly through harmonic motion and a forward transport correction.

For the kernel integral use the mesh on $[0, b_*]$ above. Its cell lengths also satisfy $\ell_c \geq c' h_0(1-\rho)$ for all sufficiently large d ; this follows from the ceiling construction because the transformed interval length tends to infinity. Set $C_\rho(y) = B_\rho(y) - I$. We first describe a circuit for $C_\rho(y)G$ at one probability node ρ .

Backward position. Apply Proposition J.1 backward from input position y at time b_* to obtain an approximation $\hat{Y}_\rho$ to $Y_\rho(y)$.

Short harmonic trajectories. At fixed probability time ρ , the Hamiltonian equations with potential H_ρ have position integral equation

$$W_z = \cos(z) Y_\rho + \sin(z) G - r \rho \int_0^z \sin(z-v) M_\rho(W_v) dv. \quad (\text{J.6})$$

For every requested angle z , interpolate on the single cell $[0, z]$. Let z_i be its nodes and $\ell_j(v)$ its Lagrange basis, and set

$$w_{ij}^{\sin} = \int_0^{z_i} \sin(z_i - v) \ell_j(v) dv.$$

Use the simultaneous Picard updates

$$\begin{aligned} W_i^{[0]} &= \cos(z_i) \hat{Y}_\rho + \sin(z_i) G, \\ W_i^{[k+1]} &= \cos(z_i) \hat{Y}_\rho + \sin(z_i) G - r \rho \sum_j w_{ij}^{\sin} M_\rho(W_j^{[k]}). \end{aligned} \quad (\text{J.7})$$

At angle zero there is no integral. The free sine and cosine terms are retained exactly.

Differentiate the transport correction. Let d_j be the weights differentiating at zero the polynomial interpolant on angle nodes $0, \psi, \dots, (m-1)\psi$. Their absolute sum is $C_m \psi^{-1}$. At every such angle node, compute $T_\rho(W_z) - W_z$ by the forward probability-flow circuit, keeping only its integral correction at the output. For exact paths,

$$\left. \frac{d}{dz}(T_\rho - \text{id})(W_z) \right|_{z=0} = (B_\rho - I)G = C_\rho G, \quad (\text{J.8})$$

since $W_0 = Y_\rho$ and $W'_0 = G$. The weighted sum with coefficients d_j therefore approximates $C_\rho G$ using exact mean nodes only.

Here and below the interpolation orders denoted by m may be increased independently. Using a common sufficiently large order merely simplifies the error displays. For the next estimates use the same number $N_p \geq 1$ of Picard layers in the backward, harmonic, and forward computations.

Source attribution. Adapted from OpenAI [Ope26, Lemma 6.2].

Lemma J.2 (Errors in the angle computation). *Let $y \sim \xi_*$ and $G \sim \mathbf{N}(0, I_d)$ independently. The exact-input angle differentiation in (J.8) has RMS truncation error at most*

$$C_m l \sqrt{d} \psi^{m-1} R^{-2m} = C_m l \sqrt{d} \delta^{2tm-4t}. \quad (\text{J.9})$$

Before applying the angle differentiation weights, the RMS errors of the computed forward correction values are at most

$$C_m \text{polylog}(d) l \sqrt{d} (h_0^m + \delta^{2tm}) + (C_m l)^{N_p} Cr D_F(x). \quad (\text{J.10})$$

Every such approximate correction is globally Lipschitz in (y, G) with constant at most $C_m l$.

Proof. For exact inputs $(Y_\rho(y), G)$, harmonic motion $\dot{W} = P$, $\dot{P} = -\nabla H_\rho(W)$ preserves $\xi_\rho \otimes \mathbf{N}(0, I_d)$. The vector field is globally Lipschitz, and its weighted divergence vanishes. Its Lie derivative is $\nabla^* A \nabla$ with constant skew coefficient whose (W, P) block is $-I$. Lemma I.4 therefore applies with $\alpha = 1$, with $a_0 = l$ for $T_\rho - \text{id}$, and with $a_0 = L$ for M_ρ . Indeed Lemma I.1 and (G.4) give the required spatial bounds; changing a one-slot split bound to a Hilbert–Schmidt bound costs only $\sqrt{d}$. To apply the angle stencil, subtract the Taylor polynomial of $z \mapsto (T_\rho - \text{id})(W_z)$ of degree $m-1$ at zero. Its derivative at zero is exact, and its remainder at each stencil node has RMS norm at most $C_m l \sqrt{d} \psi^m R^{-2m}$. Multiplication by the absolute stencil sum C_m/ψ proves (J.9). This uses pointwise RMS remainders on the stationary harmonic path.

For the harmonic state equation, interpolate $M_\rho(W_v)$ in (J.6). Since $z \leq (m-1)\psi$, its Taylor remainder, multiplied by the prefactor r , is bounded by $C_m l \sqrt{d} \psi^m R^{-2m} = C_m l \sqrt{d} \delta^{2tm}$; the additional integration factors are bounded and can be discarded. The initial free-path error is at most $Cr D_F(x)$, and Picard contracts by $C_m l$. The computed harmonic state is globally Lipschitz with bounded constant in its two inputs, by the same row-sum argument as before.

Next compare the forward probability correction started at the exact harmonic state. That state has marginal law ξ_ρ , so Proposition J.1 applies. Errors in its starting state then propagate through the approximate correction with factor $C_m l$, not a factor of order one. This also controls the backward-position error transmitted through the harmonic computation. Combining these estimates proves (J.10) and the asserted global Lipschitz bound. $\square$

J.3 The zero endpoint of the kernel integral

It remains to integrate the derivative of $C_\rho G$ against $1/\rho$. The exact identity $B'_0 = 0$ removes the apparent singularity, and we impose that identity on the interpolant as well.

On all but the first probability cell use ordinary degree- $(m-1)$ interpolation. On the first cell use $m-1$ equally spaced value nodes including both endpoints and the additional Hermite condition that the derivative at zero is zero, for a total of m conditions, with $m \geq 3$. This uniquely determines a polynomial of degree at most $m-1$: a polynomial in the null space would have a double zero at zero and the remaining $m-2$ distinct zeros. Let q_i be the resulting weights on values when the derivative of this piecewise interpolant is integrated against $1/\rho$.

Source attribution. Adapted from OpenAI [Ope26, Lemma 6.3].

Lemma J.3 (Derivative quadrature at zero). *Let $y \sim \xi_*$ and $G \sim \mathcal{N}(0, I_d)$ independently. For exact values of $C_\rho(y)G$, the RMS error of this quadrature for $\int_0^{b_*} B'_\rho(y)G d\rho/\rho$ is at most*

$$C_m \text{polylog}(d) l \sqrt{d} h_0^{m-2}.$$

The value weights satisfy

$$\sum_i |q_i| \leq C_m h_0^{-1} (1 + |\log R| + |\log h_0|). \quad (\text{J.11})$$

These weights are well defined even if the supplied values are approximate: the derivative condition at zero remains exact.

Proof. By Lemma I.1 and independence of G , the RMS norm of the order- m derivative of $C_\rho(y)G$ is at most $\text{polylog}(d) l \sqrt{d} (1 - \rho^2)^{-m}$. Here multiplication by the independent Gaussian converts the matrix norm to a Hilbert–Schmidt norm and costs at most $\sqrt{d}$.

On an ordinary cell, Taylor expansion and scaling the interpolation basis bound the derivative error by this derivative bound times $C_m \ell_c^{m-1}$. It is therefore at most $C_m \text{polylog}(d) l \sqrt{d} h_0^{m-1} / (1 - \rho)$. After integration against $1/\rho$, all ordinary cells contribute at most this prefactor times $C(1 + |\log R| + |\log h_0|)$, which fits the claimed more conservative h_0^{m-2} bound.

For the first cell, subtract the Taylor polynomial at zero of degree $m-1$. Its derivative at zero agrees with the exact zero derivative. The second derivative of the remainder, and that of the Hermite interpolant of the remainder, are bounded in RMS by $C_m \text{polylog}(d) l \sqrt{d} \ell_c^{m-2}$. Both first derivatives vanish at zero. If E is the difference between the exact function and its Hermite interpolant on this cell, then

$$\|E'(\rho)\|_2 \leq \int_0^\rho \|E''(v)\|_2 dv \leq C_m B_d l \sqrt{d} \rho \ell_c^{m-2}.$$

Consequently the integral of $E'(\rho)/\rho$ over the first cell has RMS norm at most $C_m B_d l \sqrt{d} \ell_c^{m-1}$. This is within the stated, more conservative h_0^{m-2} bound.

On an ordinary cell the absolute coefficient sum of the interpolant derivative is at most C_m/ℓ_c . Integrating and using $\ell_c \geq c'h_0(1 - \rho)$ gives a total bounded by

$$C_m h_0^{-1} \int_{\rho_{\text{first}}}^{b_*} \frac{d\rho}{\rho(1 - \rho)}.$$

This gives the logarithms in (J.11). On the first cell, scale to the unit interval. The polynomial derivative vanishes at zero and is bounded by distance to zero times the second derivative, so the integral of its absolute value divided by that distance is bounded. Scaling back gives a weight sum $C_m/\ell_c \leq C_m h_0^{-1}$. This proves the claim. $\square$

J.4 A globally Lipschitz approximate centering velocity

Let $\text{corr}_{ij}(y, G)$ denote the forward correction computed at probability node i and angle node j by the three operations above. The approximate position velocity is the explicit expression

$$\hat{v}_Y(y, G) = -\frac{1}{rs} \sum_i q_i \sum_j d_j \text{corr}_{ij}(y, G). \quad (\text{J.12})$$

Each correction retains only the integral part of the forward Picard formula. Writing its weights as w_{ijk} and its mean values as M_{ijk} gives

$$\text{corr}_{ij} = -r \sum_k w_{ijk} M_{ijk},$$

and hence

$$\hat{v}_Y = \frac{1}{s} \sum_{i,j,k} q_i d_j w_{ijk} M_{ijk}. \quad (\text{J.13})$$

The apparent factor $1/r$ has canceled algebraically. There is no additive forward state left in this formula, and all coefficients multiplying mean values are deterministic scalars. Use the exact mean expression $\hat{v}_G(y, G) = (M_*(y) - M_0)/s$ for the momentum velocity. In particular, the momentum occurs inside angle states; it is never multiplied by a separately approximated matrix at the output. *Source attribution.* Adapted from OpenAI [Ope26, Proposition 6.4].

Proposition J.4 (Kernel-action circuit). *On the stationary product input law, the RMS error of $\hat{v} = (\hat{v}_Y, \hat{v}_G)$ relative to (I.3) is bounded by $\text{polylog}(d)$ times*

$$\begin{aligned} & \frac{L}{\sigma} \sqrt{d} [h_0^{m-2} + h_0^{-1} \delta^{-4t} (h_0^m + \delta^{2tm})] \\ & + \delta^{-4t-w} (C_m l)^{N_p-1} r \frac{L}{\sigma} D_F(x). \end{aligned} \quad (\text{J.14})$$

It is globally Lipschitz, with constant at most

$$C_m \text{polylog}(d) (L/\sigma) \psi^{-1} h_0^{-1}. \quad (\text{J.15})$$

For every fixed $J \geq 1$, its RMS error can be made at most $\text{polylog}(d) \delta^J D_F(x)$ with $N_p = O(J+1)$, independently of w and the interpolation orders.

Proof. Combine Lemmas J.2 and J.3, and divide by rs . The value errors are amplified only by $C_m \psi^{-1}$ and the weight sum (J.11). The exact stencil truncation has factor $\psi^{m-1} R^{-2m} = \delta^{2tm-4t}$ and the same probability-quadrature weight sum. Crucially,

$$\frac{l}{r\sigma} = \frac{L}{\sigma}, \quad \frac{l^{N_p} r D_F(x)}{r\sigma} = l^{N_p-1} r \frac{L}{\sigma} D_F(x).$$

Thus neither division introduces a negative power of the potentially small scale r . The resulting estimate is (J.14).

Each forward correction has Lipschitz constant $C_m l$ in its inputs, and the backward and harmonic maps have bounded constants. Applying the same absolute weight sums and using $l/(r\sigma) = L/\sigma$ gives (J.15); the momentum component has the smaller Lipschitz bound CL/σ .

For the last assertion, increase interpolation orders until the first line of (J.14) is of the desired order. Under (J.2), sufficiently many $O(J+1)$ Picard layers make the second line of that order as well. The explicit common choice given below verifies independence from interpolation orders and from w . $\square$

J.5 The mean formula and the order of parameter choices

We can now discretize the centering flow itself. Interpolate on the single time cell $[0, 1]$, using ordinary degree- $(m - 1)$ interpolation and partial-integral rows w_{ij} . With anchor $Z_0 = (Y_0, G_0)$, set

$$Z_i^{[0]} = Z_0, \quad Z_i^{[k+1]} = Z_0 + \sum_j w_{ij} \widehat{v}(Z_j^{[k]}).$$

At the end keep the position components y_j and output

$$sG_0 + \sum_j \omega_j M_*(y_j), \quad \sum_j |\omega_j| \leq C_m, \quad (\text{J.16})$$

where ω_j are the full-interval integration weights.

Source attribution. Adapted from OpenAI [Ope26, Proposition 6.5].

Proposition J.5 (Mean-flow circuit). *For ideal independent anchors $Y_0 \sim \xi_*$ and $G_0 \sim \mathcal{N}(0, I_d)$, and every fixed $J \geq 1$, the circuit (J.16) can be chosen to differ in RMS by at most $\text{polylog}(d)\sigma\delta^J D_F(x)$ from the exact path integral (I.4). The approximate joint states have RMS errors at most $\text{polylog}(d)\delta^J D_F(x)$. Every integration uses $O(J + 1)$ Picard layers, with a bound independent of w and all interpolation orders.*

Proof. First compare the exact flow to polynomial integration of its exact velocity. Lemma I.4, applied to the state with one additional derivative, bounds this quadrature error by

$$C_m \sqrt{d} \left((L/s) R^{-2} \right)^{m+1}.$$

The approximation error of $\widehat{v}$ at every exact trajectory node is bounded by Proposition J.4, by stationarity. Previous state errors propagate with its global Lipschitz constant and the absolute row sums. The true centering velocity need not be globally Lipschitz in the joint variables; it is the approximate velocity's bound that is used in this recurrence.

By (J.1), $b \geq 1/2$ and $w < t$ give $b - 4t - w > 1/3$. For sufficiently large d , fixed constants and polylogarithms in (J.15) can therefore be absorbed so that the recurrence contracts by at most $\delta^{1/3}$. Its initial error is at most $C(L/s)\sqrt{d}$ by (I.8). Choose the quadrature and velocity error orders sufficiently high, and then $O(J + 1)$ Picard layers yield the state error claimed in the statement.

For the output, apply Lemma I.4 to M_* with $a_0 = L$. Its path quadrature error is at most $C_m L \sqrt{d} ((L/s) R^{-2})^m$, which can be made at most $\text{polylog}(d)\sigma\delta^J D_F(x)$. The additional error from evaluating on the approximate positions is bounded by their RMS error times $C_m L$, using the CL -Lipschitz bound for M_* . Since L/σ is small, this is within the same budget. $\square$

We collect the numerical accuracy and dependency-depth estimates before expanding the affine center formulas. This fixes the Picard depth without yet choosing the mesh exponent.

Source attribution. Adapted from OpenAI [Ope26, Corollary 6.6].

Corollary J.6 (Accuracy without increasing dependency depth). *Fix $J \geq 1$. In all the probability, harmonic, and centering integrations above, one may take at most*

$$N = \lceil 10(J + 2) \rceil$$

Picard layers. After any fixed $w \in (0, t)$ is chosen, a common fixed interpolation order satisfying

$$w(m - 3) > J + 10, \quad t(m - 3) > J + 10, \quad m > 4(J + 10) \quad (\text{J.17})$$

suffices for the following conclusions, for all sufficiently large d . Probability transport from its exact starting marginal has endpoint RMS error at most $\text{polylog}(d)\delta^J D_F(x)$; the same holds after division by an endpoint parameter bounded away from zero. The mean circuit with ideal anchors has RMS error at most $\text{polylog}(d)\sigma\delta^J D_F(x)$ relative to (I.4).

The number of layers of exact mean dependencies within either circuit is $O((N + 1)^2)$, independently of m , w , and the number of mesh cells. All mesh nodes, interpolation weights, scalar parameters, and iteration counts are deterministic functions of the prescribed parameters and dimension.

Proof. For large d , probability and harmonic Picard contractions $C_m l \leq \text{polylog}(d)\delta^b$ can be bounded by $\delta^{1/3}$. The only additional inverse powers in kernel evaluation are δ^{-4t-w} , besides the divisions already canceled in Proposition J.4. The centering iteration also contracts by $\delta^{1/3}$. The probability iteration remainder is at most $C\delta^{N/3}D_F(x)$; after the kernel weights the iteration remainder is at most $C\delta^{N/3-4t-w}D_F(x)$. Both exponents exceed J with a fixed positive margin. For the truncation terms it suffices to check the exponents

$$wm, \quad w(m - 2), \quad w(m - 1) - 4t, \quad 2tm - 4t - w, \quad (m + 1)/3.$$

The first four come respectively from probability quadrature, the zero-endpoint rule, its amplified probability-value error, and the harmonic and angle errors. For the last one use $(L/s)R^{-2} \leq \delta^{1/3}$ for large d in both outer quadratures, dividing the mean-output error by σ and using $L/\sigma \leq \delta^{1/3}$. Every displayed exponent exceeds J by (J.17). The outer Picard remainder contracts by the same $\delta^{1/3}$ factor. Thus fixed constants and polylogarithms, including those depending on m , fit within the strict exponent margins as $d \rightarrow \infty$.

Within a probability or harmonic Picard layer, all nodes depend on only the preceding layer, regardless of the number of nodes. A path through one kernel evaluation passes through a backward probability computation, a harmonic computation, and a forward probability computation, each having at most N layers. Such a path traverses at most N layers of the outer centering iteration. This gives depth $O((N + 1)^2)$; final quadratures add a bounded number of layers. All weights integrate known polynomials, or polynomials against sine or $1/\rho$ with the stated zero-endpoint condition. They require no oracle information and are allowable real constants in the model. $\square$

We choose parameters in the order

$$\text{accuracy order} \longrightarrow \text{Picard depth} \longrightarrow w \longrightarrow m.$$

The interpolation order affects the number of mean evaluations, while the depth bound remains fixed. The recursive implementation will use this separation when it chooses w .

J.6 Bounds for the finite formulas

To display a primitive's parameters, for $x' \in \mathbb{R}^d$ and $0 < r' \leq 1$ write

$$u_{x',r'} := \int_{\mathbb{R}^d} g(x' + r'z) \nu_{x',r'}(dz).$$

Thus $u_{x,r} = u$ for the primitive fixed above.

We specify the two formulas to be implemented. In this subsection $\mathcal{C}_M$ and $\mathcal{C}_S$ denote deterministic maps of their listed anchors, with exact primitive means available as analytical nodes. The parameter σ of $\mathcal{C}_M$ is the parent noise parameter, so its construction sets $s = \sigma/2$ exactly once. Define

$$\mathcal{C}_M(x, r, \sigma; y, G) = \frac{\sigma}{2}G + \sum_j \omega_j M_*(y_j), \quad (\text{J.18})$$

$$\mathcal{C}_S(x, r, \tau; y) = \rho_e^{-1} \widehat{T}_{0 \rightarrow \rho_e}(y), \quad \rho_e = (1 + \tau^2/4)^{-1/2}, \quad R \leq \tau \leq 1. \quad (\text{J.19})$$

The positions y_j in the first line are the last centering Picard iterates with anchors (y, G) , using (J.12). The second line uses the probability formula (J.3). No additional Gaussian is included in either map. The next section allocates additional noise and constructs the position anchor needed for $\mathcal{C}_M$.

A *mean dependency path* follows a mean node to a mean value used in its center expression, and then repeats. Direct expression size counts occurrences in one such expression, before expanding the center expressions of its mean nodes.

Source attribution. Adapted from OpenAI [Ope26, Theorem 6.7].

Theorem J.7 (Finite mean and sampling formulas). *Fix $J \geq 1$, the parameters (J.1), and $N = \lceil 10(J+2) \rceil$. Choose the fixed interpolation order by (J.17). For all sufficiently large d , the maps (J.18)–(J.19) have the following properties.*

(i) Deterministic scalar formulas. *Their exact mean nodes can be ordered as $M_1, \dots, M_Q$, where*

$$M_i = u_{E_i, r_i}, \quad E_i = b_i + \sum_{j < i} a_{ij} M_j, \quad E_f = b_f + \sum_j a_{fj} M_j. \quad (\text{J.20})$$

All a_{ij} and a_{fj} are deterministic scalars. The whole node list, its scales, and these coefficients can be formed without knowing a mean value or making an oracle query.

(ii) Direct terms and scales. *Each node has $r_i = r\tau_i$ and*

$$\tau_i = \sqrt{1 - \rho_i^2} \geq R/\sqrt{5}, \quad b_i = x + r(\ell_i y + m_i G).$$

For $\mathcal{C}_S$, the G term is absent, $\rho_i \leq \rho_e$, and $(\ell_i, m_i) = (\rho_i, 0)$. For $\mathcal{C}_M$, $\rho_i \leq b_$ and the pair is either $(\rho_i, 0)$ or $(\rho_i \cos z, \rho_i \sin z)$, with $|z| \leq C_m \psi$. In particular both coefficients have absolute value at most one. A node for $M_0 = u$ has center x and scale r . Every mean occurring directly in the final expression for $\mathcal{C}_M$ is M_* at a computed position and has scale exactly rR . The final direct terms are $b_f = \sigma G/2$ and $b_f = y/\rho_e$, respectively.*

(iii) Coefficient sums and direct expression size. *There are positive declared bounds A, A_f such that $\sum_{j < i} |a_{ij}| \leq A$ and $\sum_j |a_{fj}| \leq A_f$, namely*

$$\begin{aligned} A_M &= C_1(1 + \log d)^{C_1} [(r/\sigma)\delta^{-4t-c_1 w} + r^2], & A_S &= C_1 r^2, \\ A_{f,M} &= C_1, & A_{f,S} &= C_1 r, \quad c_1 = 10. \end{aligned} \quad (\text{J.21})$$

Here $C_1 \geq 1$ depends only on the fixed parameters and orders. Every direct expression has at most $B_d \delta^{-c_1 w}$ terms. Every mean dependency path has length at most $D_0 = 3N^2 + 1$; using a larger absolute multiple of $(N + 1)^2$ is harmless. This depth bound is independent of w , m , and the number of mesh cells.

(iv) Errors on the ideal anchor laws. Under (J.2), for independent $Y \sim \xi_*$ and $G \sim \mathbf{N}(0, I_d)$,

$$W_2(\text{Law}(\mathcal{C}_M(x, r, \sigma; Y, G)), \mathbf{N}(u, \sigma^2 I_d/4)) \leq B_d \sigma \delta^J D_F(x). \quad (\text{J.22})$$

For $\mathcal{C}_S$ only $l \leq B_d \delta^b$, $b \geq 1/2$, is needed. With $Y \sim \mathbf{N}(0, I_d)$,

$$W_2(\text{Law}(\mathcal{C}_S(x, r, \tau; Y)), \nu_{x,r} * \mathbf{N}(0, \tau^2 I_d/4)) \leq B_d \delta^J D_F(x). \quad (\text{J.23})$$

The coefficient and dependency assertions are deterministic, for every choice of anchors. The distributional estimates require the specified ideal anchor laws.

Proof. We verify the affine structure, coefficient bounds, depth, and ideal-law errors in turn. In each additive expansion we stop at an earlier mean value. Its own center remains a separate node, which is necessary for the local coefficient bounds.

Step 1: direct terms in the transport formulas. A probability Picard state has its input position as direct term and mean coefficients with absolute sum at most $C_m r$. A harmonic state has direct term $\cos z$ times its input position plus $\sin z$ times its input momentum, with mean coefficients of absolute sum at most $C_m r$. The sine-weighted row sums are bounded because their integration intervals have bounded length. These assertions hold at every iteration: its additive direct term is the original input, not the preceding Picard iterate.

For a kernel evaluation, the scalar expansion (J.13) has absolute coefficient sum at most $B_d \sigma^{-1} \delta^{-4t-w}$, by the derivative-quadrature, angle-stencil, and forward-integral weight bounds. The momentum formula $(M_* - M_0)/s$ has coefficient sum at most C/σ . A centering Picard update has the original anchor as direct term and bounded outer integral rows. Its position and momentum corrections thus have the same respective coefficient bounds.

Step 2: center coefficients and scales. Consider a mean evaluation inside any of these formulas. Its physical center is $x + r\rho v$ when its normalized position is v , by (G.1). Substitute its direct starting states: a backward probability state starts at the current centering position; a harmonic state starts at that backward position and the current momentum; and a forward probability state starts at the harmonic position. Stop each substitution at the mean terms just described. Only the free terms pass through this chain. Thus their anchor coefficients are exactly $(\rho, 0)$ or $(\rho \cos z, \rho \sin z)$. For sampling there is only the first probability computation, so the coefficient is $(\rho, 0)$. Multiplication by $r\rho$, with $\rho \leq 1$, bounds the center coefficient sum in the mean case by

$$B_d(r/\sigma)\delta^{-4t-w} + C_m r^2.$$

The term $C_m r^2$ accounts for the backward, harmonic, and forward integral corrections. In the sampling case the bound is simply $C_m r^2$. Enlarging these positive upper bounds to (J.21) is convenient for the implementation. The mean output is the single quadrature in (J.18), so its coefficient sum is at most C_m and its means all have probability time b_* . For sampling, dividing a probability correction by $\rho_e \geq 2/\sqrt{5}$ gives output coefficient sum at most $C_m r$. These observations also give the stated final direct terms.

The scale at probability time ρ is $r\sqrt{1-\rho^2}$. In the mean computation $\rho \leq b_*$, giving $\tau \geq R$, including equality for the output means. In the sampling computation,

$$\sqrt{1-\rho_e^2} = \frac{\tau}{\sqrt{4+\tau^2}} \geq \frac{R}{\sqrt{5}}.$$

This proves every assertion about the anchors and scales.

Step 3: expression size and dependency depth. Let $P \leq B_d \delta^{-w}$ bound the number of nodes in any one probability mesh. Angle stencils, harmonic quadratures, and outer centering quadratures have fixed numbers of nodes once m is fixed. Formula (J.13) has at most $C_m P^2$ mean terms: one factor P comes from probability-time derivative quadrature, and the other from the forward transport integral. A direct centering position has the same bound, and its momentum has only a fixed number of terms. Substituting their free states into a backward, harmonic, or forward center adds at most one backward list, one harmonic list, and one forward list. No substitution continues through a mean value. Hence $B_d \delta^{-2w}$ already bounds the direct expression size; the exponent $c_1 w = 10w$ is a uniform conservative choice.

All Picard updates at a fixed layer depend only on earlier layers. Order an internal kernel computation as backward transport, harmonic motion, and forward transport, after its input states. A mean path within this computation adds at most $3N$ nodes above a path in those inputs. There are at most N outer Picard layers, and one final mean layer in (J.18). Thus $3N^2 + 1$ bounds the depth. The sampling path has depth at most N . This gives an acyclic order for (J.20). Meshes, weights, angles and partial integral rows depend only on the prescribed scalar parameters; no exact mean is needed to form any of them.

Step 4: error under the ideal anchors. Couple the numerical mean formula and the exact centering path integral by the same ideal anchors. Corollary J.6 bounds their RMS difference by $B_d \sigma \delta^J D_F(x)$, while Proposition I.3 identifies the exact output as $N(u, \sigma^2 I_d/4)$. This coupling proves (J.22). For sampling, the exact probability transport from a standard Gaussian has law ξ_{ρ_e} . After division by ρ_e this is the law of

$$Z + \frac{\sqrt{1-\rho_e^2}}{\rho_e} G = Z + \frac{\tau}{2} G, \quad Z \sim \nu_{x,r}, \quad G \sim N(0, I_d) \text{ independent.}$$

The same-anchor RMS estimate and $\rho_e \geq 2/\sqrt{5}$ prove (J.23). □

For the implementation we use $J = K_0 + 2$. The theorem fixes N and D_0 before w ; the implementation can therefore choose w using its recursion count, and then choose m by (J.17). The formulas above introduce no random failure event. Their errors are RMS estimates under the ideal input laws, and their algebraic structure holds for every input.

K Recursive evaluation using original gradients

For the sampler in this manuscript, the executed numerical formulas are the simultaneous-iteration formulas of Section C. The present appendix records the inherited reference construction; its use in the sampler is through that explicitly stated interface substitution, including the recursive-evaluation inputs and query accounting.

We replace every conditional mean in the finite formulas by original-gradient queries. The center of a child mean may itself be an approximate computation, so its private Gaussian noise is absorbed

into the child's seeds. We first construct the deterministic shift identity and query cap, and only then prove the distributional comparison. The complete construction and proofs in this section are adapted from [Ope26].

K.1 Why Gaussian center noise helps

A seeded mean routine at scale $r > 0$ is a measurable deterministic map $Q(x; z_1, \dots, z_m) \in \mathbb{R}^d$, whose input is a center $x \in \mathbb{R}^d$ and m seed slots $z_j \in \mathbb{R}^d$. It may query a fixed oracle. A vector $v = (v_1, \dots, v_m) \in \mathbb{R}^m$ gives a *query-preserving translation* if, for every center, displacement and fixed seed list,

$$Q(x + \Delta; z + v \otimes \Delta/r) = Q(x; z), \quad (\text{K.1})$$

and the two computations ask exactly the same physical oracle queries. Here $(v \otimes a)_j = v_j a$. This is a deterministic identity; it does not require independent or Gaussian seeds.

Source attribution. Adapted from OpenAI [Ope26, Lemma 7.1].

Lemma K.1 (Absorbing a Gaussian center displacement). *Let $D_s > 0$, and let Q satisfy (K.1), with $\|v\|_2 \leq D_s$. Put*

$$n = \frac{r}{2D_s}, \quad P = \left(I_m - \frac{vv^T}{4D_s^2} \right)^{1/2}.$$

For a deterministic $x \in \mathbb{R}^d$, let $Z, G_1, \dots, G_m$ be independent standard Gaussian vectors in $\mathbb{R}^d$. Then the output of $Q(x + nZ; (P \otimes I_d)G)$ has exactly the law of $Q(x; G)$.

Proof. The eigenvalues of P lie in $[\sqrt{3}/2, 1]$. The translation identity rewrites the first output as

$$Q\left(x; PG - \frac{v \otimes Z}{2D_s}\right).$$

We suppress the tensor product with I_d on slot-space matrices. The seed array in this expression is centered Gaussian and has slot covariance

$$PP^T + \frac{vv^T}{4D_s^2} = I_m.$$

Its slots are therefore independent standard Gaussian vectors. They may remain correlated with Z ; that is harmless because the rewritten call is at the deterministic center x . $\square$

The matrix P removes exactly the seed covariance supplied by the center noise. Without this adjustment, translating the center back would increase the covariance. We next construct the translation identity for every routine, so that the same calculation can be used recursively.

K.2 The numerical input and the implementation task

We retain the primitive notation of Section F: $g = \nabla F$, $\text{Lip}(g) \leq \lambda = \delta = d^{-e}$, $0 < r \leq 1$, and $\nu_{x,r}$ is the law in (F.1). Write $u_{x,r} = \mathbb{E}_{\nu_{x,r}} g(x + rZ)$, $L = \lambda r$, $l = \lambda r^2$, and $D_F(x) = \sqrt{d} + |g(x)|$. We use the uniform moment bounds (F.6) and conditional coupling rules of Lemma F.3. Only evaluations of g will be used by the implementation.

Fix K_0, t as in Theorem F.1, and retain the numerical parameters

$$R = \delta^t, \quad b_* = (1 - R^2)^{1/2}, \quad \psi = \delta^{4t}, \quad J = K_0 + 2.$$

Theorem J.7 supplies two finite acyclic circuits, $\mathcal{C}_M$ and $\mathcal{C}_S$, whose nodes are exact conditional means. We use its notation A for the positive declared bound on a center's absolute coefficient sum, A_f for the output bound, D_0 for the mean-dependency depth, and c_1 for the absolute constant in the fan-out exponent. A center has the form $x + r(\ell_i y + m_i G) + \sum_{j < i} a_{ij} u_{x_j, r_j}$, with $r_i = r\tau_i$. All these scalar data are deterministic functions of the prescribed numerical parameters, independent of centers, anchors, and oracle responses. We will use the circuit theorem's explicit coefficient and scale bounds when constructing shifts and propagating sizes; the circuits themselves are comparison formulas, not additional runtime oracles.

For the mean circuit the parameter supplied is the parent's σ . Its underlying centering flow uses $s = \sigma/2$ once, so its direct output is $(\sigma/2)G$. Under the specified ideal anchor law, its error is

$$W_2(\text{Law}(\mathcal{C}_M), \mathbf{N}(u_{x,r}, \sigma^2 I_d/4)) \leq \text{polylog}(d) \sigma \delta^J D_F(x).$$

For the sampling circuit the direct output is y/ρ_e , where $\rho_e = (1 + \tau^2/4)^{-1/2}$, and the ideal target is $\nu_{x,r} * \mathbf{N}(0, \tau^2 I_d/4)$. These numerical statements require the anchor laws and size assumptions specified in Theorem J.7; we shall verify those assumptions at their uses.

We must implement every exact mean, including those appearing in the center expressions. The proof will use fixed-seed stability to compare an inaccurate center with an ideal noisy center. Gaussian absorption then identifies the ideal noisy-center call with a standalone call. Only after these two steps do we use the child's law guarantee.

K.3 Two routines and two different kinds of depth

We construct a mean routine $\mathbf{M}(x, r, \sigma; b, p)$ targeting $u_{x,r} + \sigma G$ and a sampling routine $\mathbf{S}(x, r, \tau; b, p)$ targeting $Z + \tau G$, for independent $Z \sim \nu_{x,r}$ and standard Gaussian G . The top call is $\mathbf{S}(x, r, 1; 1, K_0)$. Throughout, a reachable call means a descendant of this call, including the call itself. The first label b records the size conditions

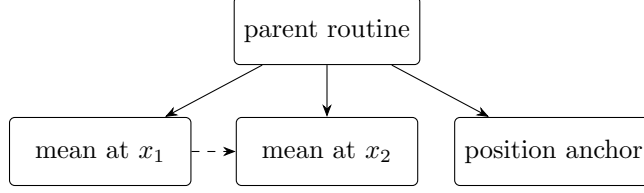
$$\max\{l, L/\sigma\} \leq \text{polylog}(d) \delta^b \quad (\mathbf{M}), \quad l/\tau \leq \text{polylog}(d) \delta^b \quad (\mathbf{S}), \quad (\text{K.2})$$

where $R \leq \tau \leq 1$. The second label p records the requested error power. The intended errors are $\text{polylog}(d) \sigma \delta^p D_F(x)$ and $\text{polylog}(d) \delta^p D_F(x)$, respectively. A call is terminal precisely when $p \leq b$; it is nonterminal when $p > b$. Every sampling routine reserves a final noise $\tau G/2$, unused in its own queries. Before adding that reserve, its target law is $\nu_{x,r} * \mathbf{N}(0, 3\tau^2 I_d/4)$.

For a nonterminal sampling call, use $\mathcal{C}_S$ with a standard Gaussian position anchor, supply additional final noise $n = \tau/\sqrt{2}$, and then add the reserve. For a nonterminal mean call, pass the parent parameter σ to $\mathcal{C}_M$ and supply final noise $n = \sqrt{3}\sigma/2$. Obtain its position anchor from an $\mathbf{S}$ call at the same center and scale with $\tau = R/b_*$, multiplying that output by b_* . An ideal such anchor has exactly the required law $b_* Z + RN$. Its momentum anchor is an independent standard Gaussian. For all sufficiently large d , $R/b_* \in [R, 1]$, so the anchor call has an admissible noise parameter.

An expression dependency and a routine call play different roles. For example, suppose a parent circuit contains $E_f = a u_{x_2, r_2}$ and $x_2 = x + c u_{x_1, r_1}$. Evaluating E_f invokes two new mean routines: one to help supply x_2 , and one at that supplied center. Both are children of the *same parent routine*. Their own internal circuits begin the next routine level. The two nested affine expressions do not create two successive losses in the parent's accuracy label. Figure 2 makes this distinction explicit.

To evaluate an expression, we expand its mean dependencies as an uncached tree, duplicating every mean occurrence together with its entire center-supplying subtree. All mean calls at any expression tier are direct children of the routine whose circuit is being evaluated.



Solid arrows: routine calls. Dashed arrow: mean dependency.

Figure 2: The two depths used in the recursive implementation. A mean needed to supply another mean’s center is still a direct child of the same parent routine. Its expression dependency does not create an extra routine level. Each occurrence has its own fresh seed blocks.

The children of a parent labeled (b, p) receive labels as follows:

child’s role	child labels	
term of E_f in an M parent	$(b + t, p)$	(K.3)
term of E_f in an S parent	(b, p)	
any other term, or the M position anchor	$(b - 10t, p - 1/4)$.	

In the last row the labels are always those of the parent routine, not successively decreased while descending its expression tree.

Source attribution. Adapted from OpenAI [Ope26, Lemma 7.2].

Lemma K.2 (Bounded routine height). *Starting at $(b, p) = (1, K_0)$, the rules (K.3) give a finite routine tree of height at most a constant $H = H(K_0, t)$. Every reachable label satisfies $b > 1/2$.*

Proof. Before $4K_0 + 1$ decreasing transitions, the total loss in b is less than $10t(4K_0 + 1) < 1/4$, so $b > 3/4$. At that many transitions, $p \leq K_0 - (4K_0 + 1)/4 < 0$, forcing termination. Therefore no branch can have more such transitions. Positive transitions increase b by t , while a nonterminal satisfies $b < p \leq K_0$; their total number is consequently bounded in terms of K_0, t and the already bounded total decrease in b . A neutral transition is an S-to-M transition. Another neutral transition requires first reaching another S, which can occur only through a decreasing transition. This bounds the neutral transitions as well and proves both claims. $\square$

A routine is a deterministic function of its center, scalar parameters, oracle, and a finite ordered list of d -dimensional seed slots. Standalone use fills these slots by independent standard Gaussians. In addition, each routine will have a deterministic vector v of scalar coefficients, one per slot, with

$$\|v\|_2 \leq D_s, \quad D_s := \delta^{-t}. \quad (\text{K.4})$$

For *every* fixed seed list, changing x to $x + \Delta$ and adding $v \otimes \Delta/r$ to the seeds will preserve every physical query point. The output of M will be unchanged; the normalized output of S will decrease by Δ/r . These are pathwise properties, not statements restricted to Gaussian seeds.

Within each call write $u = u_{x,r}$ and $\nu = \nu_{x,r}$. At a terminal call define

$$\begin{aligned} \mathbf{M} &= g(x + rZ) + \sigma G, \\ \mathbf{S} &= Z - rg(x + rZ) + (\sqrt{3}\tau/2)G_1 + (\tau/2)G_2, \end{aligned}$$

with separate seed slots. The last term in S is its reserved noise. Taking $v = -1$ on the Z slot and zero elsewhere proves the shift property and $\|v\|_2 = 1$.

K.4 Evaluating an expression at a noisy center

Fix a parent circuit and its anchors, and write $M_i = u_{x_i, r_i}$ for its exact mean nodes. For each expression $E = b_0 + \sum_i a_i M_i$, the operation $\text{Eval}(E, n)$ will approximate its exact value plus an independent $N(0, n^2 I_d)$ noise. Assign disjoint seed blocks to every term occurrence, its center-supplying computation, and each residual noise; keep all these blocks separate from the anchors. Thus the term blocks are independent conditional on the anchors in a standalone call.

Here is the recursive evaluation rule, assuming the child routines and their shift vectors have already been constructed. Take $A' = A$ for a center expression and $A' = A_f$ at the output, and put $\sigma_i = n/(2A')$. For the term of scale $r_i = r\tau_i$, first evaluate its center expression with noise

$$n_i = \frac{r_i}{2D_s}. \quad (\text{K.5})$$

At this supplied center run the child $\mathbf{M}$ with scale r_i , noise σ_i , and labels from (K.3). If its shift vector is v_i , apply to its fresh seed block G_i the deterministic slot-space transformation

$$P_i = \left(I - \frac{v_i v_i^T}{4D_s^2} \right)^{1/2}, \quad \text{and use seeds } (P_i \otimes I_d)G_i. \quad (\text{K.6})$$

Denote its output by Q_i . Return

$$b_0 + \sum_i a_i Q_i + \left(n^2 - \sum_i a_i^2 \sigma_i^2 \right)^{1/2} G_E, \quad (\text{K.7})$$

where G_E is a separate slot. The square root is defined, since

$$\sum_i a_i^2 \sigma_i^2 \leq \left(\sum_i |a_i| \right)^2 \frac{n^2}{4(A')^2} \leq n^2/4.$$

If there are no terms, return $b_0 + nG_E$. The mean-node ordering makes the expression recursion finite. At the parent's output use the value of n specified above; add the reserved noise for $\mathbf{S}$. A routine supplied with correlated seeds performs these same deterministic operations, without any change of rule.

At an ideal supplied center $x_i + n_i Z$, Lemma K.1 now identifies this child with its standalone law at x_i . We still owe the construction of the shift vectors and a proof that the noisy supplied centers are accurate. We prove these in that order.

K.5 Query-preserving shifts and the deterministic cap

We construct the shifts required by Lemma K.1 and bound their norms. The same bottom-up construction proves that all routines are well defined and have a deterministic query cap.

For a nonterminal $\mathbf{S}$, shift its Gaussian position anchor by $-\rho_e \Delta/r$. For a nonterminal $\mathbf{M}$, we require its position anchor y_* to shift by $-\Delta/(b_* r)$ and leave its momentum anchor unchanged. To obtain this, apply the shift of its anchor $\mathbf{S}$ call and additionally shift that call's reserved seed by the coefficient $-2R/b_*$ times Δ/r . The ordinary shift changes its output by $-\Delta/r$. Since the reserved standard deviation is $R/(2b_*)$, the additional change is $-R^2 \Delta/(b_*^2 r)$. Multiplication by b_* makes the total change

$$-\left(b_* + \frac{R^2}{b_*} \right) \frac{\Delta}{r} = -\frac{\Delta}{b_* r}.$$

The extra slot is used in no query, so query invariance is preserved.

For a term of scale $r\tau_i$, the direct part of its supplied center then changes by $k_i\Delta$, where

$$k_i = 1 - \ell_i \rho_e \quad (\text{S}), \quad k_i = 1 - \ell_i / b_* \quad (\text{M}).$$

These coefficients satisfy $|k_i|/\tau_i \leq C$. For S, $\rho_i \leq \rho_e$ gives $1 - \rho_i \rho_e \leq 2(1 - \rho_i) \leq 2\tau_i$. For M, its two possible coefficients obey

$$0 \leq 1 - \rho_i \cos z / b_* \leq C(1 - \rho_i) + Cz^2,$$

with $z = 0$ when there is no cosine. Use $1 - \rho_i \leq \tau_i$, $\tau_i \geq cR$, and $z^2/R \leq C\delta^{7t} \rightarrow 0$.

Shift that child's untransformed seed block by

$$\frac{k_i}{\tau_i} P_i^{-1} v_i \otimes \frac{\Delta}{r}. \quad (\text{K.8})$$

If (K.4) holds for the child, the smallest eigenvalue of P_i is at least $\sqrt{3}/2$, so $\|P_i^{-1}\|_{\text{op}} \leq 2$. Formula (K.8) induces exactly the child's shift for center displacement $k_i\Delta$. Leave all residual noise slots fixed. Induction from leaf expressions upward shows that every mean output is unchanged, and therefore that the actual supplied-center change is precisely its direct-term change. All queries are preserved. The parent's final direct term is unchanged for M and changes by $-\Delta/r$ for S, giving the required output behavior.

The center-supplying subtree and the invoked child have disjoint slots. In particular, the factor k_i/τ_i in (K.8) multiplies only the shift of that child, not the shifts of routines used to supply its center. Those are other direct children of the same parent and are charged once in its total shift vector.

Source attribution. Adapted from OpenAI [Ope26, Lemma 7.3].

Lemma K.3 (Well-defined construction and query cap). *One can choose $w > 0$, depending only on K_0, t and the fixed Picard counts, and then choose the interpolation orders, so that the entire construction is well defined for all sufficiently large d . Every routine reached from a top call satisfies (K.4). Its total number of seed slots and oracle queries is at most $\text{polylog}(d)\delta^{-t}$ on every execution.*

Proof. Let D_0 be the dependency-depth bound in Theorem J.7, and let F_0 be its direct fan-out bound. Expanding a circuit into an expression tree costs at most $(D_0 + 1)(1 + F_0)^{D_0}$ nodes. Its size is thus $\text{polylog}(d)\delta^{-C_2 w}$, where C_2 depends on the Picard counts but not on w or the interpolation orders. This bounds its direct child invocations and its own anchor and residual slots. Expanding through the height H from Lemma K.2 gives $\text{polylog}(d)\delta^{-C_3 w}$ slots and terminal queries. Interpolation orders change only fixed multiplicative constants in these bounds, not C_2 or C_3 .

The terminal shift has norm 1. Assuming the shift bound for the children, the bounded ratios and inverse norms above show that a parent's shift norm is at most a fixed constant times one plus the sum of its direct children's shift norms, including the anchor call. The extra reserved-slot shift has bounded coefficient. The triangle inequality is sufficient even if ordinary and extra anchor shifts affect the same slot. Iterating through the fixed routine height gives the same bound $\text{polylog}(d)\delta^{-C_3 w}$, after increasing C_3 if necessary.

Choose $w > 0$ with

$$C_3 w < t/2, \quad w < t, \quad c_1 w < t,$$

and only then choose the fixed interpolation orders needed in Theorem J.7. For sufficiently large d , the last shift bound is smaller than $D_s = \delta^{-t}$. This is a valid bottom-up induction: it proves each child's bound before using its inverse transformation in the parent. The count bound is also at most $\text{polylog}(d)\delta^{-t}$.

To make the parameter order explicit, first fix the Picard counts, hence D_0, H, C_3 , then w , then the interpolation orders. The universal scalar D_s fixes every center-noise budget. The circuit shapes, labels, requested noises, and slot counts are determined by these scalar parameters before any shifts are computed. Compute child shifts before their matrices and the parent shift. No center value, oracle response, numerical-error estimate, or assumption (K.2) enters this finite construction or the count. Only terminal routines query g , once each; hence the bound is a cap on every execution, not an expected query count. $\square$

K.6 Terminal errors and propagation of scales

Termination and query counts no longer depend on the accuracy analysis. We now prove the terminal bounds and verify that the numerical size conditions hold for every reachable child.

Source attribution. Adapted from OpenAI [Ope26, Lemma 7.4].

Lemma K.4 (Terminal accuracy). *In standalone use, a terminal call satisfying (K.2) obeys*

$$\begin{aligned} W_2(\text{Law}(\mathbf{M}), \mathbf{N}(u, \sigma^2 I_d)) &\leq \text{polylog}(d) \sigma \delta^p D_F(x), \\ W_2(\text{Law}(\mathbf{S}_{\text{pre}}), \nu_{x,r} * \mathbf{N}(0, 3\tau^2 I_d/4)) &\leq \text{polylog}(d) \delta^p D_F(x), \end{aligned}$$

where $\mathbf{S}_{\text{pre}}$ is the output before adding its reserved noise $\tau G_2/2$. The sampling bound also holds after adding that same independent reserve to both laws. The fixed-seed center Lipschitz constant is at most λ for $\mathbf{M}$ and $r\lambda$ for $\mathbf{S}$.

Proof. For the standard Gaussian seed Z , the primitive moment bounds give

$$\|g(x + rZ) - u\|_{L^2} \leq CL(\sqrt{d} + r|g(x)|) \leq CLD_F(x).$$

Since $L/\sigma \leq \text{polylog}(d)\delta^b$ and $p \leq b$, this is within the mean error budget; couple the added Gaussian identically.

Let z_0 be the mode of $\nu_{x,r}$, so $z_0 = -rg(x + rz_0)$. Then

$$\|Z - rg(x + rZ) - (Z + z_0)\|_{L^2} \leq l\|Z - z_0\|_{L^2} \leq ClD_F(x).$$

Also $W_2(\mathbf{N}(z_0, I_d), \nu_{x,r}) \leq Cl\sqrt{d}$. Apply the synchronous comparison in Lemma F.2 with Gaussian potential $H_0(z) = |z - z_0|^2/2$ (contraction constant 1) and tilted potential $H(z) = |z|^2/2 + F(x + rz)$. Their gradient difference is $r[g(x + rz) - g(x + rz_0)]$, whose $L^2(\nu_{x,r})$ norm is at most $l\|Z - z_0\|_{L^2(\nu_{x,r})} \leq Cl\sqrt{d}$. This proves the displayed Wasserstein bound under exactly the C^2 and bounded-Hessian hypotheses of that lemma. Adding the same independent comparison noises proves the sampling error $ClD_F(x) \leq \text{polylog}(d)\tau\delta^b D_F(x)$, which is at most the requested bound because $\tau \leq 1$ and $p \leq b$. Omitting G_2 proves the pre-reserved assertion. The Lipschitz bounds follow directly from the two displayed terminal formulas. $\square$

Source attribution. Adapted from OpenAI [Ope26, Lemma 7.5].

Lemma K.5 (Size conditions descend through the routine tree). *Starting with a top call satisfying (K.2), every descendant satisfies that condition with its assigned labels, up to a uniform polylogarithmic factor. At each nonterminal circuit,*

$$\lambda A \leq \text{polylog}(d)\delta^{1/3}. \tag{K.9}$$

For nonfinal expression terms within an $\mathbf{M}$ parent, $\sigma_i \leq C\sigma$; within an $\mathbf{S}$ parent, $r\sigma_i \leq C$.

Proof. For a final term of an M parent, the output uses only means of scale rR , so $r_i = rR$, while σ_i is a fixed positive multiple of σ . Thus L_i/σ_i gains a factor R and l_i gains R^2 , allowing labels $(b+t, p)$. For a final term of an S parent, σ_i is a fixed positive multiple of τ/r . Therefore $L_i/\sigma_i \leq Cl/\tau$ and $l_i \leq l$, allowing the unchanged labels.

For a nonfinal term, let $r\tau_{\text{prev}}$ be the scale of the node whose center is currently being supplied. Its expression is evaluated with budget $r\tau_{\text{prev}}/(2D_s)$, irrespective of the tier at which it occurs. Consequently

$$\sigma_i = \frac{r\tau_{\text{prev}}}{4D_s A}, \quad \frac{L_i}{\sigma_i} \leq \frac{CD_s \lambda A}{R}.$$

The coefficient estimates give $\lambda A \leq \text{polylog}(d)\delta^{b-4t-c_1w}$ for an M parent. For an S parent, $\lambda A = C_1 l \leq \text{polylog}(d)\delta^b$, since $\tau \leq 1$. Using $D_s/R = \delta^{-2t}$ and $c_1w < t$ costs less than the allowed $10t$ decrease in b . Also $l_i \leq l$. For the position-anchor S call of M, $l/\tau \leq l/R$, which has the same allowance. This proves the size conditions by the bounded routine height. All accumulated fixed and polylogarithmic factors are still one polylogarithmic factor.

As $b > 1/2$ and $4t + c_1w < 5t < 1/6$, the coefficient bounds imply (K.9). Finally, the positive declared bound for A includes a fixed multiple of r/σ in an M parent, so $\sigma_i \leq r/(4D_s A) \leq C\sigma$. For S, $A = C_1 r^2$ gives $r\sigma_i \leq 1/(4C_1 D_s) \leq C$. $\square$

K.7 Stability for arbitrary fixed seeds

Source attribution. Adapted from OpenAI [Ope26, Lemma 7.6].

Lemma K.6 (Fixed-seed stability). *For every seed array, including correlated or transformed arrays, the compiled M output is a function of x with Lipschitz constant at most $\text{polylog}(d)\lambda$; the normalized S output has constant at most $\text{polylog}(d)r\lambda$. If the anchors of a nonterminal M are treated as separate inputs, its Lipschitz constant in the position anchor is at most $\text{polylog}(d)L$.*

Proof. Induct on routine height, with the terminal case supplied by Lemma K.4. First hold the anchors fixed. A direct center term has Lipschitz constant 1 in x and at most Cr in a position anchor. The center-supplying expression combines child outputs with total absolute coefficient at most A . Using the child M bound, the contribution of earlier supplied-center sensitivities is at most $\text{polylog}(d)\lambda A$ times their maximum. By (K.9), this is small. Induction through the bounded expression depth therefore bounds center sensitivities by a constant times their direct sensitivities, with an allowed polylogarithmic factor.

At the final expression the coefficient sum is A_f . Its direct term has no center dependence when anchors are held fixed; for M it also has no position-anchor dependence. Hence its center sensitivity is at most $\text{polylog}(d)A_f\lambda$, and its M position-anchor sensitivity at most $\text{polylog}(d)A_f\lambda r$. These are the asserted bounds with anchors treated as separate inputs. The S Gaussian anchor is indeed center independent. The M position anchor is supplied by S at the same center, whose center Lipschitz constant by induction is $\text{polylog}(d)r\lambda$. Its additional contribution is at most $\text{polylog}(d)Lr\lambda = \text{polylog}(d)L^2$, and $L^2 \leq \lambda$ for $r \leq 1$ and $\lambda \leq 1$. All seed transformations depend only on scalar parameters, not on x . The proof uses only fixed-seed identities and Lipschitz bounds, and therefore requires no seed independence. $\square$

K.8 Standalone accuracy and the primitive theorem

We prove the law guarantees by induction on routine height. A standalone call uses independent standard Gaussian seeds. A child in the actual computation receives transformed seeds, so we invoke its standalone bound only after applying the absorption identity.

Source attribution. Adapted from OpenAI [Ope26, Proposition 7.7].

Proposition K.7 (Accuracy of the compiled routines). *For every call reached from the top labels, the standalone routines satisfy*

$$\begin{aligned} W_2(\text{Law}(\mathbf{M}), \mathbf{N}(u, \sigma^2 I_d)) &\leq \text{polylog}(d) \sigma \delta^p D_F(x), \\ W_2(\text{Law}(\mathbf{S}_{\text{pre}}), \nu_{x,r} * \mathbf{N}(0, 3\tau^2 I_d/4)) &\leq \text{polylog}(d) \delta^p D_F(x), \end{aligned}$$

where $\mathbf{S}_{\text{pre}}$ denotes the output before its reserved noise. The latter bound also holds after adding the reserved noise to both laws, and

$$\|\text{Law}(\mathbf{S}) - \nu_{x,r} * \mathbf{N}(0, \tau^2 I_d)\|_{\text{TV}} \leq \text{polylog}(d) \frac{\delta^p D_F(x)}{\tau}.$$

Proof. We induct on the finite routine height. The terminal case is Lemma K.4. The actual outputs have finite second moments because their computations are finite compositions of affine maps and globally Lipschitz oracle evaluations. The ideal outputs have the moments in (F.6). Thus each coupling below is well defined.

For a nonterminal $\mathbf{M}$, first couple its approximate $\mathbf{S}$ position anchor to an ideal anchor, independently of all remaining seed blocks. The child labels are $(b - 10t, p - 1/4)$. By induction and Lemma K.6, replacing this anchor costs at most

$$\text{polylog}(d) L \delta^{p-1/4} D_F(x).$$

After division by the mean error unit $\sigma \delta^p D_F(x)$, the cost is at most $\text{polylog}(d) \delta^{b-1/4}$, hence is bounded for large d . We may now assume that the position anchor has law $b_* Z + RN$, where $Z \sim \nu_{x,r}$ and N is standard Gaussian, and that the momentum anchor is an independent standard Gaussian. These are exactly the input laws in Theorem J.7.

Let x_i be the centers of the deterministic exact-mean circuit under these anchors, and put $D_i = \sqrt{d} + |g(x_i)|$. Their RMS values over anchors obey

$$\max_i \|D_i\|_{L^2} \leq \text{polylog}(d) D_F(x). \quad (\text{K.10})$$

Indeed, both position and momentum anchors have RMS at most $CD_F(x)$. Use $|g(x_i)| \leq |g(x)| + \lambda|x_i - x|$ and the circuit's direct terms. An exact primitive mean at a preceding center has magnitude at most CD_j . Their contribution to this maximum-of-RMS recurrence is at most $C\lambda A$ times its previous maximum. This is small by (K.9); the direct terms cost at most $CD_F(x)$. Induction through the node ordering proves (K.10) without a maximum inside the expectation.

Condition on the anchors. For an expression E , write ϵ_E for the W_2 error of $\text{Eval}(E, n)$ relative to its deterministic exact value plus $\mathbf{N}(0, n^2 I_d)$. Couple the supplied center of its i th term to $x_i + n_i Z$ with its conditional W_2 error ϵ_{E_i} . Use the same transformed child seeds on these two centers, drawn independently of this coupling. The fixed-seed Lipschitz bound costs at most $\text{polylog}(d) \lambda \epsilon_{E_i}$. At the ideal noisy center, Lemma K.1 identifies the child's law with its standalone law at x_i . The induction hypothesis therefore supplies the additional error $\text{polylog}(d) \sigma_i \delta^{p_i} D_i$.

Different term occurrences have disjoint blocks, so their outputs are independent conditional on the anchors. Product couplings compare them to independent noised exact means. Together with the independent residual in (K.7), the ideal sum has exactly covariance $n^2 I_d$. We conclude that

$$\epsilon_E \leq \sum_i |a_i| [\text{polylog}(d) \lambda \epsilon_{E_i} + \text{polylog}(d) \sigma_i \delta^{p_i} D_i]. \quad (\text{K.11})$$

In (K.11), the first term uses pathwise stability, and the second uses standalone accuracy after absorption. This order avoids assigning ideal laws to the actual intermediate states.

Take RMS in the anchors and expand (K.11) through the finite expression tree. At its root, direct-child labels have $p_i = p$ and

$$\sum_i |a_i| \sigma_i \leq A_f \frac{n}{2A_f} = n/2.$$

Thus their intrinsic errors fit the requested budget. At tier $j \geq 2$, absolute coefficient sums and Lipschitz factors contribute at most

$$A_f (\text{polylog}(d) \lambda A)^{j-1}.$$

All such children have the single accuracy-label loss $p_i = p - 1/4$, not one loss per expression tier. Their noise units are at most $C\sigma$ in an M parent and C/r in an S parent by Lemma K.5; these units combine with $A_f = C_1$ or $C_1 r$, respectively. Already the first factor $\text{polylog}(d) \lambda A \leq \text{polylog}(d) \delta^{1/3}$ pays for the loss $\delta^{-1/4}$, leaving a positive power $\delta^{1/12}$. The bounded expression depth allows all further tiers and polylogarithmic factors to be summed. Leaf affine expressions have zero error. This proves the required final evaluation error.

Conditionally on the anchors, the ideal final evaluation is its deterministic exact-mean circuit value plus a Gaussian of the specified fixed covariance. Its noise may therefore be taken independent of the anchors. By Lemma K.5, Theorem J.7 applies. It compares the ideal mean circuit to $\mathbf{N}(u, \sigma^2 I_d/4)$ and the ideal sampling circuit to $\nu_{x,r} * \mathbf{N}(0, \tau^2 I_d/4)$, with the respective errors $\text{polylog}(d) \sigma \delta^J D_F(x)$ and $\text{polylog}(d) \delta^J D_F(x)$. Since $J = K_0 + 2$ and $p \leq K_0$, both errors fit the requested bounds. Couple the independent evaluation noise identically in each comparison. For M, its variance $3\sigma^2/4$ raises the Gaussian target variance from $\sigma^2/4$ to σ^2 . For S, its variance $\tau^2/2$ raises the Gaussian convolution variance from $\tau^2/4$ to $3\tau^2/4$. This closes the standalone induction. Adding the common reserved noise preserves the W_2 bound; Gaussian smoothing by its standard deviation $\tau/2$ gives the stated total variation bound by (F.7). $\square$

Proof of Theorem F.1. Use the top S call with $(b, p, \tau) = (1, K_0, 1)$. Because $l = \delta r^2 \leq \delta$, its size condition holds with constant 1. Proposition K.7 gives both W_2 and total variation error at most $\text{polylog}(d) \delta^{K_0} D_F(x)$ from $\nu_{x,r} * \mathbf{N}(0, I_d)$, uniformly in x and $0 < r \leq 1$. Lemma K.3 gives at most $\text{polylog}(d) \delta^{-t}$ evaluations of g on every execution. The output has the form $X_{\text{pre}} + G/2$, where the reserved standard Gaussian G is independent of X_{pre} and all queries, and the pre-reserved conclusion of Proposition K.7 gives $W_2(\text{Law}(X_{\text{pre}}), \nu_{x,r} * \mathbf{N}(0, 3I_d/4)) \leq \text{polylog}(d) \delta^{K_0} D_F(x)$. The exact means and numerical comparison laws serve only to analyze this finite algorithm; its only oracle operations are the terminal evaluations. $\square$

L The mean-root specialization

This appendix specializes the finite mean formula and recursive evaluation of OpenAI [Ope26, Theorem 6.7, Lemmas 7.2–7.3 and 7.5–7.6, and Proposition 7.7] to the stated mean-root output. The underlying two-type recursive construction is inherited.

We prove Proposition E.1 from the explicit constructions in Sections J and K. This appendix records the change of root and the final Gaussian reserve, because Theorem F.1 is a noisy sampling result rather than a standalone mean theorem.

L.1 The recursive computation

We use the mean routine $M(x, r, \sigma; b, p)$ and sampling routine $S(x, r, \tau; b, p)$ as defined in [Section K](#), with label rules [\(K.3\)](#). Their mean and sampling size conditions are respectively

$$\max\{\delta r^2, \delta r/\sigma\} \leq B_d \delta^b, \quad \delta r^2/\tau \leq B_d \delta^b.$$

A call is terminal when $p \leq b$. For a nonterminal parent, final-expression mean children of a mean routine receive labels $(b + t, p)$, final-expression mean children of a sampling routine receive (b, p) , and all other children, including the mean routine's position anchor, receive $(b - 10t, p - 1/4)$. All expression-tree children are children of that parent routine; following an affine supplying expression does not repeatedly lower the same parent labels.

The ideal mean-circuit theorem, [Theorem J.7](#), produces a finite affine conditional-mean expression whose law is within $B_d \sigma \delta^{p+2} D_F(x)$ in W_2 of $N(m(x), \sigma^2 I/4)$ under the mean size condition. Its prescribed Picard and interpolation orders depend on the fixed labels and parameters. The shift and query-cap, size-propagation, fixed-seed stability, and two-type coupling results, [Lemmas K.3, K.5 and K.6](#) and [Proposition K.7](#), provide the following properties for the recursive evaluation: deterministic seed slots; the exact shift [\(E.4\)](#); shift norm at most δ^{-t} ; cap $B_d \delta^{-t}$; center Lipschitz constant at most $B_d \delta$; and the intended W_2 errors $B_d \sigma \delta^p D_F(x)$ for means and $B_d \delta^p D_F(x)$ for sampling. These properties are uniform in the center and scalar scales when their declared size conditions hold. They do not assert exact unbiasedness or a joint guarantee for exported internal records.

L.2 Starting at a mean routine

Start the same recursion with $M(x, r, \sigma; 1, K_M)$. Its root is legal by [\(E.2\)](#). To check termination independently of the old root type, consider a branch. Before $4K_M + 1$ decreasing transitions, the total reduction of b is less than $10t(4K_M + 1) < 1/4$. At that many transitions, $p < 0$ and the call is terminal. Thus $b > 3/4$ along every nonterminal part of a branch. The number of increasing transitions is bounded because each increases b by t while a nonterminal call has $b < p \leq K_M$. A neutral transition is from S to M ; another such transition requires a decreasing transition to a new S . Hence the height is bounded by a constant depending only on K_M, t .

The proofs of [Lemmas K.2 and K.3](#) depend on this height, the fixed circuit orders, and scalar size formulas, rather than on a sampling root. The size-propagation induction applies with the same root label $b = 1$. The two-type coupling induction can therefore be run at the mean root, yielding [\(E.3\)](#), with the same shift, cap, and fixed-seed stability conclusions. This changes neither a child definition nor a probability law within the recursion.

L.3 Reserving a fresh Gaussian

At a nonterminal mean root, the affine evaluator in [Section K](#) uses final Gaussian budget $n_0 = \sqrt{3}\sigma/2$. If its final scalar coefficients are a_i , with declared absolute sum A_f , each child receives noise $\sigma_i = n_0/(2A_f)$. The independent residual variance is

$$v = n_0^2 - \sum_i a_i^2 \sigma_i^2 \geq 3n_0^2/4 = 9\sigma^2/16.$$

Split it into independent variances $v - \sigma^2/4$ and $\sigma^2/4$, retaining the second until after all queries. This is positive, does not change the output law, and adds no query. At a terminal mean root, the output is $\nabla F(x + rZ) + \sigma G$, so the same splitting is immediate.

For the pre-reserve estimate, repeat the coupling proof of [Proposition K.7](#) before the last independent addition. Fixed-seed stability is unchanged on subtracting this additive noise. Every center coupling and every independent product coupling of child means is unchanged. The ideal affine evaluation covariance drops from $3\sigma^2/4$ to $\sigma^2/2$. The ideal circuit itself approximates $\mathbf{N}(m(x), \sigma^2 I/4)$ by [Theorem J.7](#). Adding the ideal evaluation noise gives covariance $3\sigma^2/4$ before the reserve, with total error $B_d \sigma \delta^{K_M} D_F(x)$. No deconvolution of a Wasserstein bound is used. Finally convolution by the reserved $\sigma G/2$ preserves the W_2 bound and gives TV error at most twice that bound divided by σ . This proves all assertions of [Proposition E.1](#).

M Source correspondence and attribution

The analytic modules in [Sections F to K](#) adapt Sections 2–7 of OpenAI, *Subpolynomial query complexity for well-conditioned log-concave sampling* (September 26, 2026) [[Ope26](#)]. The correspondence below uses the numbering of the [pinned source version](#). These inherited results and their proofs are not claimed as new results of this manuscript. OpenAI is the immediate source of the adapted statement and proof text, not necessarily the original source of every standard mathematical ingredient; the earlier-source citations are retained. The assumptions and qualifications remain in the statements; the table records provenance rather than asserting verbatim identity of the expositions.

In particular, the independent Gaussian reserve and the sampling pre-reserve comparison in [Theorem F.1](#) and [Proposition K.7](#) are already part of OpenAI’s Theorem 2.1 and Proposition 7.7. The underlying finite mean and sampling formulas in [Theorem J.7](#) are adapted from its Theorem 6.7.

The mean-root specialization in [Section L](#) starts the inherited recursion at its mean routine and proves the stated mean-output guarantees. It is a specialization of the construction in OpenAI [[Ope26](#), Sections 6–7], rather than an independent invention of the underlying noisy-mean or sampling machinery. The posterior routine in [Proposition E.3](#) adapts the conditional-descent construction of [[Ope26](#), Section 8.2], with the arbitrary-center estimates and terminal rule stated here; related posterior descent appears in [[CCLZ26c](#), Algorithm 3.3 and Section 6.3]. The retained-noise strengthening and simultaneous-iteration organization in [Sections A and C](#) are manuscript-specific refinements of that inherited construction. The gap-shifted corrector and modified-energy method are attributed to Fan, Li, and Lu [[FLL26](#), Section 2, equations (22)–(23), and Section 3, Lemma 1]; the discrete latent-cycle and dormant-transition estimates are developed in this manuscript.

This manuscript	OpenAI [Ope26]	Result or construction
Theorem F.1	Theorem 2.1	Noisy near-quadratic sampling
Lemma F.2	Lemma 2.2	Uniformly convex laws
Lemma F.3	Lemma 2.3	Conditional comparisons and smoothing
Lemma G.1	Lemma 3.1	All-split cumulant bound
Proposition G.2	Proposition 3.2	Gaussian jet bounds
Proposition G.3	Proposition 3.3	Probability transport
Lemma H.1	Lemma 4.1	Material derivative identities
Lemma H.2	Lemma 4.2	A single contraction
Definition H.3	Definition 4.3	Tensor expressions and their weights
Theorem H.4	Theorem 4.4	Tensor divergence bound
Lemma H.5	Lemma 4.5	Hilbert–Schmidt divergence estimate
Corollary H.6	Corollary 4.6	Derivatives along transported paths
Lemma I.1	Lemma 5.1	Transport derivatives
Lemma I.2	Lemma 5.2	The centering kernel
Proposition I.3	Proposition 5.3	A noisy mean from a path integral
Lemma I.4	Lemma 5.4	Stationary time derivatives
Proposition J.1	Proposition 6.1	Probability-flow circuit
Lemma J.2	Lemma 6.2	Errors in the angle computation
Lemma J.3	Lemma 6.3	Derivative quadrature at zero
Proposition J.4	Proposition 6.4	Kernel-action circuit
Proposition J.5	Proposition 6.5	Mean-flow circuit
Corollary J.6	Corollary 6.6	Accuracy without increasing dependency depth
Theorem J.7	Theorem 6.7	Finite mean and sampling formulas

This manuscript	OpenAI [Ope26]	Result or construction
Lemma K.1	Lemma 7.1	Absorbing a Gaussian center displacement
Lemma K.2	Lemma 7.2	Bounded routine height
Lemma K.3	Lemma 7.3	Well-defined construction and query cap
Lemma K.4	Lemma 7.4	Terminal accuracy
Lemma K.5	Lemma 7.5	Size conditions descend through the routine tree
Lemma K.6	Lemma 7.6	Fixed-seed stability
Proposition K.7	Proposition 7.7	Accuracy of the compiled routines

Adaptation and license notice. The adapted exposition reorganizes the source material, reconciles notation, namespaces labels and citations, and omits unrelated outer-sampler material. These are modifications of the cited OpenAI exposition; the additional mean-root, affine-composition, and outer-sampler arguments are identified separately above. The source correspondence records the immediate provenance of the inherited statements and proofs.

Adapted OpenAI material is reused under the [Apache License 2.0](#). A full copy of the license accompanies the PDFs in the [release repository](#). Existing citations and attribution to other prior work are retained. Reuse of the source does not imply OpenAI endorsement of this manuscript.

References

- [AW15] Radosław Adamczak and Paweł Wolff. Concentration inequalities for non-Lipschitz functions with bounded derivatives of higher order. *Probability Theory and Related Fields*, 162:531–586, 2015. doi: [10.1007/s00440-014-0579-3](https://doi.org/10.1007/s00440-014-0579-3). URL <https://arxiv.org/abs/1304.1826v1>. Statement numbering cited from arXiv:1304.1826v1.
- [Ans04] Michael Anshelevich. Appell polynomials and their relatives. *International Mathematics Research Notices*, 2004(65):3469–3531, 2004. doi: [10.1155/S107379280413345X](https://doi.org/10.1155/S107379280413345X). URL <https://arxiv.org/abs/math/0311043v3>. Preprint version: arXiv:math/0311043v3.
- [BGL14] D. Bakry, I. Gentil and M. Ledoux. *Analysis and Geometry of Markov Diffusion Operators*. Springer, 2014. doi:[10.1007/978-3-319-00227-9](https://doi.org/10.1007/978-3-319-00227-9).
- [CCLZ26a] Fan Chen, Sinho Chewi, Jianfeng Lu, and Matthew S. Zhang. *Accelerated High-Accuracy Sampling from a Warm Start via the Proximal Bouncy Particle Sampler*. [arXiv:2609.06905v1](https://arxiv.org/abs/2609.06905v1), September 7, 2026.
- [CCLZ26b] Fan Chen, Sinho Chewi, Jianfeng Lu, and Matthew S. Zhang. Sharp Gaussian divergence and Bernstein–Markov inequalities, 2026. URL <https://arxiv.org/abs/2609.32851v1>. Version 1, September 26, 2026.
- [CCLZ26c] F. Chen, S. Chewi, J. Lu, and M. S. Zhang. *Smoothed Picard Hamiltonian Monte Carlo*. [arXiv:2609.06906v2](https://arxiv.org/abs/2609.06906v2), September 29, 2026.
- [CFP19] Thomas A. Courtade, Max Fathi, and Ashwin Pananjady. Existence of Stein kernels under a spectral gap, and discrepancy bounds. *Annales de l’Institut Henri Poincaré, Probabilités et Statistiques*, 55(2):777–790, 2019. doi: [10.1214/18-AIHP898](https://doi.org/10.1214/18-AIHP898). URL <https://arxiv.org/abs/1703.07707v2>.
- [DMS15] Jean Dolbeault, Clément Mouhot, and Christian Schmeiser. *Hypocoercivity for linear kinetic equations conserving mass*. *Transactions of the American Mathematical Society* 367 (2015), 3807–3828. [arXiv:1005.1495](https://arxiv.org/abs/1005.1495).
- [FLL26] Z. Fan, B. Li, and J. Lu. *Sharp hypocoercive convergence estimates for underdamped Langevin dynamics via the modified L^2 method*. [arXiv:2604.10068v1](https://arxiv.org/abs/2604.10068v1), April 11, 2026.
- [KS17] Pravesh K. Kothari and Jacob Steinhardt. Better agnostic clustering via relaxed tensor norms, 2017. URL <https://arxiv.org/abs/1711.07465v1>. Version 1, November 20, 2017.
- [LSV18] Yin Tat Lee, Zhao Song, and Santosh S. Vempala. *Algorithmic Theory of ODEs and Sampling from Well-Conditioned Logconcave Densities*. [arXiv:1812.06243v1](https://arxiv.org/abs/1812.06243v1), December 15, 2018.
- [MCF15] Yi-An Ma, Tianqi Chen, and Emily B. Fox. A complete recipe for stochastic gradient MCMC. In *Advances in Neural Information Processing Systems*, volume 28, 2015. URL <https://proceedings.neurips.cc/paper/2015/hash/9a4400501febb2a95e79248486a5f6d3-Abstract.html>.
- [Ope26] OpenAI. *Subpolynomial query complexity for well-conditioned log-concave sampling*. September 26, 2026. [Pinned manuscript and sources](#). The analytic and recursive-implementation appendices adapt its results and proofs.
- [OV00] F. Otto and C. Villani. Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *Journal of Functional Analysis* 173 (2000), 361–400. doi:[10.1006/jfan.1999.3557](https://doi.org/10.1006/jfan.1999.3557).
- [Vil09] C. Villani. *Optimal Transport: Old and New*. Springer, 2009. doi:[10.1007/978-3-540-71050-9](https://doi.org/10.1007/978-3-540-71050-9).